

37MURA

Crossover Outdoor Heat Pump System

Sizes 18K to 60K

SERVICE MANUAL

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SAFETY CONSIDERATIONS

Installing, starting up, and servicing air-conditioning equipment can be hazardous due to system pressures, electrical components, and equipment location (roofs, elevated structures, etc.). Only trained, qualified installers and service mechanics should install, start-up, and service this equipment.

Untrained personnel can perform basic maintenance functions such as coil cleaning. All other operations should be performed by trained service personnel.

When working on the equipment, observe precautions in the product literature and on tags, stickers, and labels attached to the equipment.

Follow all safety codes. Wear safety glasses and work gloves. Keep a quenching cloth and fire extinguisher nearby when brazing. Use care in handling, rigging, and setting bulky equipment.

Read this manual thoroughly and follow all warnings or cautions included in the literature and attached to the unit. Consult local building codes and National Electrical Code (NEC) for special requirements. Recognize safety information. This is the safety-alert symbol . When you see this symbol on the unit and in instructions or manuals, be alert to the potential for personal injury.

Understand these signal words: **DANGER**, **WARNING**, and **CAUTION**. These words are used with the safety-alert symbol. **DANGER** identifies the most serious hazards which **will** result in severe personal injury or death. **WARNING** signifies hazards which **could** result in personal injury or death. **CAUTION** is used to identify unsafe practices which **may** result in minor personal injury or product and property damage. **NOTE** is used to highlight suggestions which **will** result in enhanced installation, reliability, or operation.



WARNING

ELECTRICAL SHOCK HAZARD

Failure to follow this warning could result in personal injury or death.

Before installing, modifying, or servicing system, main electrical disconnect switch must be in the **OFF** position. There may be more than 1 disconnect switch. Lock out and tag switch with a suitable warning label.



WARNING

EXPLOSION HAZARD

Failure to follow this warning could result in death, serious personal injury, and/or property damage. Never use air or gases containing oxygen for leak testing or operating refrigerant compressors.

Pressurized mixtures of air or gases containing oxygen can lead to an explosion.



CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation. Do not bury more than 36 in. (914 mm) of refrigerant pipe in the ground. If any section of pipe is buried, there must be a 6 in. (152 mm) vertical rise to the valve connections on the outdoor units.

If more than the recommended length is buried, refrigerant may migrate to the cooler buried section during extended periods of system shutdown. This causes refrigerant slugging and could possibly damage the compressor at start-up.

INTRODUCTION

This service manual provides the necessary information to service, repair, and maintain the 37MURA family of heat pumps. This manual has an "APPENDICES" with data required to perform troubleshooting. Use the "TABLE of CONTENTS" to locate a desired topic.

MODEL NUMBER NOMENCLATURE

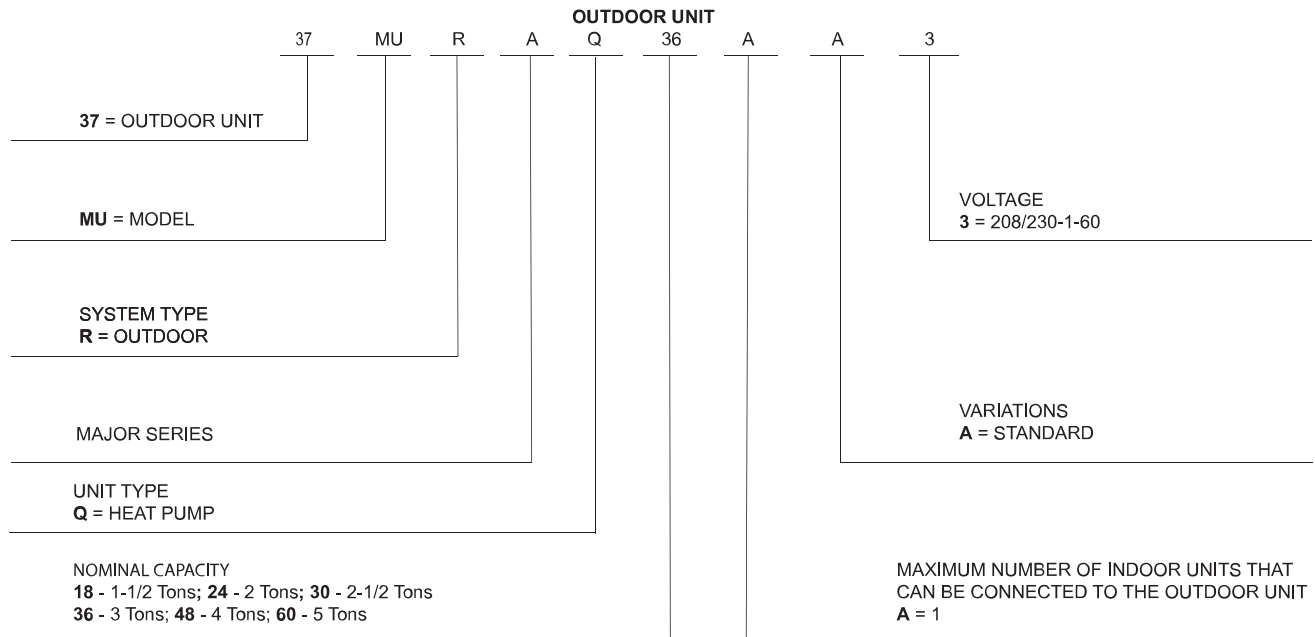


Fig. 1 —Nomenclature

For the Carrier/Bryant brands, the
coded by a two digit week, a two
followed with a 5 digit sequential number. The sequential number for MIDEA is between 10001 and 15000.

b. The SERIAL NUMBER will be as follows:

SERIAL NUMBER will be date
digit year followed by "V"

01 17 V 10001

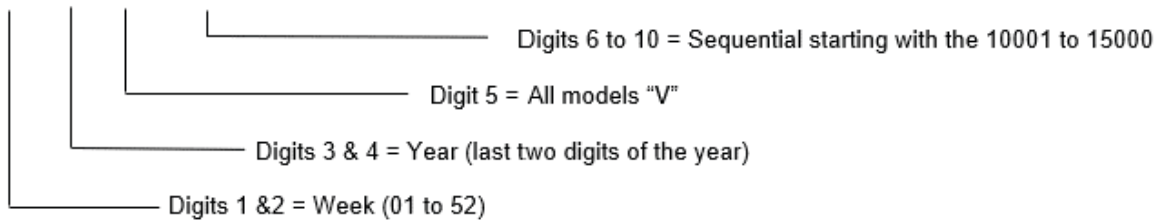


Fig. 2 —Serial Number Nomenclature



Use of the AHRI Certified
TM Mark indicates a
manufacturer's
participation in the
program For verification
of certification for individual
products, go to
www.ahridirectory.org.

DIMENSIONS

Table 1 – Dimensions and Weights

System Size		18K	24K	30K	36K	48K	60K
		(208/230 V)	(208/230 V)	(208/230 V)	(208/230 V)	(208/230 V)	(208/230 V)
Height (H)	in	21.81	26.5	31.89	31.89	38.39	38.39
	mm	554	673	810	810	975	975
Width (W)	in	31.69	35.04	37.24	37.24	38.58	38.58
	mm	805	890	946	946	980	980
Depth (D)	in	12.99	13.46	16.14	16.14	16.34	16.34
	mm	330	342	410	410	415	415
Weight -Net	lbs.	77.16	102.29	141.76	153.22	192.9	192.9
	kg	35	46.4	64.3	69.5	87.5	87.5
Packaging							
Height (H)	in	24.21	29.13	34.84	34.84	42.52	42.52
	mm	615	740	885	885	1080	1080
Width (W)	in	36.02	39.17	42.91	42.91	45.08	45.08
	mm	915	995	1090	1090	1145	1145
Depth (D)	in	14.57	15.67	19.68	19.68	19.68	19.68
	mm	370	398	500	500	500	500
Weight-Gross	lbs.	83.77	109.13	152.34	163.58	224.21	224.21
	kg	38	49.5	69.1	74.2	101.7	101.7

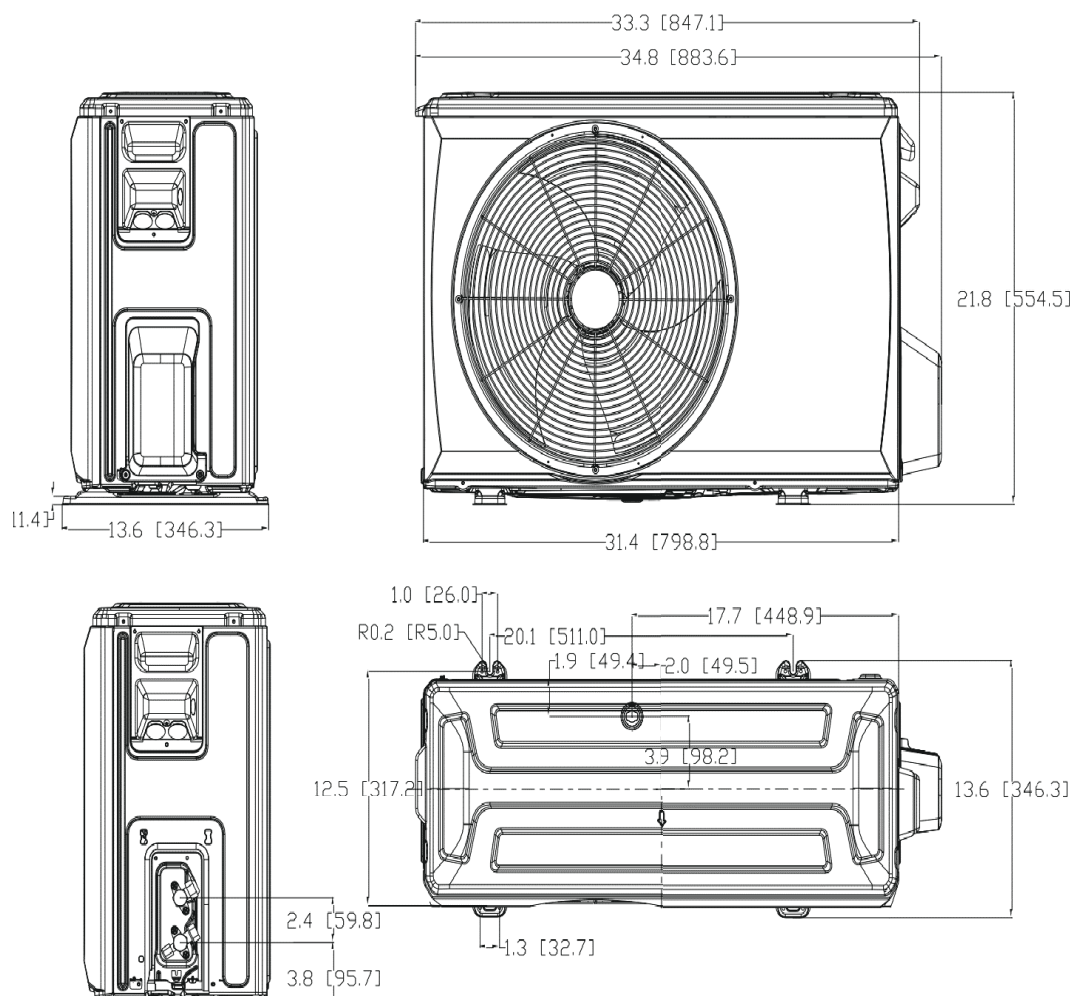


Fig. 3 —Dimensions (18K)

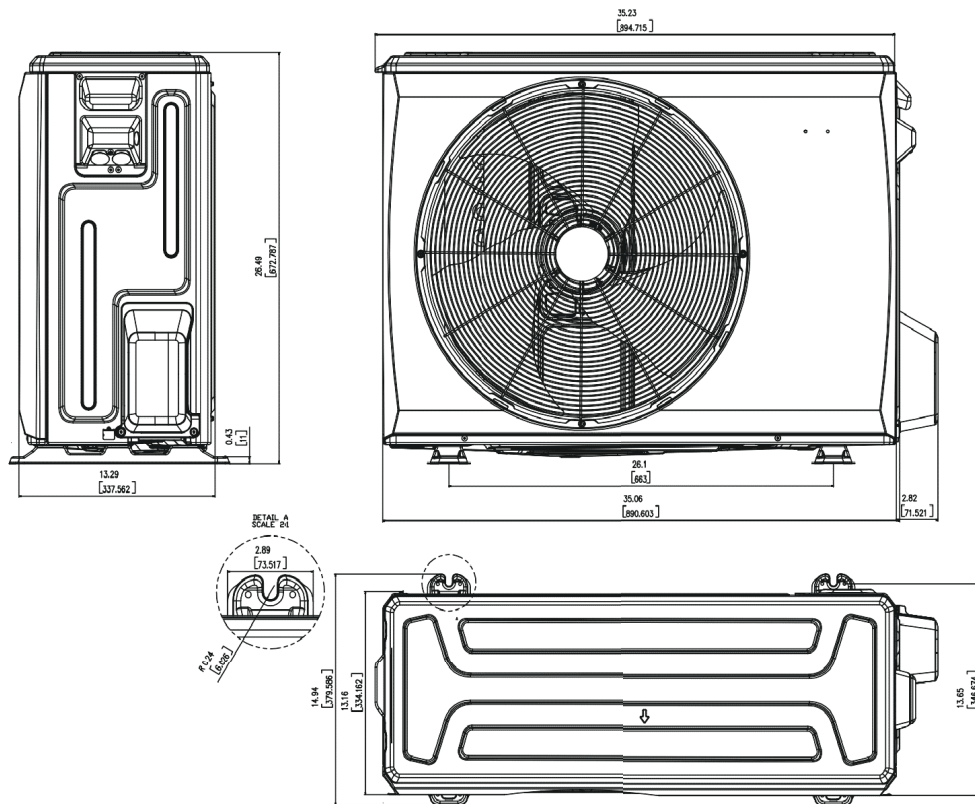


Fig. 4 —Dimensions (24K)

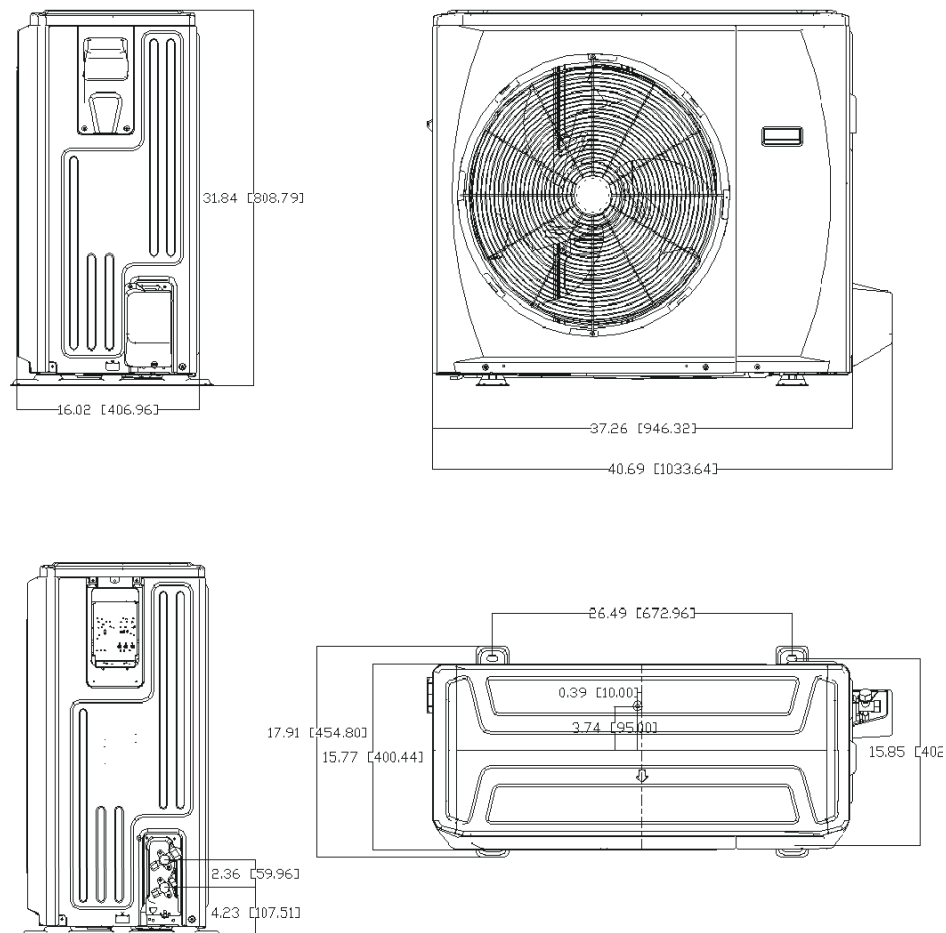


Fig. 5 —Dimensions (30K - 36K)

DIMENSIONS (CONT.)

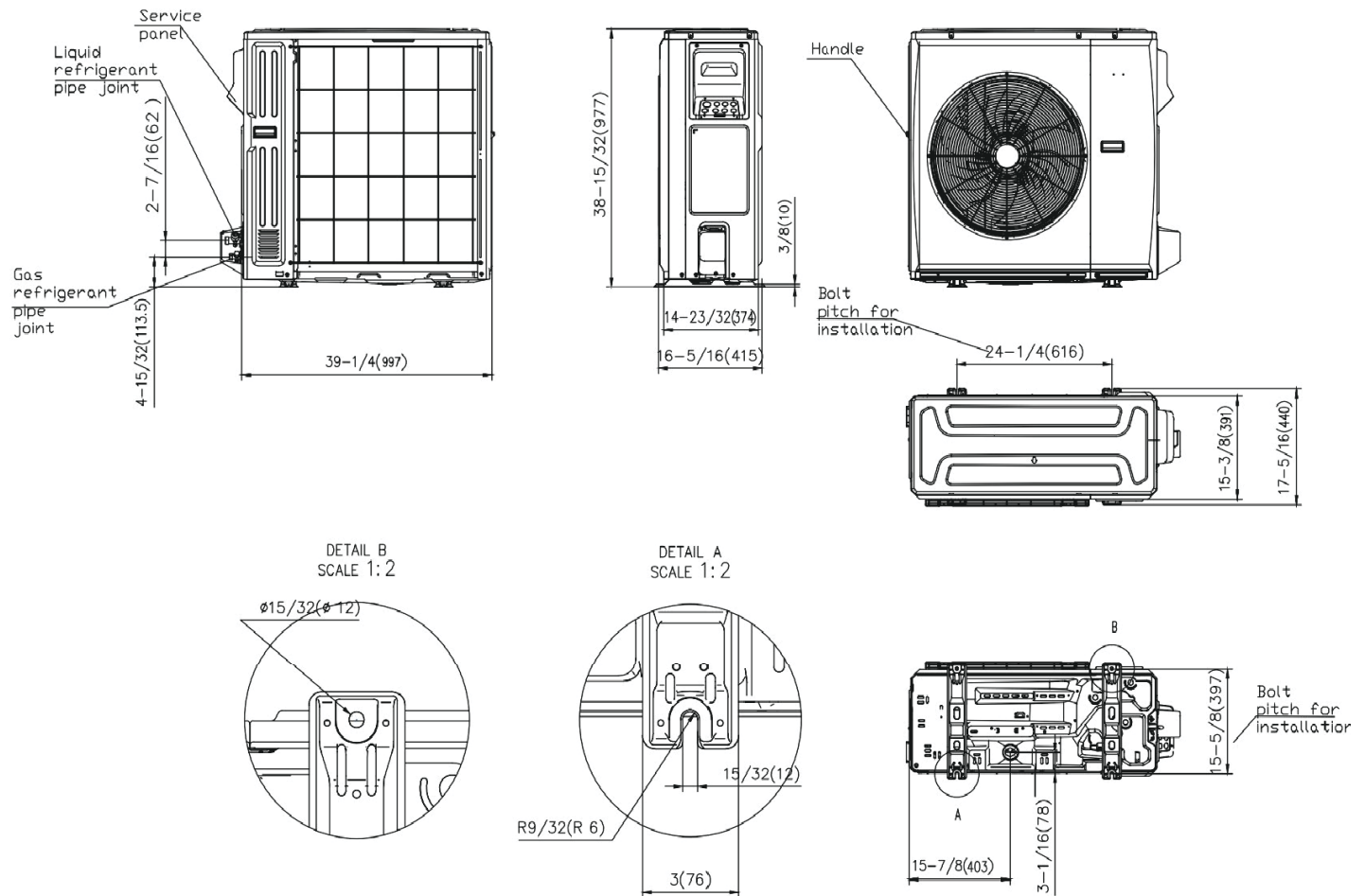


Fig. 6 —Dimensions (48K - 60K)

WIRING

ELECTRICAL WIRING INSTALLATION

Wiring for the outdoor unit must conform to NEC (National Electrical Code) or CEC (Canadian Electrical Code) and local codes. Use Electrical Data table MCA (minimum circuit amps) and MOCP (maximum over current protection) to correctly size the wires and the disconnect fuse or breakers respectively.

All field wiring construction should be finished by a qualified electrician.

Air conditioning equipment must be grounded according to the local electrical codes.

Provide electrical disconnect per local codes.

NOTE: DO NOT connect the power wire to the terminal of the signal wire. Connection of power to any other terminal other than L1 or L2 will cause damage to the control board.

Any control signal cable should be run separately from the power wiring.

Use of metallic conduit or shielded cable is recommended. Maintain a distance of 12 inches(300mm) from the power wiring.

NOTE: DO NOT run the power wiring and control wiring in the same conduit.

Size the wiring in accordance to the NEC / CEC. Select different colors for different wire according to relevant regulations.

Ensure that the wire color of the outdoor and the terminal number are the same as those of the indoor unit.

CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Be sure to comply with local codes while running wire from the indoor unit to the outdoor unit.

Wires should be sized based on NEC and local codes.

Every wire must be connected firmly. Loose wiring may cause the terminal to overheat or result in unit malfunction. A fire hazard may also exist. Ensure all wiring is tightly connected.

No wire should touch the refrigerant tubing, compressor or any moving parts.

Disconnecting means must be provided and shall be located within sight and readily accessible from the air conditioner.

WIRING DIAGRAMS

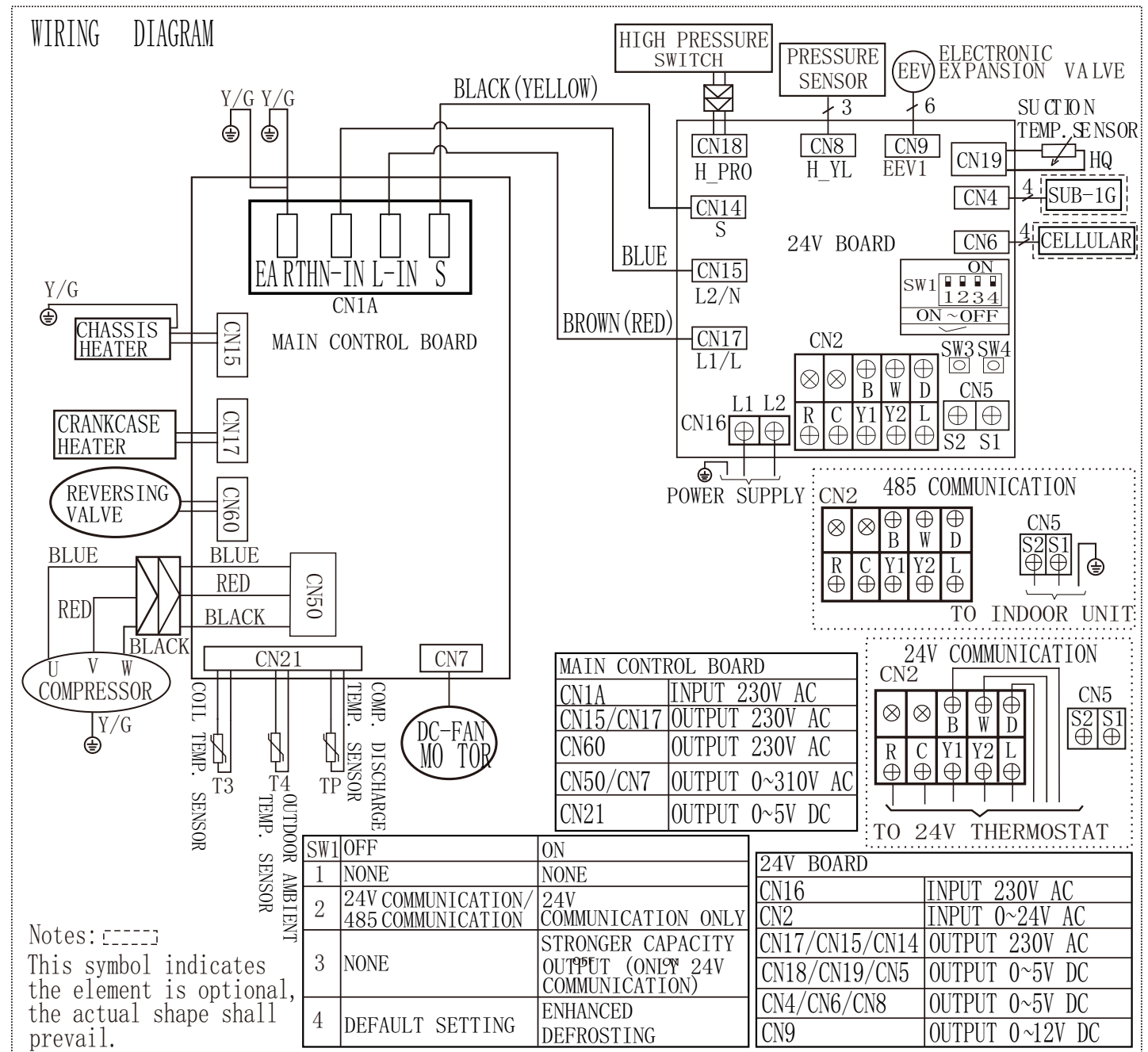


Fig. 7 —Indoor Unit Wiring Diagram (18K)

Table 2 – Terminal Functions and Definitions

TERMINAL MAIN PCB	FUNCTION DEFINITION	TERMINAL 24V COMMUNICATION PLATE	FUNCTION DEFINITION
CN1A	Input 208/230VAC, external power input terminal	CN16	Input 208/230VAC, 24V small board power input terminal
CN15	Output 208/230VAC, chassis electric heating port	CN17/15/14	Output 208/230VAC, communication port with main board
CN17	Output 208/230VAC, compressor electric heating port	CN18	Output 5V, pressure sensor port
CN60	Output 208/230VAC, four-way valve port	CN19	Output 5V, pressure sensor port
CN50	Output 310V, compressor port	CN5	Output 5V, 485 communication port
CN7	Output 310V, DC fan port	CN9	Output 12V, electronic expansion valve port
CN21	Output 5V, sensor port	CN2	Input 24V, 24V communication port

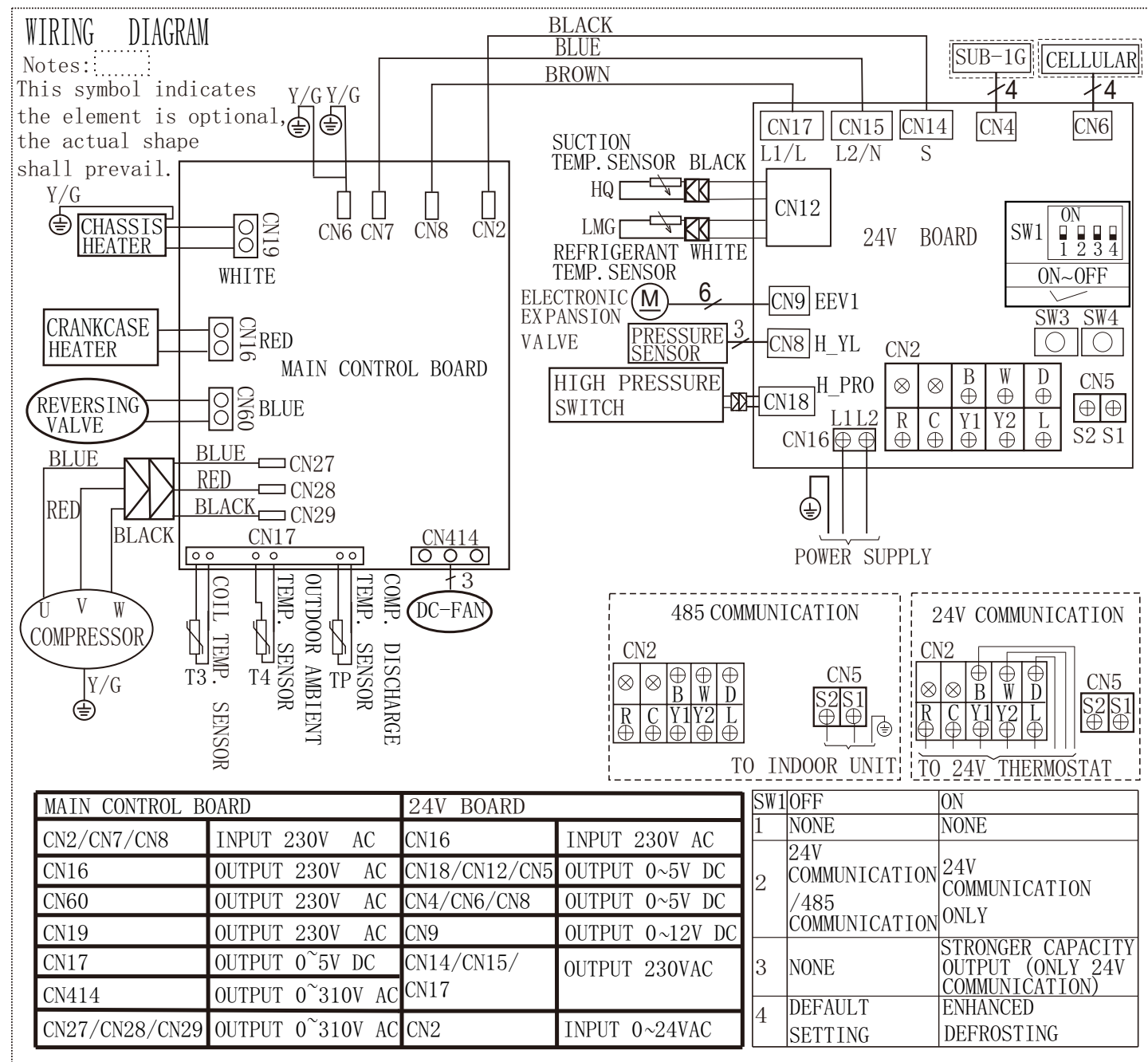


Fig. 8 —Indoor Unit Wiring Diagram (24K)

Table 3 – Terminal Functions and Definitions

TERMINAL MAIN PCB	FUNCTION DEFINITION	24V COMMUNICATION PLATE	FUNCTION DEFINITION
CN2/7/8	Input 208/230VAC, external power input terminal	CN16	Input 208/230VAC, 24V small board power input terminal
CN19	Output 208/230VAC, chassis electric heating port	CN17/15/14	Output 208/230VAC, communication port with main board
CN16	Output 208/230VAC, compressor electric heating port	CN18	Output 5V, pressure sensor port
CN60	Output 208/230VAC, four-way valve port	CN19	Output 5V, pressure sensor port
CN27/28/29	Output 310V, compressor port	CN5	Output 5V, 485 communication port
CN414	Output 310V, DC fan port	CN9	Output 12V, electronic expansion valve port
CN17	Output 5V, sensor port	CN2	Input 24V, 24V communication port

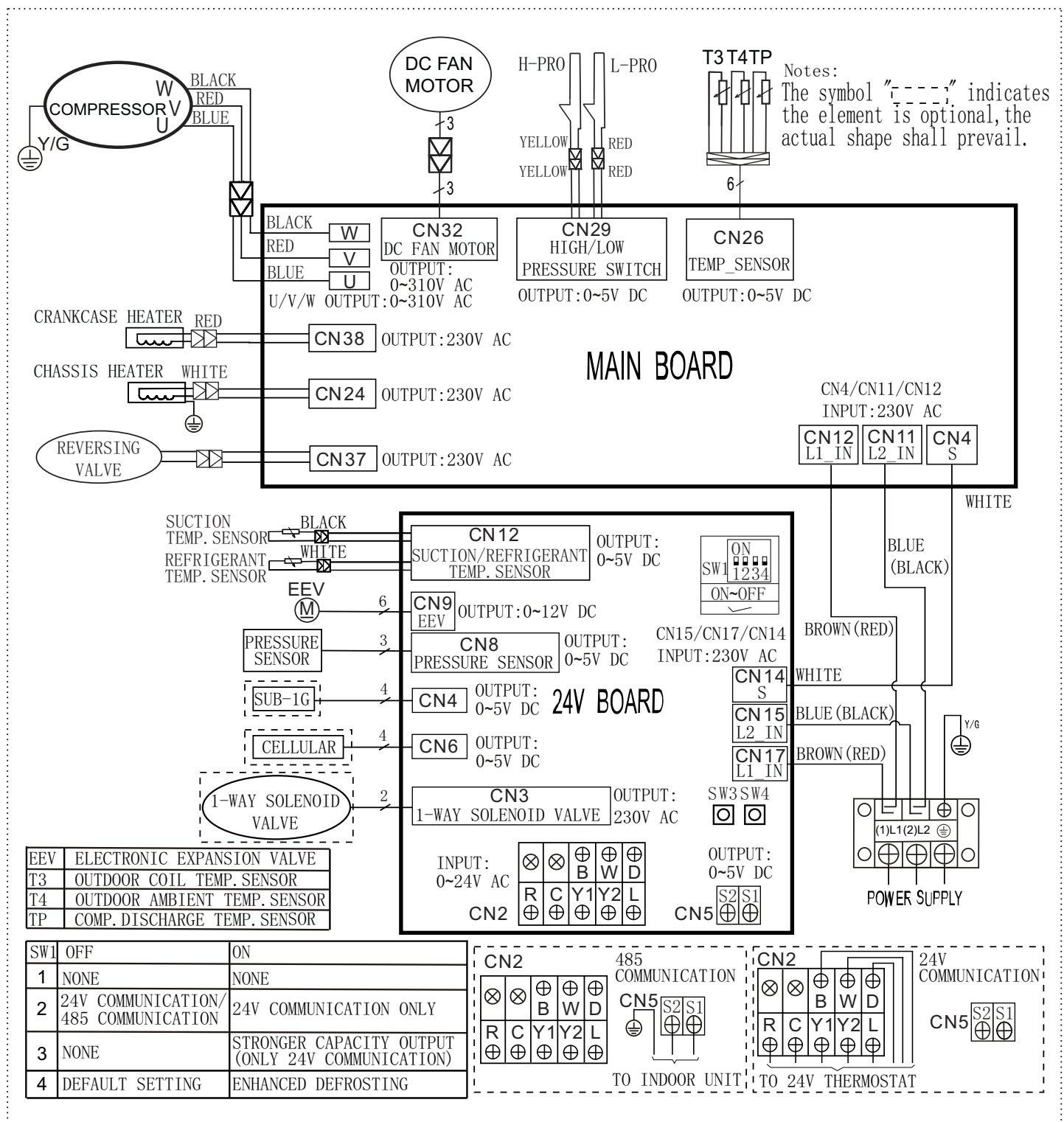


Fig. 9 —Indoor Unit Wiring Diagram (30K)

Table 4 – Terminal Functions and Definitions

TERMINAL MAIN PCB	FUNCTION DEFINITION	TERMINAL 24V COMMUNICATION PLATE	FUNCTION DEFINITION
CN11/CN12	Input 208/230VAC, external power input terminal	CN15/CN17	Input 208/230VAC, 24V small board power input terminal
CN4	Input 208/230VAC, communication port with 24V small board	CN14	Input 208/230VAC, communication port with main board
CN26	Output 5V, T3/T4/TP Sensor port	CN12	Output 5V, return air sensor, condensation protection sensor port
CN29	Output 5V, high and low switch port	CN9	Output 12V, electronic expansion valve port
CN32	Output 310V, DC fan port	CN8	Output 5V, pressure sensor port
W/U/V	Output 310V, compressor port	CN2	Input 24V, 24V communication port
CN38	Output 208/230VAC, compressor electric heating port	CN5	Output 5V, 485 communication port
CN24	Output 208/230VAC, chassis electric heating port		
CN37	Output 208/230VAC, four-way valve port		

- Notes: 1. 36K regular heat model does not have the COMP. TOP OLP TEMP. SENSOR.
 2. The symbol "-----" indicates the element is optional, the actual shape shall prevail.

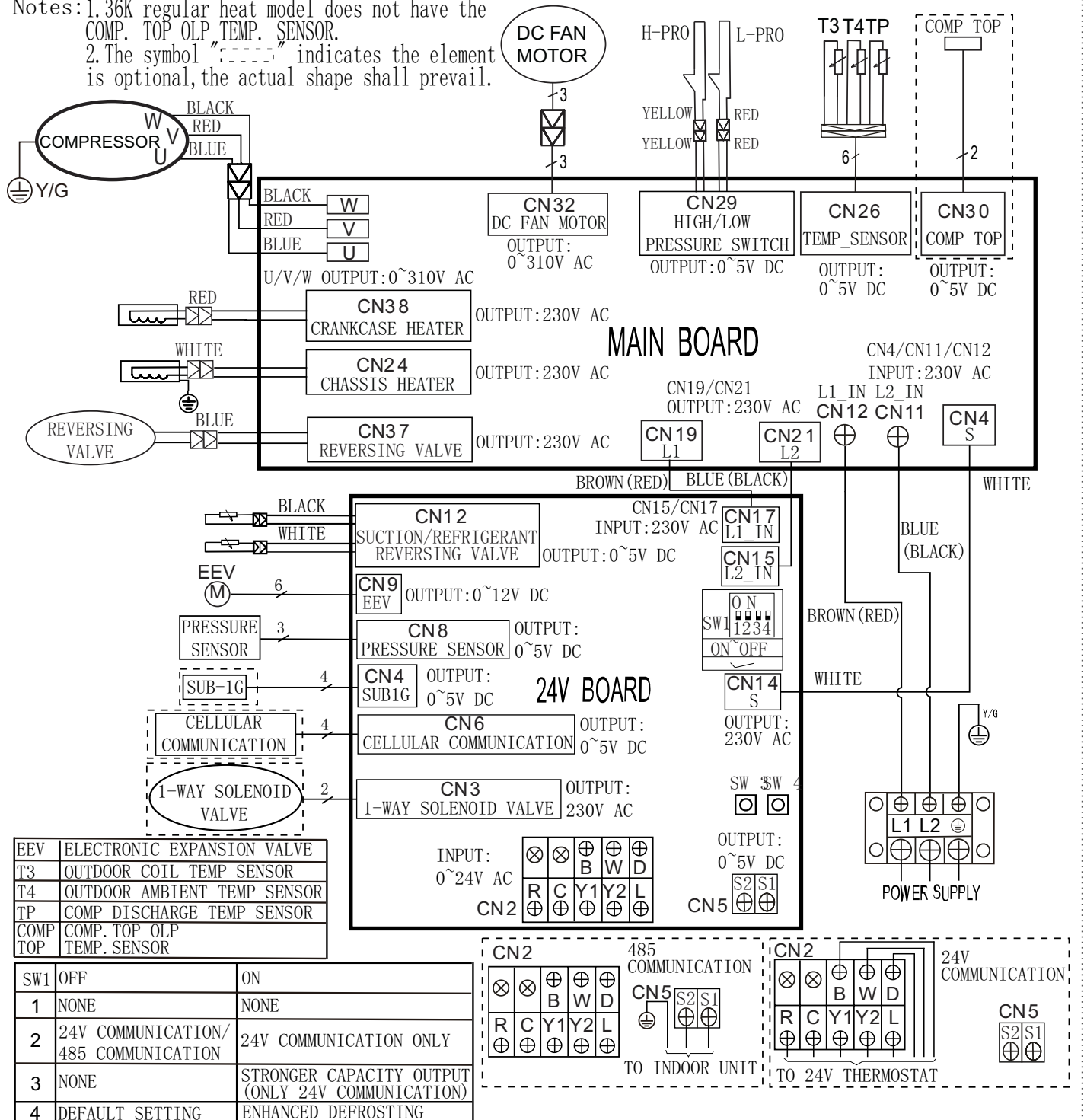


Fig. 10 —Indoor Unit Wiring Diagram (36K)

Table 5 – Terminal Functions and Definitions

TERMINAL MAIN PCB	FUNCTION DEFINITION	TERMINAL 24V COMMUNICATION PLATE	FUNCTION DEFINITION
CN11/CN12	Input 208/230VAC, external power input terminal	CN15/CN17	Input 208/230VAC, 24V small board power input terminal
CN4	Input 208/230VAC, communication port with 24V small board	CN14	Input 208/230VAC, communication port with mainboard
CN26	Output 5V, T3/T4/TPSensor port	CN12	Output 5V, return air sensor, condensation protection sensor port
CN29	Output 5V, high and low switch port	CN9	Output 12V, electronic expansion valve port
CN32	Output 310V, DC fan port	CN8	Output 5V, pressure sensor port
W/U/V	Output 310V, compressor port	CN2	Input 24V, 24V communication port
CN38	Output 208/230VAC, compressor electric heating port	CN5	Output 5V, 485 communication port
CN24	Output 208/230VAC, chassis electric heating port	CN3	Output 208/230VAC, single-way solenoid valve port
CN37	Output 208/230VAC, four-way valve port		
CN19/CN21	Output 208/230VAC, external unit supplies power to 24V small board output terminal		

Notes:

1. COMP. TOP OLP TEMP. SENSOR and EEV2 are only for 36K hyper heat model.
2. The symbol "-----" indicates the element is optional, the actual shape shall prevail.

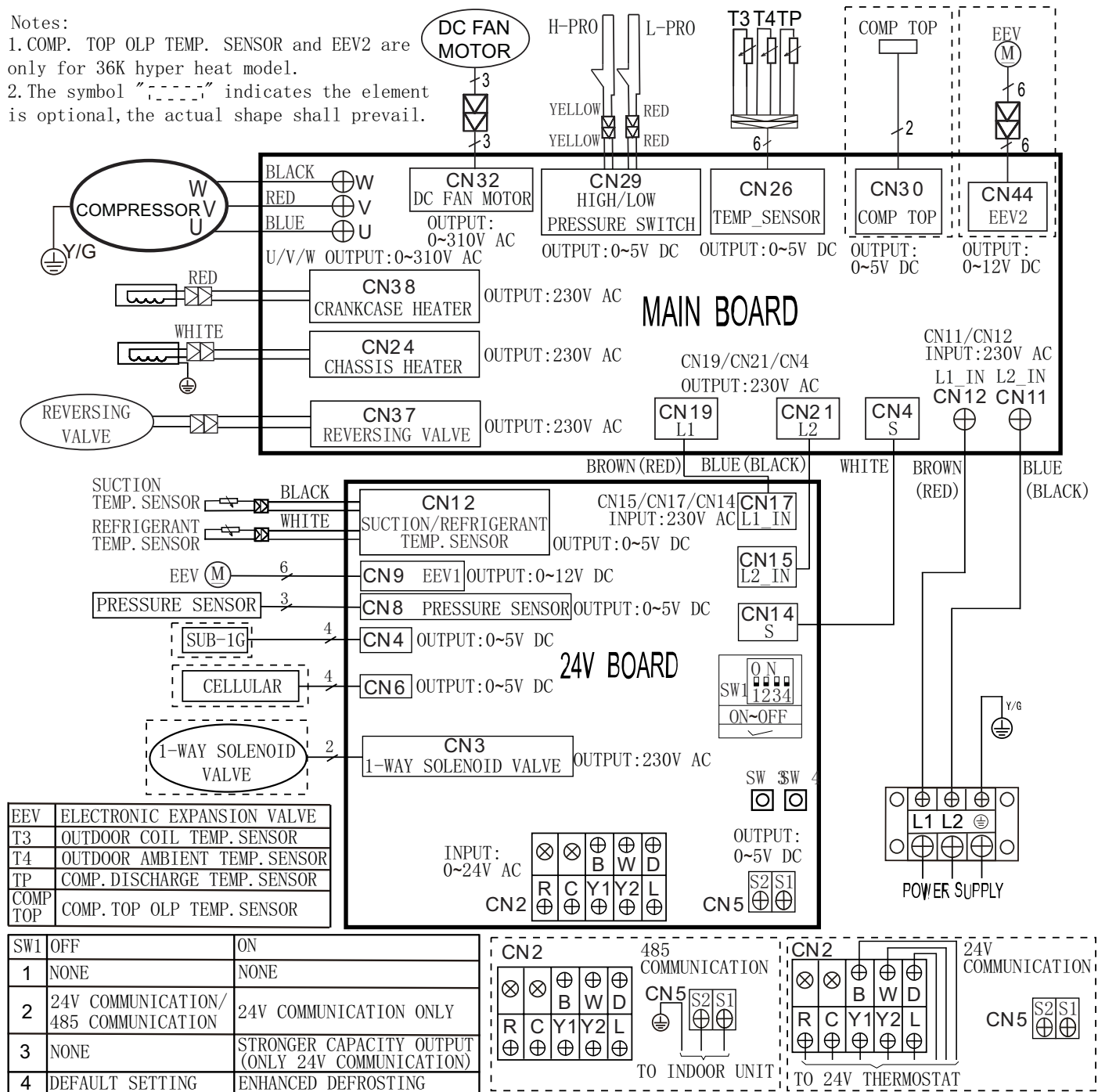


Fig. 11 —Indoor Unit Wiring Diagram (48K and 60K)

Table 6 – Terminal Functions and Definitions

TERMINAL MAIN PCB	FUNCTION DEFINITION	TERMINAL 24V COMMUNICATION PLATE	FUNCTION DEFINITION
CN11/CN12	Input 208/230VAC, external power input terminal	CN15/CN17	Input 208/230VAC, 24V small board power input terminal
CN4	Input 208/230VAC, communication port with 24V small board	CN14	Input 208/230VAC, communication port with main board
CN26	Output 5V, T3/T4/TP Sensor port	CN12	Output 5V, return air sensor, condensation protection sensor port
CN29	Output 5V, high and low switch port	CN9	Output 12V, electronic expansion valve port
CN32	Output 310V, DC fan port	CN8	Output 5V, pressure sensor port
W/U/V	Output 310V, compressor port	CN2	Input 24V, 24V communication port
CN38	Output 208/230VAC, compressor electric heating port	CN5	Output 5V, 485 communication port
CN24	Output 208/230VAC, chassis electric heating port	CN3	Output 208/230VAC, single-way solenoid valve port
CN37	Output 208/230VAC, four-way valve port		
CN19/CN21	Output 208/230VAC, external unit supplies power to 24V small board output terminal		
CN44	Output 12V, electronic expansion valve port		

REFRIGERANT CYCLE DIAGRAM

18K

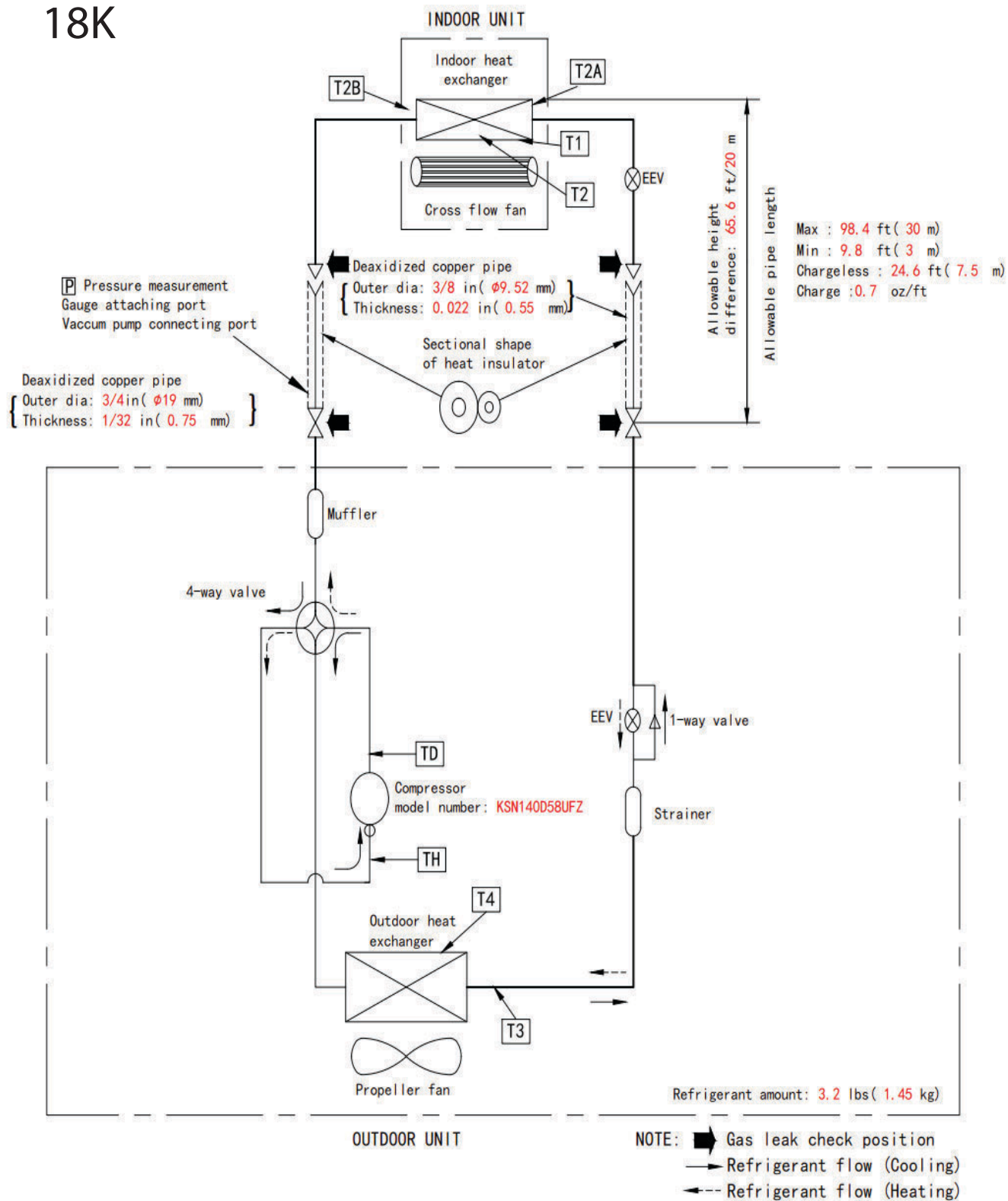


Fig. 12 —Refrigerant Cycle Diagram (Size 18K)

24K

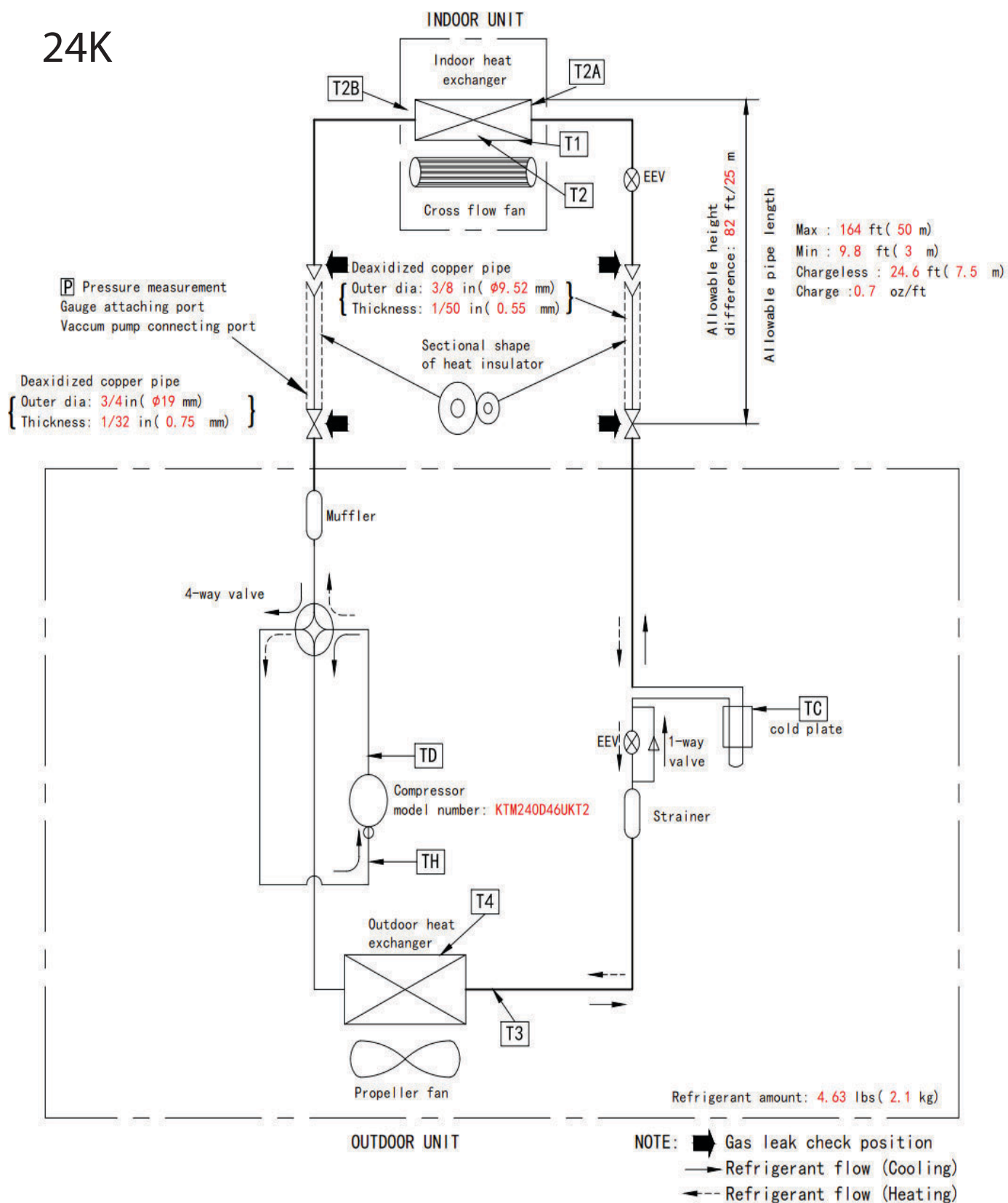


Fig. 13 —Refrigerant Cycle Diagram (Size 24K)

30K

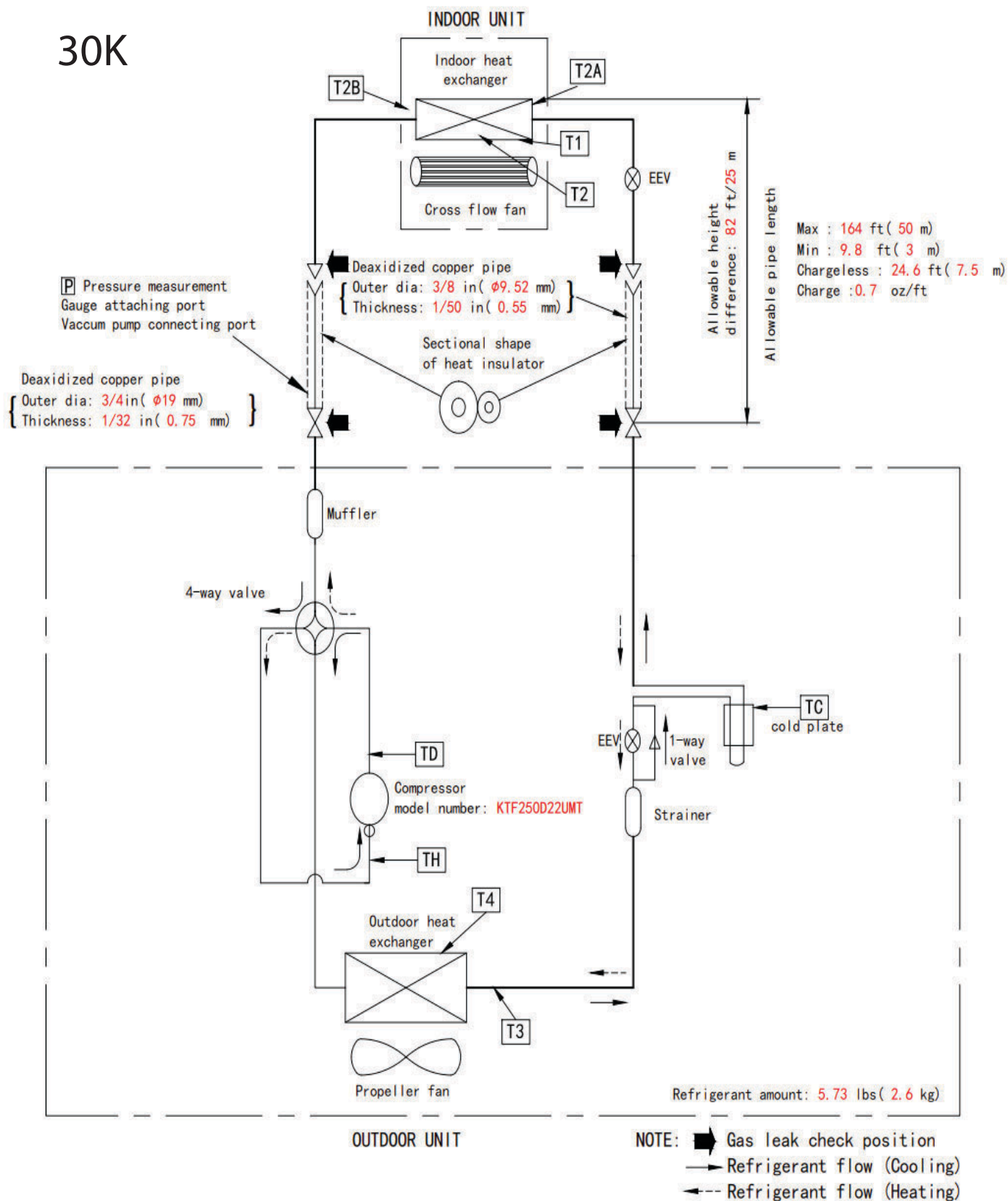


Fig. 14 —Refrigerant Cycle Diagram (Size 30K)

36K

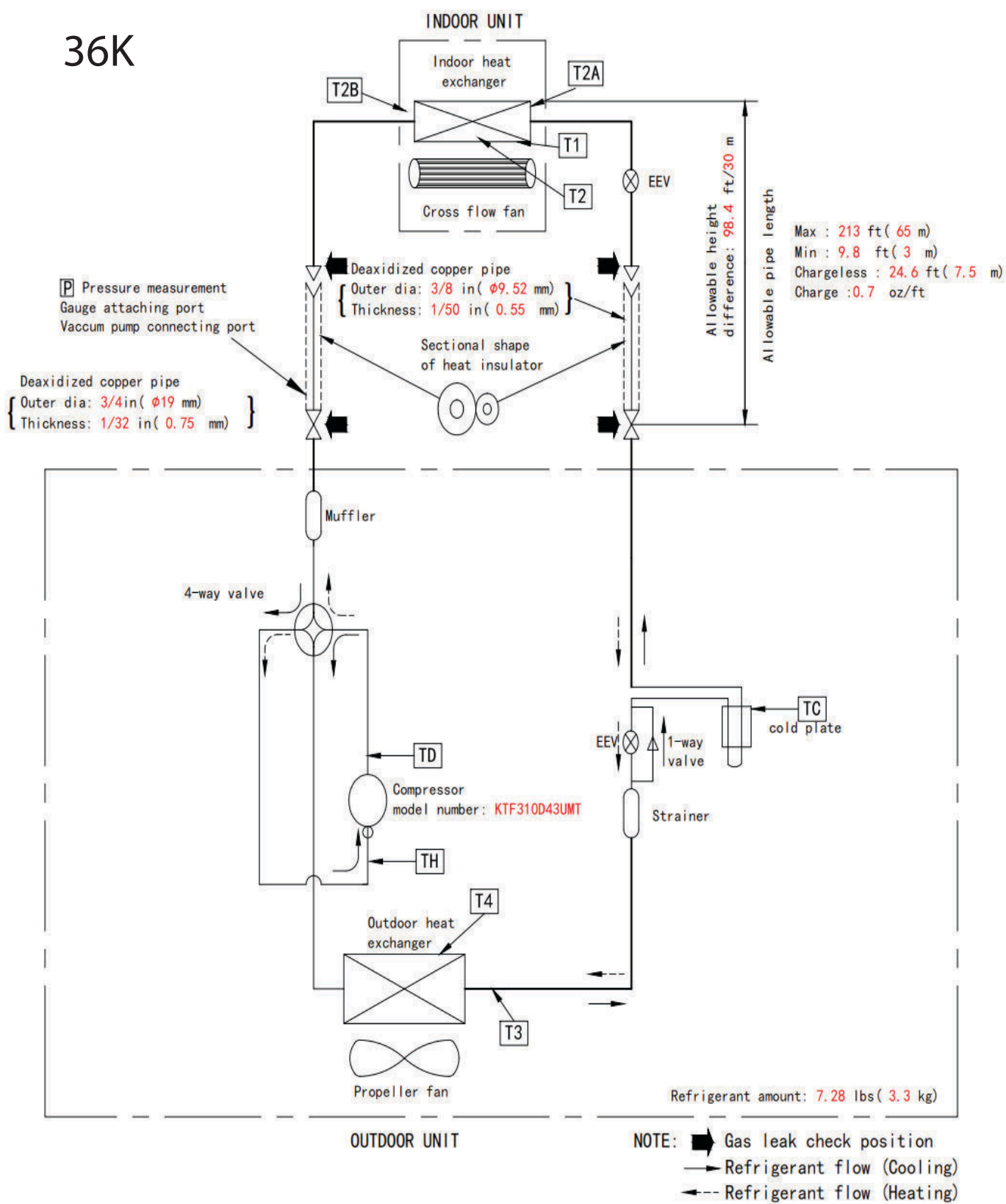


Fig. 15 —Refrigerant Cycle Diagram (Size 36K)

48K/60K

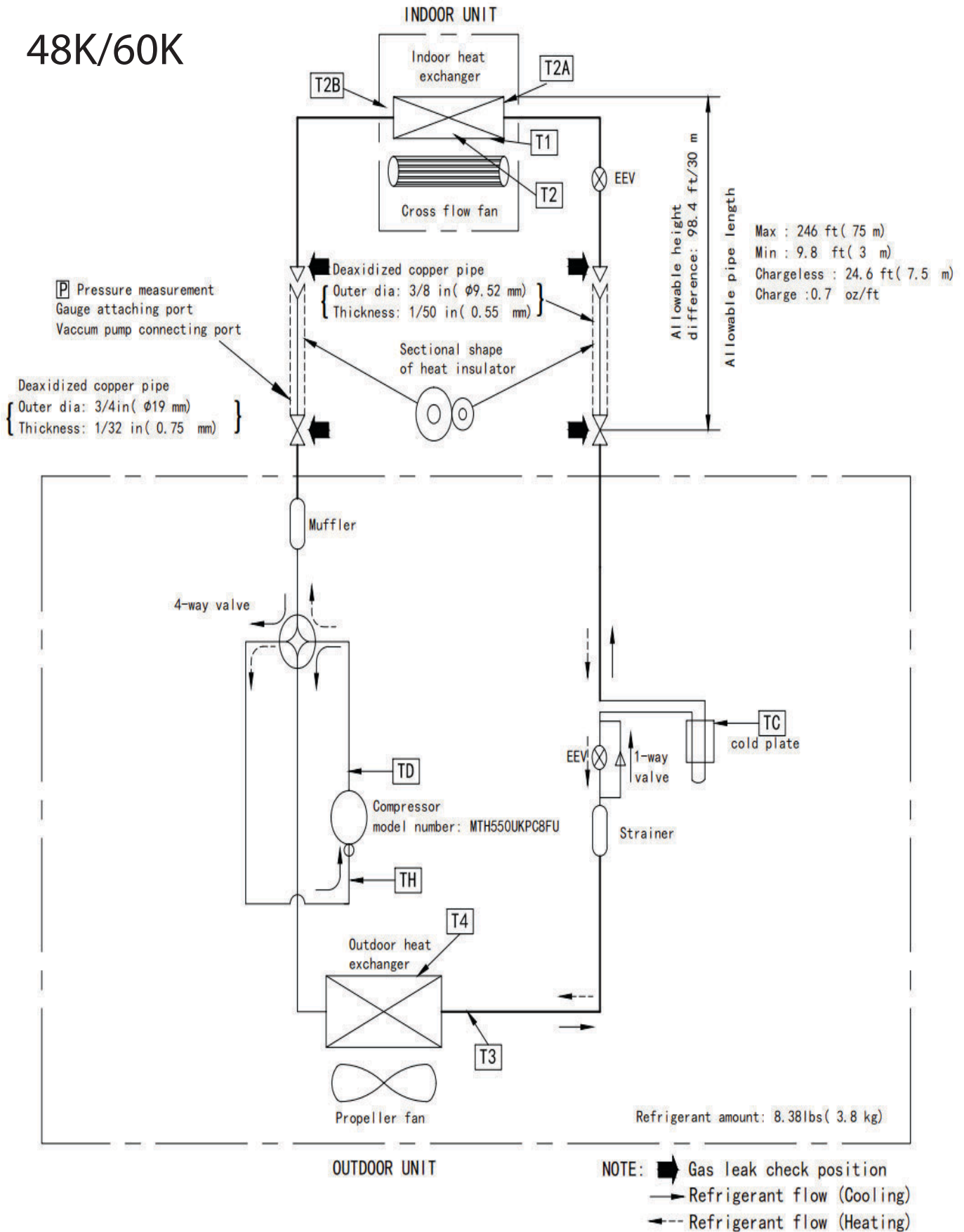


Fig. 16 —Refrigerant Cycle Diagram (Sizes 48K and 60K)

REFRIGERANT LINES

General Refrigerant Line Sizing

1. The outdoor units are shipped with a full charge of R-454B refrigerant. All charges, line sizing, and capacities are based on runs of 25ft. (7.6 m). For runs over 25 ft. (7.6 m), consult [Table 7](#), below, for the proper charge adjustments.
2. The minimum refrigerant line length between the indoor and outdoor units is 10 ft. (3 m).
3. Refrigerant lines should not be buried in the ground. If it is necessary to bury the lines, not more than 36 in (914 mm) should be buried. Provide a minimum 6in (152 mm) vertical rise to the service valves to prevent refrigerant migration.
4. Suction line must be insulated. Use a minimum of 1/2in. (12.7 mm) thick insulation. Closed-cell insulation is recommended in all long-line applications.
5. Special consideration should be given to isolating interconnecting tubing from the building structure. Isolate the tubing so vibration or noise is not transmitted into the structure.

NOTE: [Table 7](#) displays the following maximum lengths allowed.

Table 7 — Max Length and Drop Height Based on Model

System Size		18K	24K	30K	36K	48K	60K
208/230V							
Min. Piping Length	ft. (m)	10 (3)					
Standard Piping Length	ft. (m)	24.6 (7.5)					
Max. outdoor-indoor height difference (OU higher than IU)	ft. (m)	65.6 (20)	82 (25)	82 (25)	98.4 (30)	98.4 (30)	98.4 (30)
Max. outdoor-indoor height difference (IU higher than OU)	ft. (m)	65.6 (20)	82 (25)	82 (25)	98.4 (30)	98.4 (30)	98.4 (30)
Suction Pipe (size - connection type)	in (mm)	ø3/4" (19)	ø3/4" (19)	ø3/4" (19)	ø3/4" (19)	ø3/4" (19)	ø3/4" (19)
Liquid Pipe (size-connection)	in (mm)	ø3/8" (9.52)					
Refrigerant Type	Type	R454B					
Charge Amount	lb. (kg)	3.2(1.45)	5.73(2.6)	6.61 (3.0)	7.28(3.3)	8.38(3.8)	8.38(3.8)
Additional Refrigerant Charge (when Pipe length > 24.6 ft)	Oz/ft (g/m)	0.7(65)	0.7(65)	0.7(65)	0.7(65)	0.7(65)	0.7(65)
Total Maximum Piping Length per system	ft. (m)	98.42 (30)	164.04 (50)	164.04 (50)	246 (75)	246 (75)	246 (75)

1.The charge amount listed in [Table 7](#) is for piping runs up to 25 ft. (7.6 m).

2.For piping runs longer than 25 ft. (7.6 m), add the refrigerant up to the allowable length as specified in [Table 7](#).

SYSTEM EVACUATION AND CHARGING



CAUTION

UNIT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Never use the system compressor as a vacuum pump.

Refrigerant tubes and the indoor coil should be evacuated using the recommended 500 micron deep vacuum method. The alternate triple evacuation method may be used if the procedure outlined below is followed.

NOTE: Always break a vacuum with dry nitrogen.

USING VACUUM PUMP

1. Completely tighten flare nuts A, B, C, D. Connect the manifold gage charge hose to a charge port of the low side service valve (see Fig. 24).
2. Connect the charge hose to vacuum pump.
3. Fully open the low side of manifold gage (see Fig. 17).
4. Start the vacuum pump.
5. Evacuate using either the deep vacuum or triple evacuation method.
6. After evacuation is complete, fully close the low side of manifold gage and stop the vacuum pump operation.
7. The factory charge contained in the outdoor unit is good for up to 25 ft. (8 m) of line length. For refrigerant lines longer than 25 ft. (8 m), add refrigerant, up to the allowable length.
8. Disconnect the charge hose from the charge connection of the low side service valve.
9. Fully open service valves B and A.
10. Securely tighten the service valve caps.

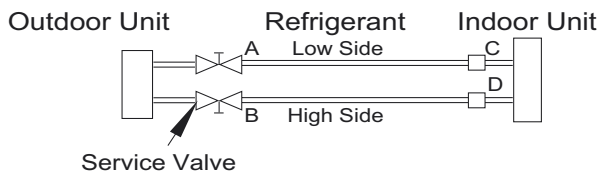


Fig. 17 —Service Valve

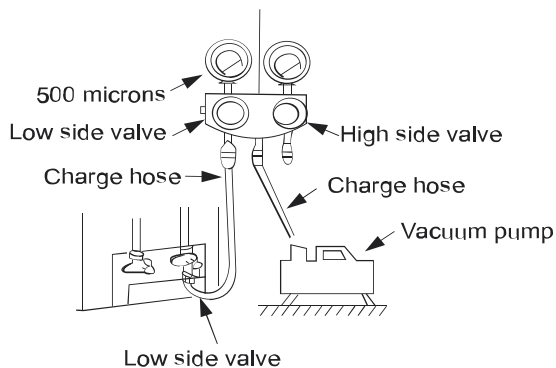


Fig. 18 —Manifold

EVACUATION

Evacuation of the system will remove air or nitrogen (non-condensables) as well as moisture. A proper vacuum will assure a tight, dry system before charging with refrigerant. The two methods used to evacuate a system are the deep vacuum method and the triple vacuum method.

DEEP VACUUM METHOD

The deep vacuum method requires a vacuum pump capable of pulling a vacuum of 500 micron and a vacuum gauge capable of accurately measuring this vacuum depth. This method is the most positive way of assuring a system is free of air and moisture (see Figure 19).

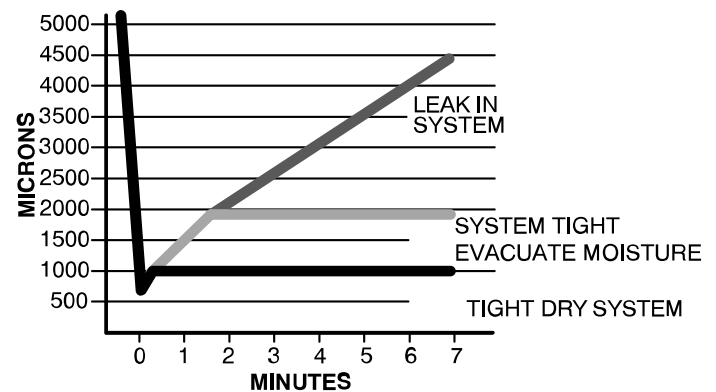


Fig. 19 —Deep Vacuum Graph

TRIPLE EVACUATION METHOD

The triple evacuation method should be used when vacuum pump is not capable of pumping down to 500 microns and system does not contain any liquid water. Refer to Fig. 20 and proceed as follows:

1. Attach refrigeration gauges and evacuate system down to 28 inches of mercury and allow pump to continue operating for an additional 15 minutes.
2. Close service valves and shut off vacuum pump.
3. Connect a nitrogen cylinder and regulator to system and flow nitrogen until system pressure is 2 psig.
4. Close service valve and allow system to stand for 1 hour. During this time, dry nitrogen will be able to diffuse throughout the system absorbing moisture.
5. Repeat this procedure as indicated in Fig. 20. System is now free of any contaminants and water vapor.

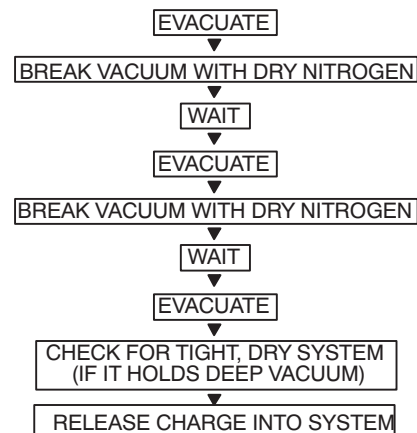


Fig. 20 —Triple Evacuation Method

FINAL TUBING CHECK

Check to be certain factory tubing on both the indoor and outdoor unit has not shifted during shipment. Ensure tubes are not rubbing against each other or any sheet metal. Pay close attention to the feeder tubes, making sure wire ties on feeder tubes are secure and tight.

ELECTRONIC FUNCTIONS

Abbreviation:

Table 8 – Unit element abbreviations

Abbreviation	Element
T1	Indoor room temperature
T2	Coil temperature of evaporator
T3	Coil temperature of condenser
T4	Outdoor ambient temperature
TP	Compressor discharge temperature
Tsc	Adjusted setting temperature
CDIFTEMP	Cooling shutdown temperature
HDIFTEMP2	Heating shutdown temperature
TCDE1	Exit defrost temperature1
TCDE2	Exit defrost temperature2 (maintain for a period of time)
TIMING_DEFROST_ TIME_ADD	Enter defrost time
EE_TIME_DEFROST7_ST RONG	Enter enhanced defrost time
TCDE1_ADD_STRONG	Exit enhanced defrost temperature1
TCDE2_ADD_STRONG	Exit enhanced defrost temperature2 (maintain for a period of time)

In this manual, such as CDIFTEMP, HDIFTEMP2, TCDE1, TCDE2, TIMING_DEFROST_TIME_ADD...etc., they are well-setting parameter of EEPROM.

FAN Mode

When fan mode is activated:

- Outdoor fan and compressor are stopped

COOLING Mode

Compressor Control

Reach the configured temperature:

When the compressor runs continuously for less than 120 minutes.

- If the following conditions are satisfied, the compressor ceases operation.
- Calculated frequency (fb) is less than minimum limit frequency (FminC).
- Compressor runs at FminC more than ten minutes.
- Indoor room temperature(T1) is lower than or equal to (Tsc-CDIFTEMP 0.9°F/-0.5°C)

When the compressor runs continuously for more than 120 minutes.

If the following conditions are satisfied, the compressor ceases operation.

- Calculated frequency (fb) is less than minimum limit frequency (FminC).
- Compressor runs at FminC more than 10 minutes.
- When indoor room temperature (T1) is lower than or equal to (Tsc-CDIFTEMP).

NOTE: CDIFTEMP is EEPROM setting parameter. It is 4°F (2°C) usually.

If one of the following conditions is satisfied, not judge protective time.

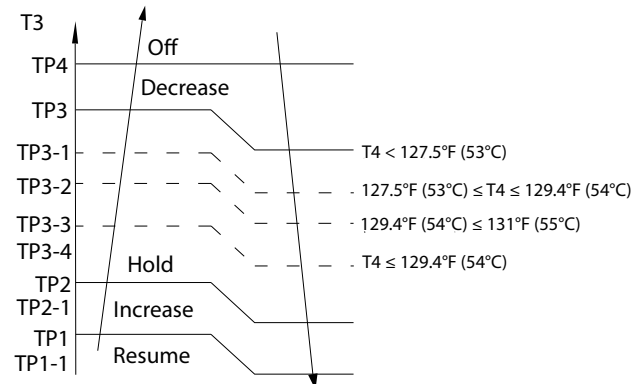
- Compressor running frequency is more than test frequency.

- When compressor running frequency is equal to test frequency, outdoor ambient temperature (T4) is more than 59°F (15°C) or outdoor ambient sensor (T4) fault.
- Change setting temperature.
- Turning on/off turbo or sleep function
- Various frequency limit shutdown occurs.

Outdoor Fan Control

- The outdoor unit will be run at different fan speed according to outdoor ambient temperature(T4) and compressor frequency.
- For different outdoor units, the fan speeds are different.

Condenser Temperature Protection



When the condenser temperature exceeds a configured value, the compressor ceases operation.

Heating Mode (Heat Pump Modes)

Compressor Control

Reach the configured temperature

- If the following conditions are satisfied, the compressor ceases operation.
- Calculated frequency (fb) is less than minimum limit frequency (FminH).
- Compressor runs at FminH more than 10 minutes.
- T1 is higher than or equal to Tsc+ HDIFTEMP2.

NOTE: HDIFTEMP2 is EEPROM setting parameter. It is 4°F (2°C) usually.

- If one of the following conditions is satisfied, not judge protective time.
- Compressor running frequency is more than test frequency.
- Compressor running frequency is equal to test frequency, outdoor ambient temperature (T4) is more than 59°F (15°C) or Outdoor ambient sensor (T4) fault.
- Change setting temperature.
- Turning on/off turbo or sleep function

When the current is higher than the predefined safe value, surge protection is activated, causing the compressor to cease operations.

Outdoor Fan Control

- The outdoor unit will be run at different fan speed according to outdoor ambient temperature (T4) and compressor frequency.
- For different outdoor units, the fan speeds are different.

Defrosting mode

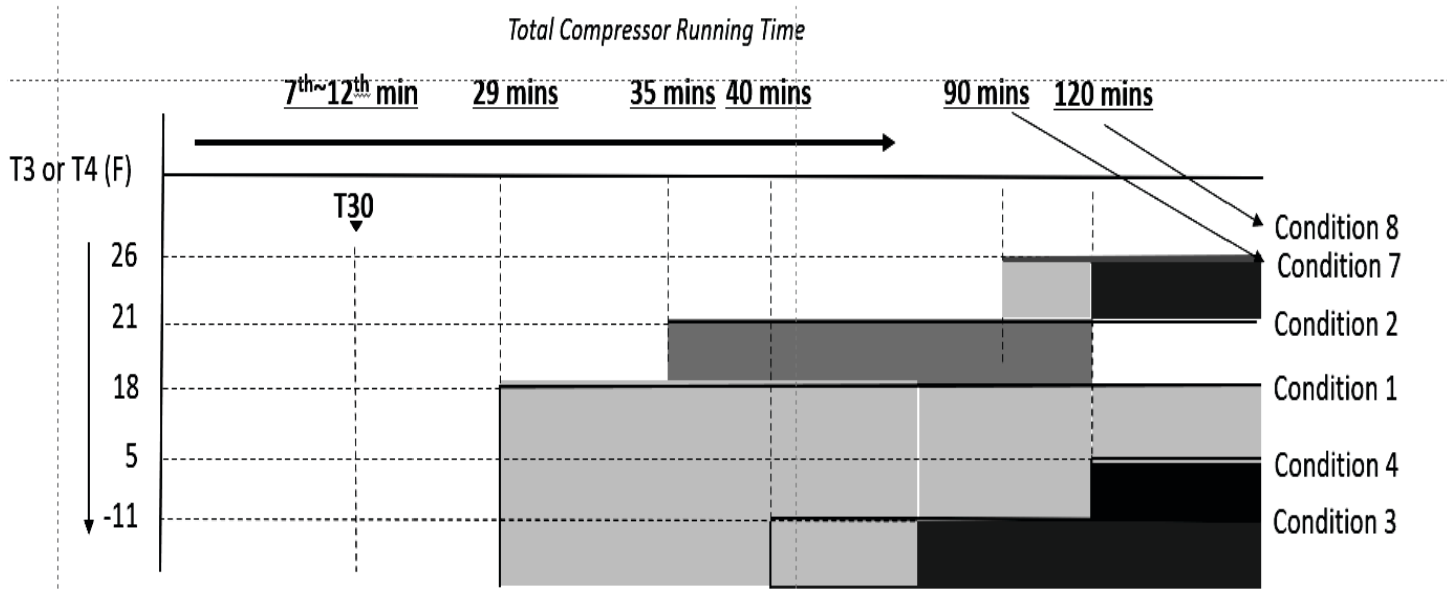
The unit enters defrosting mode according to the temperature value of condenser temperature (T3) and outdoor ambient temperature (T4) as well as the compressor running time.

- In defrosting mode, the compressor continues to run, the indoor and outdoor motor will cease operation, the defrost light of the indoor unit will turn on, and the “dF” symbol is displayed.
- If any one of the following conditions is satisfied, defrosting ends and the machine switches to normal heating mode:
- Condenser temperature (T3) rises above TCDE1.
- Condenser temperature (T3) maintained above TCDE2 for 80 seconds.
- Unit runs for 15 minutes consecutively in defrosting mode.
- If Outdoor ambient temperature (T4) is lower than or equal to -7.6°F (-22°C) and compressor running time is more than TIMING_DEFROST_TIME, if any one of the following conditions is satisfied, defrosting ends and the machine switches to normal heating mode:
- Unit runs for 10 minutes consecutively in defrosting mode.
- Condenser temperature (T3) rises above 50°F/ 10°C.
- If any one of the following conditions is satisfied, the unit enters defrosting mode

- If condenser temperature (T3) or outdoor ambient temperature (T4) is lower than -3°C for 30 seconds, Ts-T1 is lower than 5°C and compressor running time is more than EE_TIME_DEFROST7_ADD.
- If condenser temperature (T3) or outdoor ambient temperature (T4) is lower than -3°C for 30 seconds and compressor running time is more than EE_TIME_DEFROST7_ADD+30 minutes.
- If any one of the following conditions is satisfied, defrosting ends and the machine switches to normal heating mode:
 - Condenser temperature (T3) rises above TCDE1 7.2°F/+4°C.
 - Condenser temperature (T3) maintained above TCDE2 7.2°F/+4°C for 80 seconds.
 - Unit runs for 15 minutes consecutively in defrosting mode.

Enhanced Defrost

Enhanced Defrost is a feature that is enabled by turning on dipswitch SW4 on the 24 volt ODU board. This feature is designed to be used for problem areas of defrosting (near water sources and areas of high humidity during cold outdoor conditions). When enabled, the total heating cumulative run time will be reduced to 40 minutes, allowing for defrosting to occur at the 40th minute providing the T3 or T4 temperature is 26F or cooler at that time. The defrost termination temperature at T3 will also increase to 77F to ensure better completion of defrost prior to terminating defrost.



Condition 1	Total compressor running time is 29 mins	$T3 \leq 18F$ and $T4 > -8F$
Condition 2	Total compressor running time is 35 mins	$T3 \leq 21F$ and $T4 > -8F$
Condition 3	Total compressor running time is 29 mins	$T3 \leq -11F$ and last for 3 mins, and $T4 > -8F$
Condition 4	Total compressor running time is 120 mins	$T3 \leq 5F$ and $T4 > -8F$
Condition 5	Cumulative running 30 mins	$T4 - T3 > (0.5T4 + 5F)$ and $T3 \leq 10F$, $T4 > -8F$
Condition 6	Cumulative running 8 hours	$T4 \leq -8F$, with $T4$ operating without malfunction
Condition 7	Cumulative running time 90 mins and diff. of $T_s - T_1 \leq 9F$ Cumulative running time 120 minutes (if $T_s - T_1$ diff. above not applicable)	$T3$ or $T4 \leq 26F$ last for 30 seconds
*Condition 8	If enhanced defrost is ON, cumulative running time is 40 minutes (T30 does not apply) *Only applies to Crossover ODU's. Enable enhanced defrost by turning on dipswitch 4 in the ODU.	$T3$ or $T4 \leq 26F$ last for 30 seconds

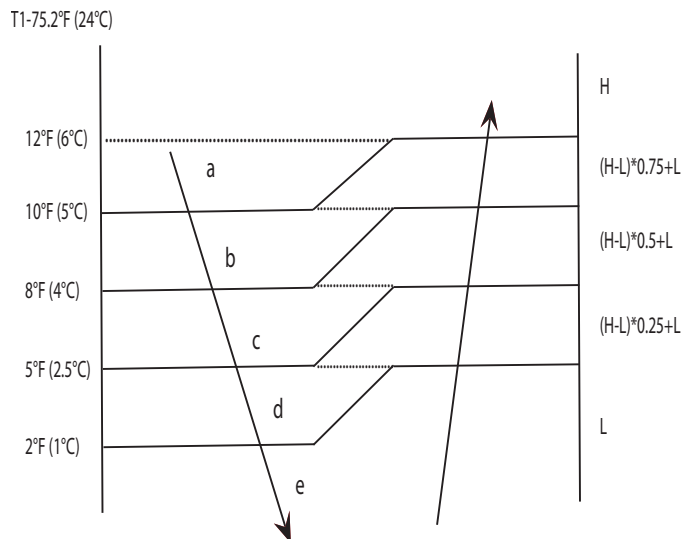
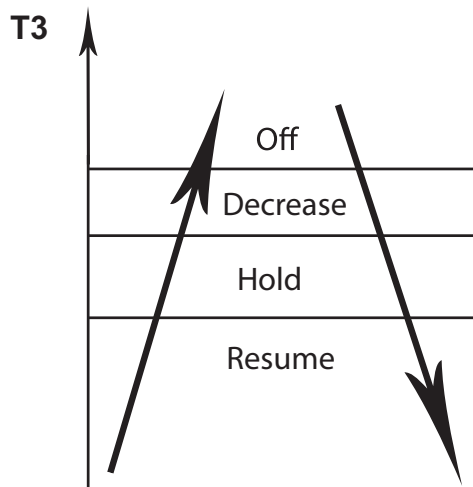
Indoor Fan Running Rules:

- In the COOLING mode, the indoor fan runs continuously and the user can select any of the following speeds: HIGH, MEDIUM, LOW and AUTO.
- When the setting temperature is reached, if the compressor stops operating, the indoor fan motor runs in the minimum or setting speed (see Fig. 19).

Setting Fan Speed	T1-Td °F (°C)		Actual Fan Speed
H	8°F (4.5°C)	A	H+ (H+=H+G)
	5°F (3.0°C)	B	H (=H)
	3°F (1.5°C)	C	H- (H-=H-G)
M	8°F (4.5°C)	D	M+ (M+=M+Z)
	5°F (3.0°C)	E	M (M=M)
	3°F (1.5°C)	F	M- (M-=M-Z)
L	8°F (4.5°C)	G	L+ (L+=L+D)
	5°F (3.0°C)	H	L (L=L)
	3°F (1.5°C)	I	L- (L-=L-D)

Fig. 21 —Indoor Fan Running Rules

The AUTO fan adheres to the following rules (see Fig. 20):

**Fig. 22 —AUTO Mode****Compressor Temperature Protection****Fig. 23 —Compressor Temperature Protection**

- **Off:** Compressor stops
- **Decrease:** Decrease the running frequency to the lower level
- **Hold:** Keep the current frequency
- **Resume:** No limitation for frequency

When the condenser temperature is higher than the setting value, the compressor stops.

Evaporator Temperature Protection

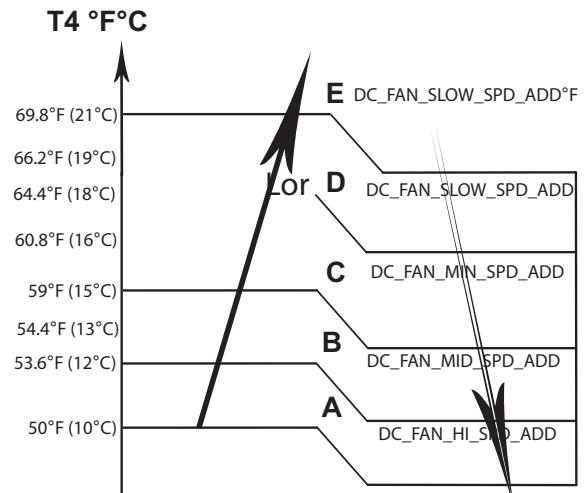
When the evaporator temperature is lower than the setting value the compressor stops.

HEATING Mode**Compressor Running Rules:**

- When $T1-Ts > -f'T$, the compressor stops.
- When $T1-Ts < -f'T-1.5$, the compressor is on. $f'T$ is the programmed parameter for temperature compensation.
- When the AC runs in the MUTE mode, the compressor runs with a low frequency.
- When the current is more than the setting value, the current protection function activates and the compressor stops.

Outdoor Fan Running Rules:

The outdoor unit runs at a different fan speed according to T4. For different outdoor units, the fan speeds differ.

**Fig. 24 —Outdoor Fan Running Rules**

OUTDOOR UNIT POINT CHECK FUNCTION

- A check switch is included on the adapter board.
- Push SW4 to check the unit's status while running. The digital display shows the following codes each time the SW4 is pushed.

Number of Presses	Display	Remark		
00	Normal display	Displays running frequency, running state, or malfunction code Defrosting mode: "dF" or alternative displays between running frequency and "dF" (each appears for 0.5s.) Forced cooling mode: the LED displays "FC" or alternative displays between running frequency and "FC" (each appears for 0.5s).		
01	Indoor unit capacity demand code	Actual data*HP*10 If capacity demand code is higher than 99, light the decimal point of the high digit tube. (For example, the digital display tube show "5.0",so 5.0 multiplied by 10 to become 50, then added to 100 to become 150, so actual capacity demand=150/10=15. the digital display tube show "60",so actual capacity demand=60/10=6.0) GA algorithm models display "--"		
02	The frequency after the capacity requirement adapter	If the value is higher than 99, light the decimal point of the high digit tube.		
03	Room temperature (T1)	If the temp. is lower than 0oC, the digital display tube will show "0". If the temp. is higher than 70oC, the digital display tube will show "70".		
04	Indoor unit evaporator temperature (T2)	If the temp. is lower than -9oC, the digital display tube will show "-9".If the temp. is higher than 70oC, the digital display tube will show "70". If the indoor unit is not connected, the digital display tube will show: "--"		
05	Condenser pipe temp.(T3)			
06	Outdoor ambient temp.(T4)			
07	Compressor discharge temp. (TP)	The display value is between 0~199oC. If the temp. is lower than 0oC, the digital display tube will show "0".If the temp. is higher than 99oC,light the decimal point of the high digit tube. (For example, the digital display tube show "0.5",so 0.5 multiplied by 10 to become 5, then added to 100 to become 105oC.)		
08	AD value of current	The display value is a hex number. For example, the digital display tube shows "Cd", so C*161+d*160=12*16+13=205, it means AD value is 205.		
09	AD value of voltage			
10	Indoor unit running mode code	Standby:0,Cooling:1, Heating:2, Fan only 3, Drying:4, Forced cooling:6, Defrost:7; ECO:9; Forced defrosting :A; Self clean: B		
11	Outdoor unit running mode code			
12	EXV open angle	Actual data/4. If the value is higher than 99, light the decimal point of the high digit tube. For example, the digital display tube show "2.0",so 2.0 multiplied by 10 to become 20, then added to 100 to become 120,it means the EXV open angle is 120×4=480p.)		
13	Frequency limit symbol	Bit 7	Frequency limit caused by IGBT radiator	The display value is a hex number. For example, the digital display show 2A, the corresponding binary is 101010, so Bit5=1, Bit3=1, and Bit1=1. This means that a frequency limit may be caused by T3 or the current.
		Bit 6	Reserved	
		Bit 5	Reserved	
		Bit 4	Frequency limit caused by low temperature of T2.(LH00)	
		Bit 3	Frequency limit caused by T3.(LC01)	
		Bit 2	Frequency limit caused by TP.(LC02)	
		Bit 1	Frequency limit caused by current(LC03)	
		Bit 0	Frequency limit caused by voltage (LC05)	
14	Outdoor unit fan speed	If it is higher than 99, light the decimal point of the high digit tube. (For example, the digital display tube show "2.0", so 2.0 multiplied by 10 to become 20, then added to 100 to become 120. This value is multiplied by 8, and it is the current fan speed: 120*8=960		

Number of Presses	Display	Remark
15	The average value of the temperature values detected by the high and low pressure sensors in the last 10 seconds of the compressor frequency calculation period	The displayed value is the actual value plus 60 (that is, when the displayed value is 10, the actual value is -50). When the displayed value is higher than 99, light the decimal point of the high digit tube. (if it displays 2.0, so 2.0 multiplied by 10 to become 20, then added to 100 to become 120, the actual value is 60)
16	The temperature value detected by the high and low pressure sensor	When there is no pressure sensor, it is displayed as --
17	AD value detected by the high and low pressure sensor	The display value is a hex number. For example, the digital display tube shows "Cd", so $C*161+d*160=12*16+13=205$, it means AD value is 205. When there is no pressure sensor, it is displayed as --
18	The currently running communication protocol version	00-99

GENERAL TROUBLESHOOTING

SAFETY CAUTION



WARNING

Be sure to turn off all power supplies or disconnect all wires to avoid electric shock. While checking indoor/outdoor PCB, equip yourself with anti-static gloves or wrist strap to avoid damage to the board.



WARNING

Electricity remains in capacitors even when the power supply is off. Ensure the capacitors are fully discharged before troubleshooting.

NOTE: If using the inverter test tool for troubleshooting, shut off power, remove the electrical panel and locate the cable that is already connected to the test port on the outdoor unit. Connect the test tool to the cable with the connector provided with the test tool. After the maintenance is completed, insert the female end back into the port.

For the R454B single zone ODU with capacity less than 24K, there will be the test tool connector. For 24K HH and 30K-60K single zone ODU, there is a diagnosis/check board which has digital display on it, you can read the parameters from it directly.

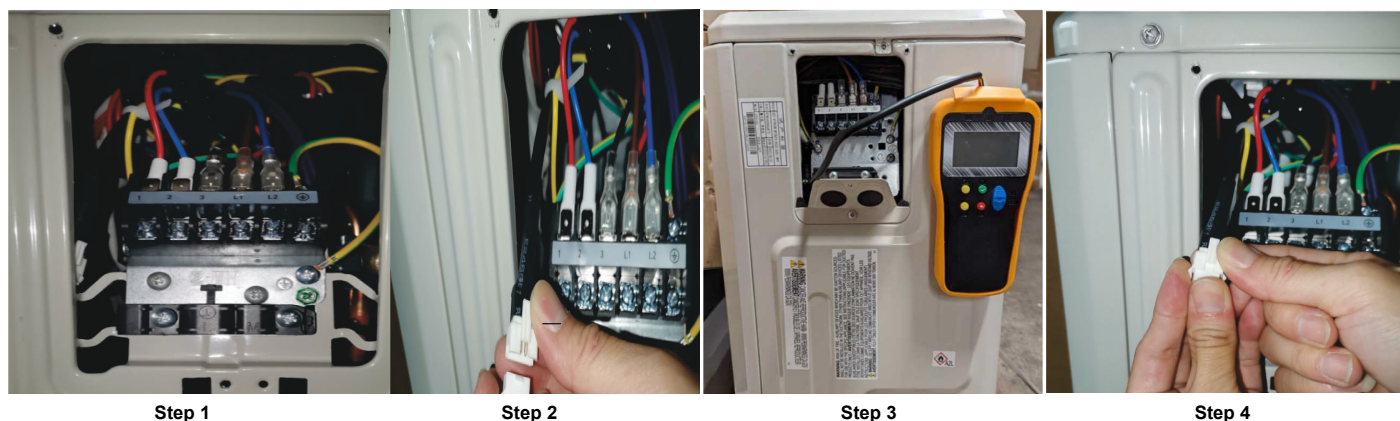


Fig. 25 —Inverter Test Tool Maintenance

Connect the Dr. SMART tool to the white terminal as shown in Step 3 above.



Fig. 26 —Dr. SMART Tool

NOTE: These pictures is for reference only. Actual appearance may vary.

ERROR CODES

Error Display

Display	Indoor/Outdoor Code	Malfunction or Protection	Solution (Refer to Page)
d ^f	N/A	Defrosting	Normal Display, not error code
FC	N/A	Forced cooling	
FH CC	IDU	Refrigerant Sensor Error	page 55
EC 07	ODU	Fan speed out of control	page 32
EC 51	ODU	EEPROM parameter error	page 30
EH 00 / EH0A	IDU & ODU	EEPROM Malfunction Error Diagnosis and Solution	page 30
EC 52	ODU	Coil temp. sensor(T3) error	page 35
EC 53	ODU	Ambient temp. sensor(T4) error	page 35
EC 54	ODU	COMP. discharge temp. sensor (TP) error	page 35
EC 55	ODU	IPM module temperature sensor malfunction	page 36
EL 0C	IDU & ODU	System Lacks Refrigerant Diagnosis and Solution	page 37
EH b3	IDU	Communication Malfunction Between Wire and Master Control	page 54
EH bA	IDU & ODU	Communication Malfunction Between Wire and Master Control	page 54
EH C1	IDU	Refrigerant Sensor Detects Leakage	page 56
EH C2	IDU	Refrigerant Sensor is Out of Range and Leakage is Detected	page 56
EH 3b	IDU & ODU	External Fan DC Bus Voltage is Too High	page 54
EH 0b	IDU	IDU Main Control Board and Display Board Communication Error Diagnosis and Solution	page 57
EH 0E	IDU	Water-Level Alarm Malfunction Diagnosis and Solution	page 38
EC 57	ODU	Refrigerant pipe temperature sensor error	page 35
EC 5C	ODU	Suction Transducer Failure	page 34
EL 01	IDU & ODU	Communication error	page 31
EL 16	ODU	Communication malfunction between adapter board and ODU main board	page 58
FL 09	IDU & ODU	Mismatch between the new and old platforms diagnosis and solution	page 58
PC 00	ODU	IPM module protection	page 39
PC 01	ODU	ODU Voltage Protection Diagnosis and Solution	page 40
PC 02	ODU	Compressor top (or IPM) temp. protection	page 41
PC 04	ODU	Inverter Compressor Drive Error Diagnosis and Solution	page 44
PC 06	ODU	Discharge temperature protection of compressor	page 41
PC 08	ODU	Outdoor overcurrent protection	page 50
PC 0A	ODU	High temperature protection of condenser	page 51
PC 0F	ODU	PFC module protection	page 52
PC 0L	ODU	Low Ambient Temperature Protection	page 54
PC 10	ODU	Low AC voltage protection	page 53
PC 11	ODU	Main control board DC bus high voltage protection	page 53
PC 12	ODU	Main control board DC bus high voltage protection /341 MCE error	page 53
PC 03	IDU	Pressure Protection (low or high pressure)	page 42
PC 30	ODU	System high pressure protection	page 42
PC 31	ODU	Low Pressure Protection	page 42
PC 40	ODU	Communication error between ODU main chip and compressor driven chip	page 45
PC 41	ODU	Compressor current sampling failure	page 46

Display	Indoor/Outdoor Code	Malfunction or Protection	Solution (Refer to Page)
PC 42	ODU	Compressor start failure of outdoor unit	page 50
PC 43	ODU	Compressor lack phase protection	page 47
PC 44	ODU	Zero speed protection	page 50
PC 45	ODU	IR chip drive failure	page 48
PC 46	ODU	Compressor speed has been out of control	page 50
PC 49	ODU	Compressor overcurrent failure	page 50
LC 06	ODU	High temperature protection of Inverter module (IPM)	page 41
PH 90	IDU & ODU	High temperature protection of evaporator	page 59
PH 91	IDU & ODU	Low temperature protection of evaporator	page 60

TROUBLESHOOTING BY ERROR CODE

EH 00/ EH 0A / EC 51 (EEPROM Malfunction Error Diagnosis and Solution)

Description: Indoor or outdoor PCB main chip does not receive feedback from EEPROM chip.

Recommended parts to prepare: Indoor PCB, Outdoor PCB

Troubleshooting and repair:

Troubleshooting and repair:

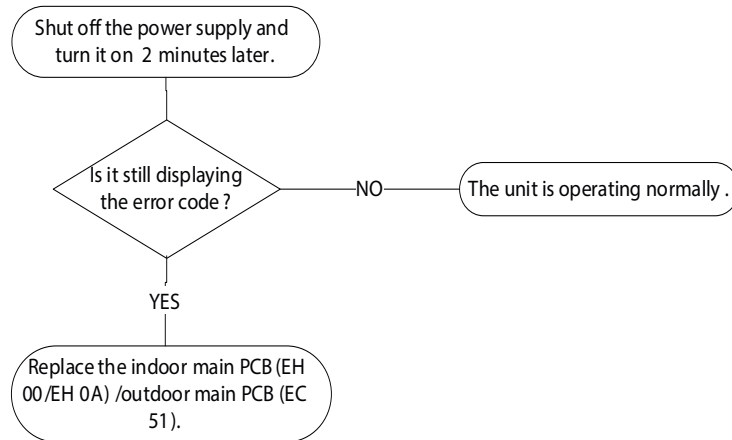


Fig. 27 —EC 51

Remarks:

EEPROM: A read-only memory whose contents can be erased and reprogrammed using a pulsed voltage.

The location of the EEPROM chip on the outdoor PCB is shown in the following image:

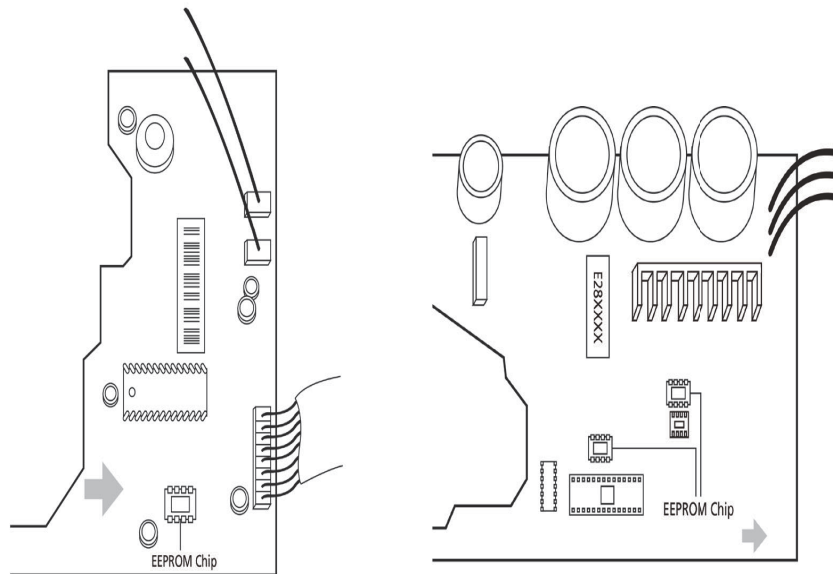


Fig. 28 —Location of EPROM Chip

NOTE: This picture is only for reference, actual appearance may vary.

IMPORTANT: Troubleshooting and repair of compressor driven chip EEPROM parameter error sand communication errors between outdoor main chip and compressor driven chip are same as EC 51.

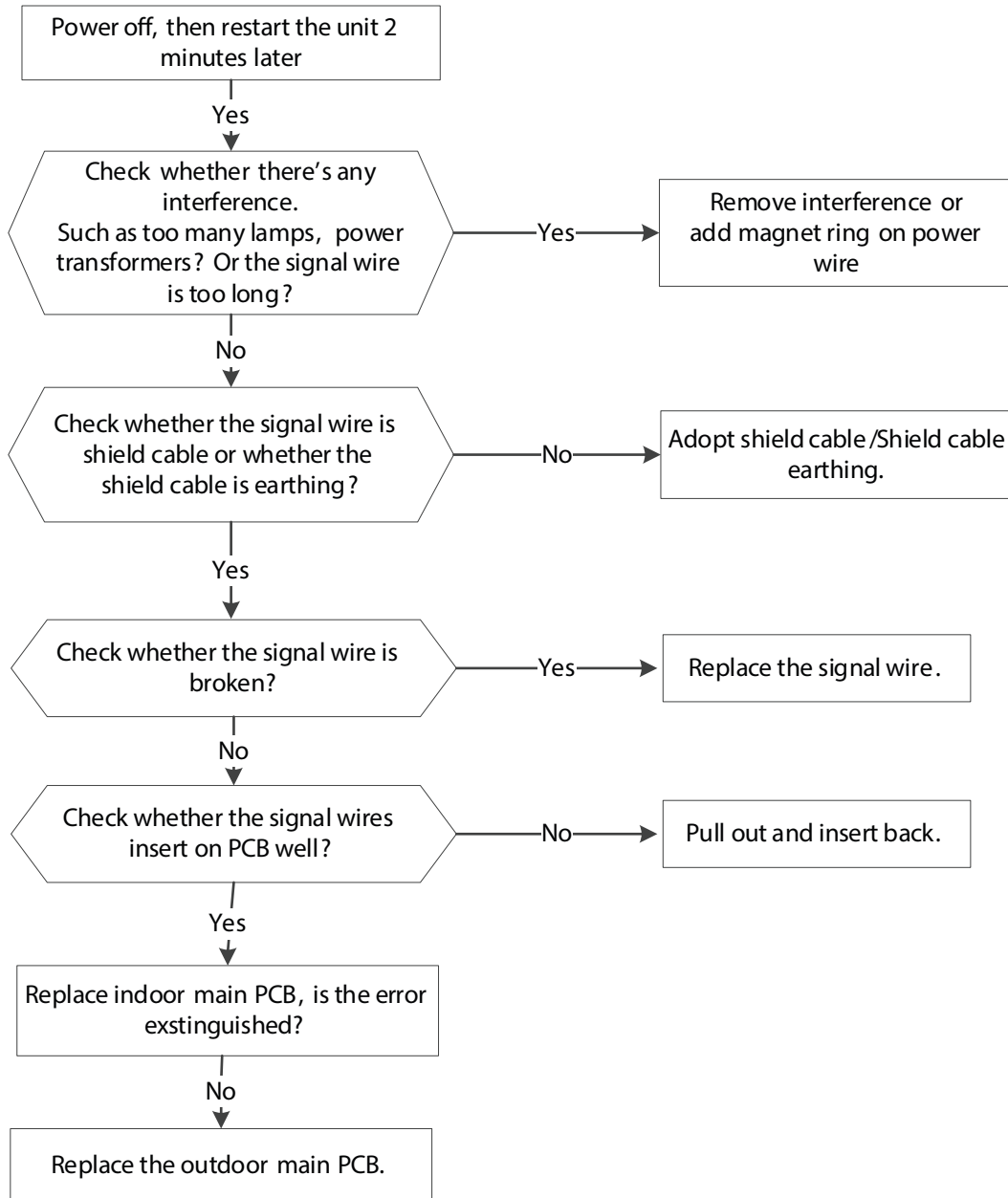
EL 01 (Indoor and Outdoor Unit Communication Error Diagnosis and Solution)

Description: Indoor unit can not communicate with outdoor unit.

Recommended parts to prepare: Signal Wires, Magnetic Ring, Indoor PCB, Outdoor PCB

Troubleshooting and repair: RS 485 Communication

:

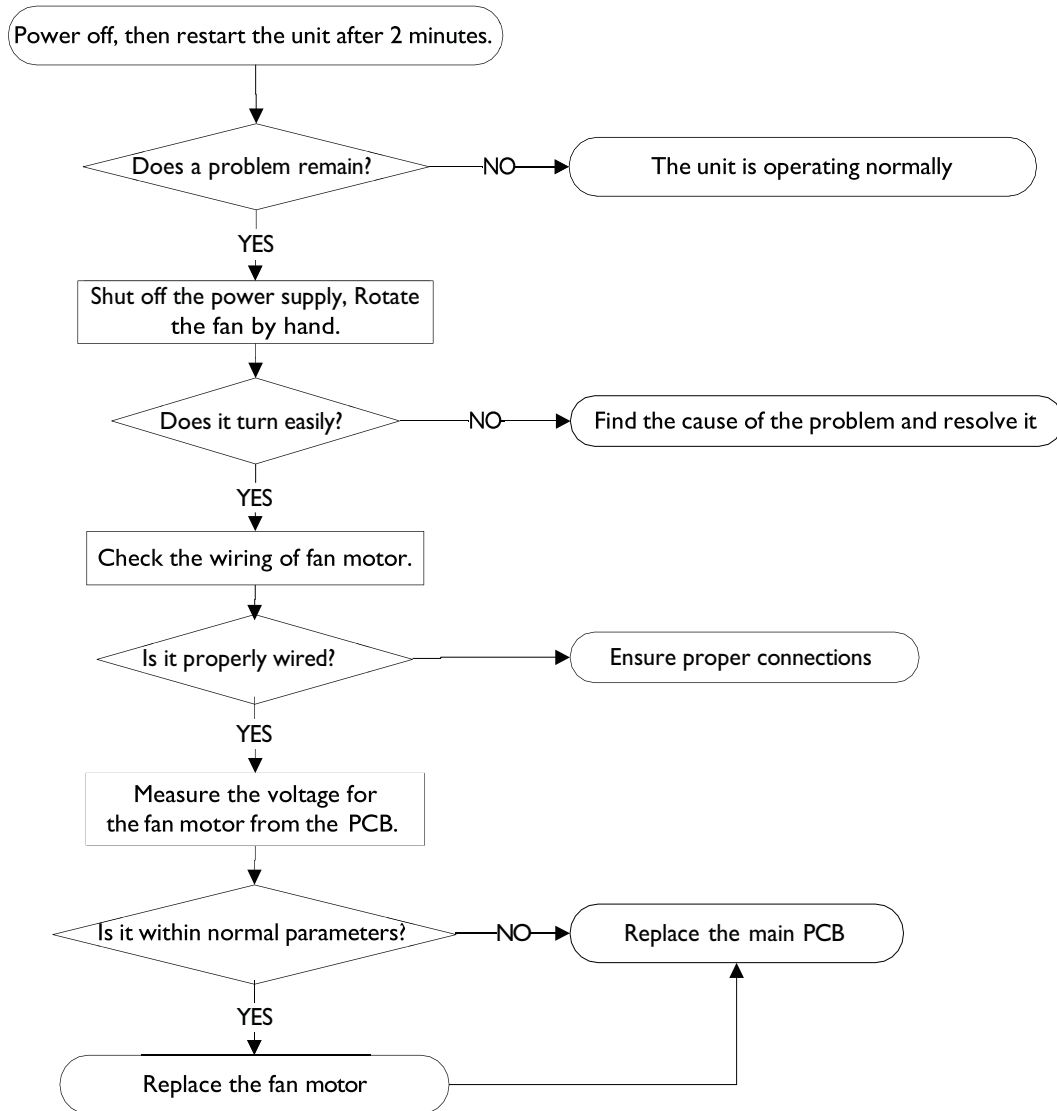


EH03 / EC 07 (Fan Speed Is Operating Outside of Normal Range Diagnosis and Solution)

Description: When indoor / outdoor fan speed keeps too low or too high for a certain time, the LED displays the failure code and the AC turns off.

Recommended parts to prepare: Connection wires, Fan assembly, Fan motor, PCB

Troubleshooting and repair:



Index

- Indoor or Outdoor DC Fan Motor (control chip is in fan motor)
With the power on and when the unit is in standby, measure the voltage of pin1- pin3, pin4- pin3 in the fan motor connector. If the value of the voltage is not in the range shown in below table, the PCB needs to be replaced.

No.	Color	Signal	Voltage
1	Red	Vs/Vm	192V~380V
2	—	—	—
3	Black	GND	0V
4	White	Vcc	13.5-16.5V
5	Yellow	Vsp	0~6.5V
6	Blue	FG	13.5-16.5V

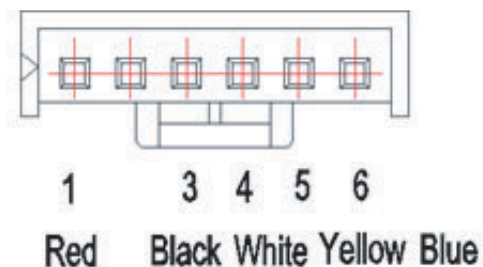


Fig. 29 —Fan Motor Connector, Pin Location

- Outdoor DC Fan Motor (control chip is in outdoor PCB)
Release the UVW connector. Measure the resistance of U-V, U-W, V-W. If the resistances are not equal to each other, the fan motor needs to be replaced. Otherwise, the PCB needs to be replaced.

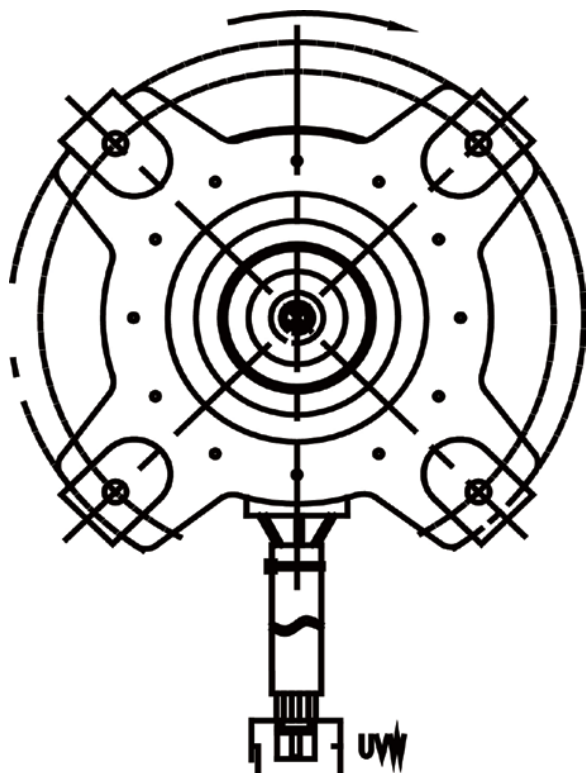


Fig. 30 —UVW Connector

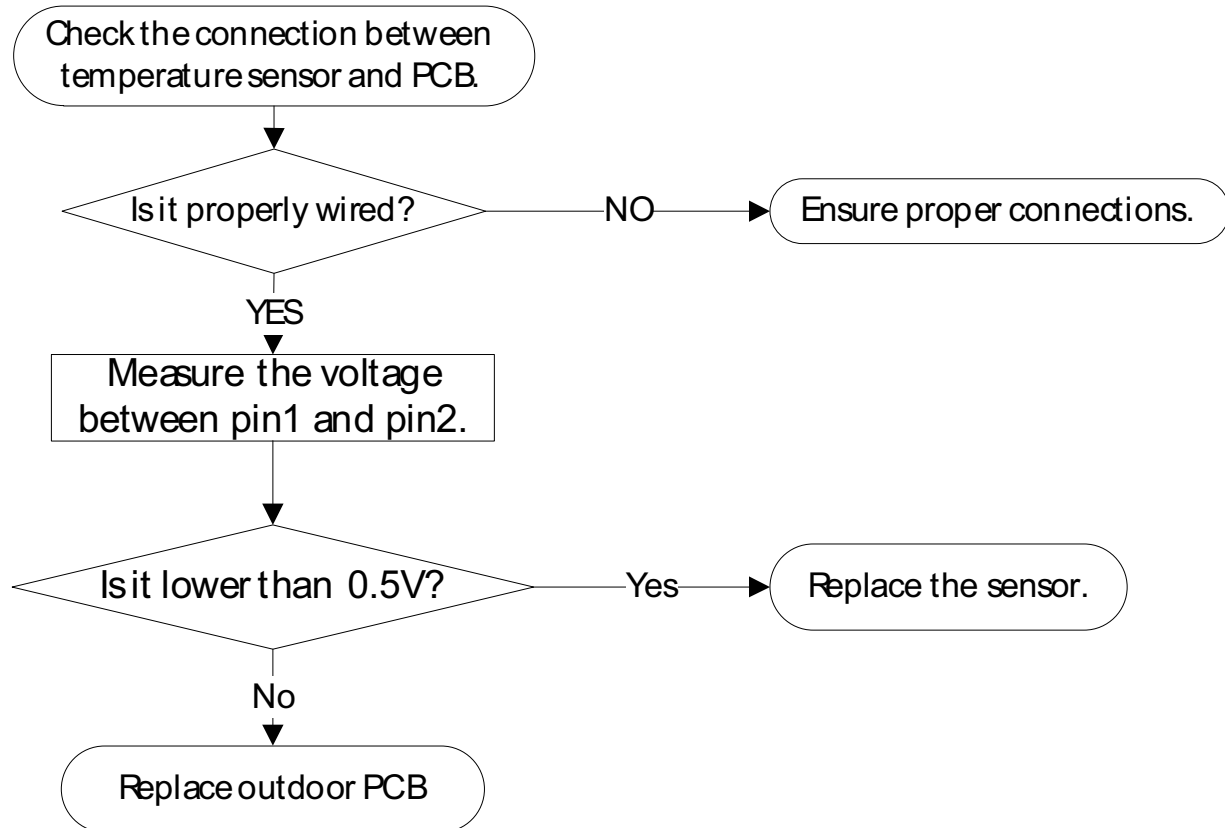
EC 5C (Suction Transducer is in open circuit or has short circuited) diagnosis and solution

Description: If the sampling voltage is lower than 2V or higher than 254V, the LED displays the failure code.

Recommended parts to prepare:

- Connection wires
- Sensor
- Outdoor PCB

Troubleshooting and repair:



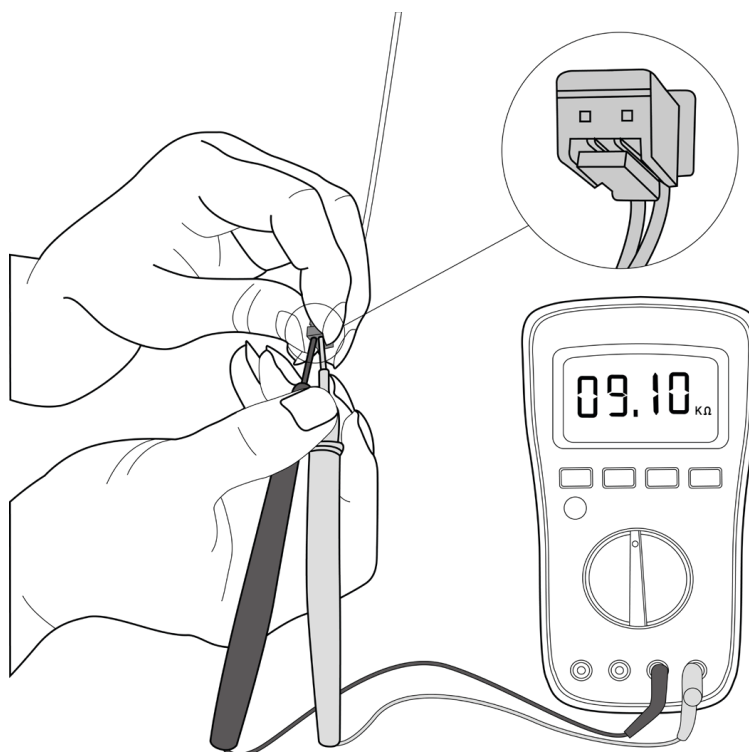
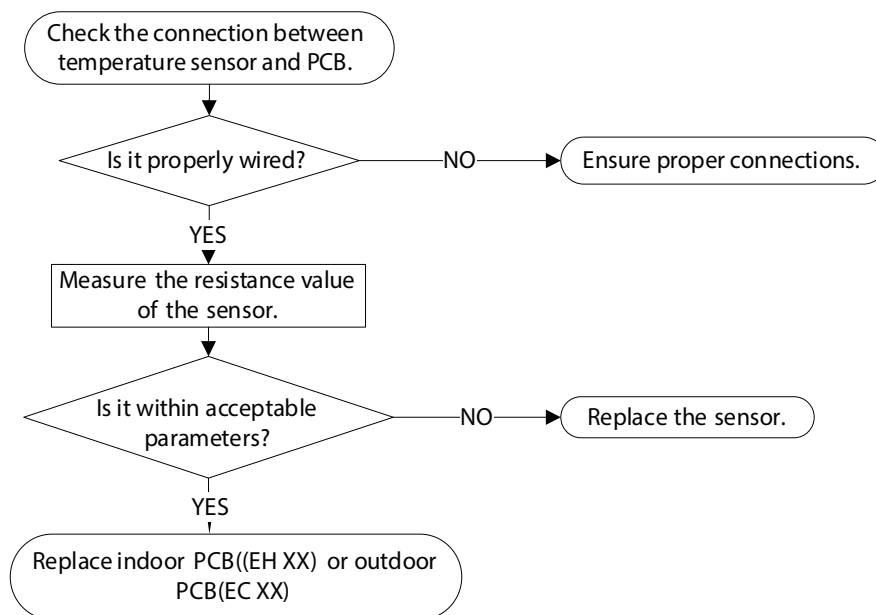
EC 53/ EC 52/ EC 54/ EC 56/ EC 57/ EC 50 (Open Circuit or Short Circuit of Temperature Sensor Diagnosis and Solution)

Description: If the sampling voltage is lower than 0.06V or higher than 4.94V, the LED displays the failure.

Recommended parts to prepare: Connection wires, Sensors, PCB

Troubleshooting and repair:

Refer to Appendix, page 88.



NOTE: This picture and the value are only for reference, actual appearance and value may vary.

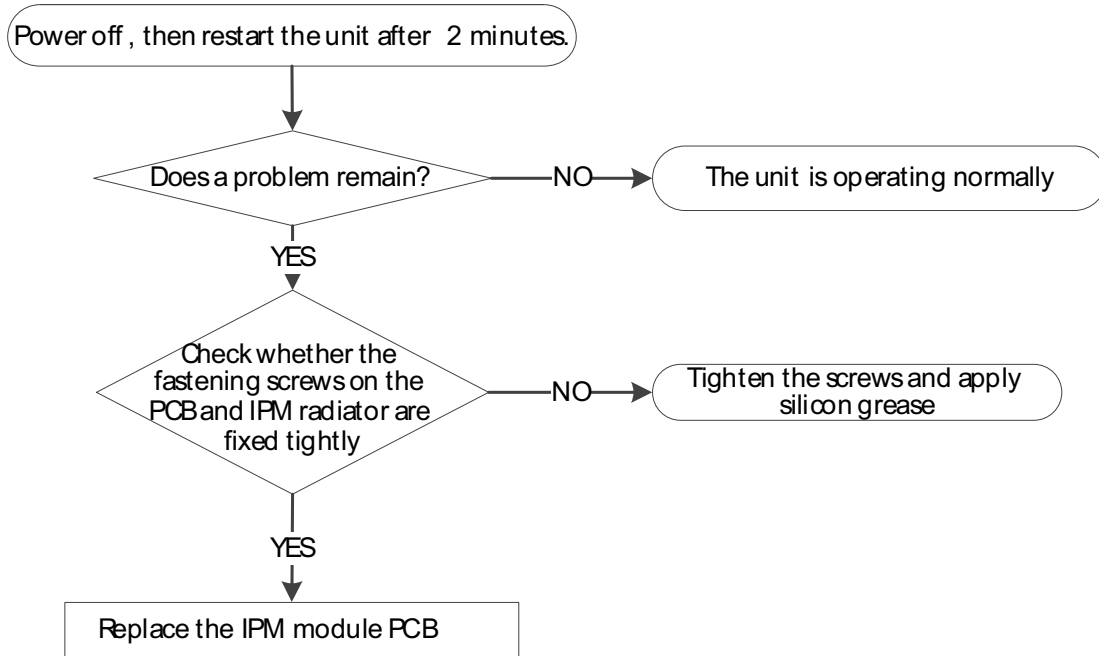
EC 55 (ODU IPM module temperature sensor malfunction diagnosis and solution)

Description: If the sampling voltage is 0V or 5V, the LED displays the failure code.

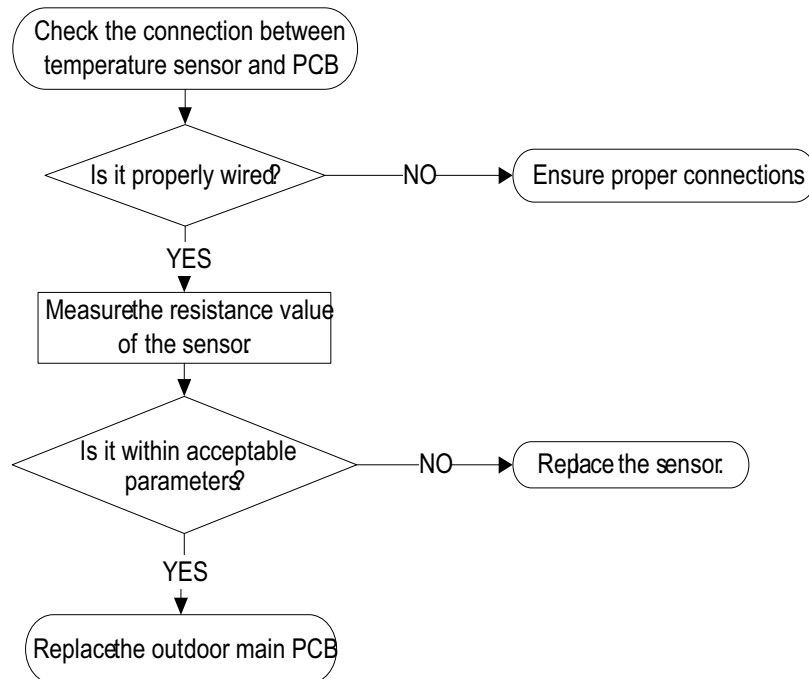
Recommended parts to prepare:

- IPM module PCB
- Connection wires
- Sensors
- Outdoor main PCB

Troubleshooting and repair: If the radiator has no sensor, follow the steps below to resolve:



If the radiator has a sensor(TH), follow the steps below:

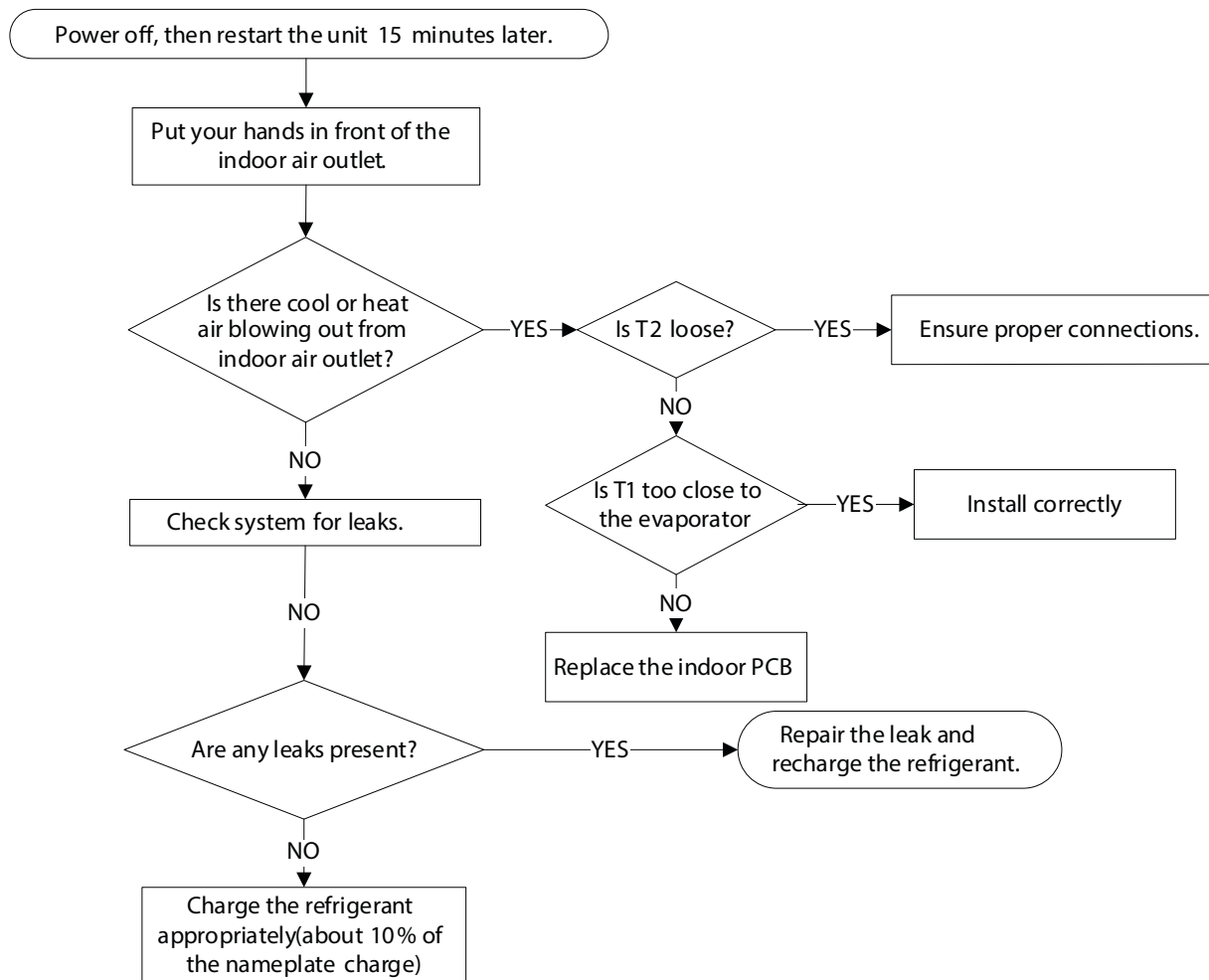


EL 0C (System Lacks Refrigerant Diagnosis and Solution)

Description: Judging the abnormality of the refrigeration system according to the number of compressor stops and the changes in operating parameters caused by excessive exhaust temperature.

Recommended parts to prepare: Indoor PCB, Additional refrigerant

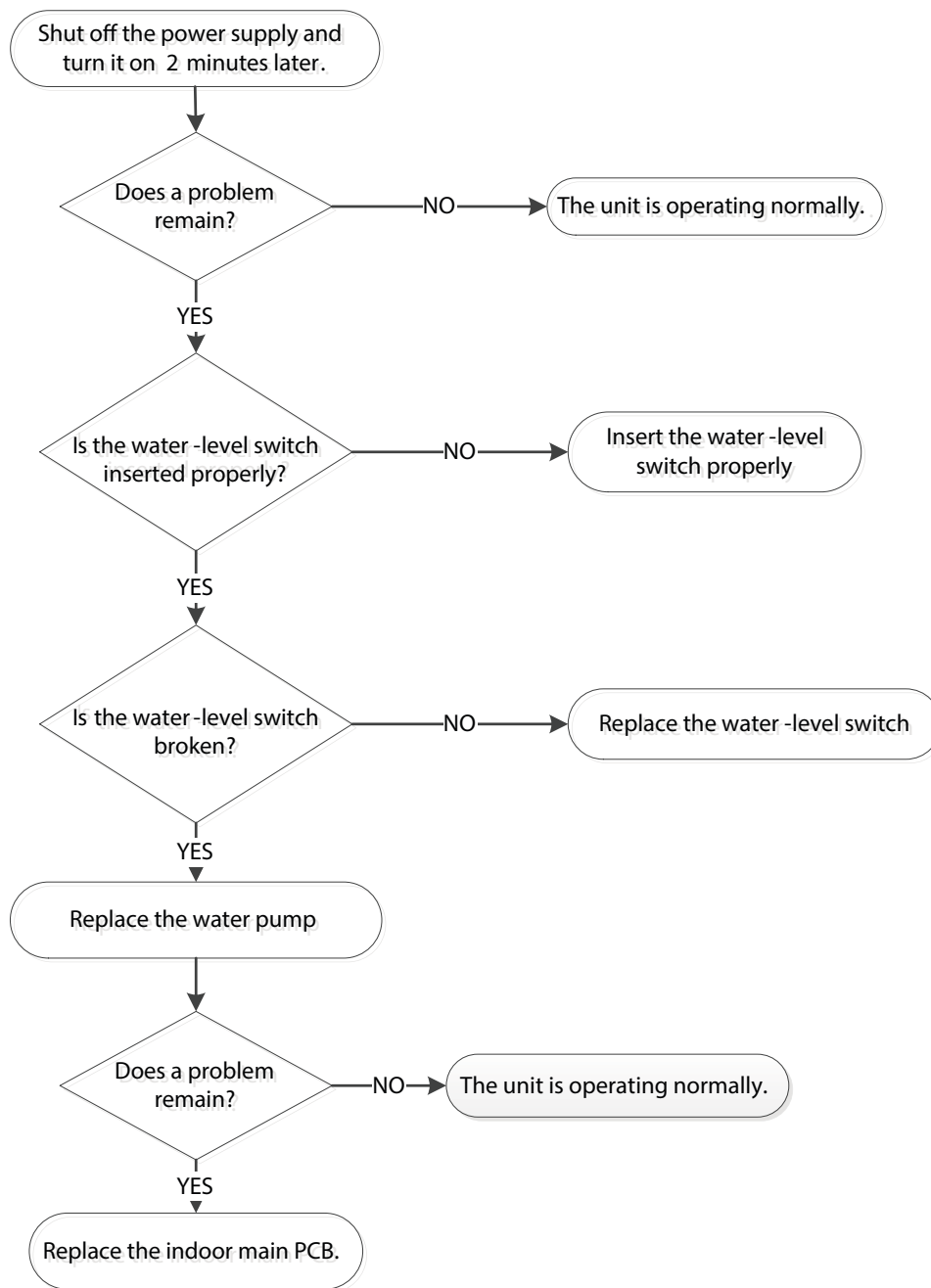
Troubleshooting and repair:



EH 0E (Water-Level Alarm Malfunction Diagnosis and Solution)

Description: If the sampling voltage is not 5V, the LED displays the failure code.

Recommended parts to prepare: Connection wires, Water-level switch, Water pump, Indoor PCB

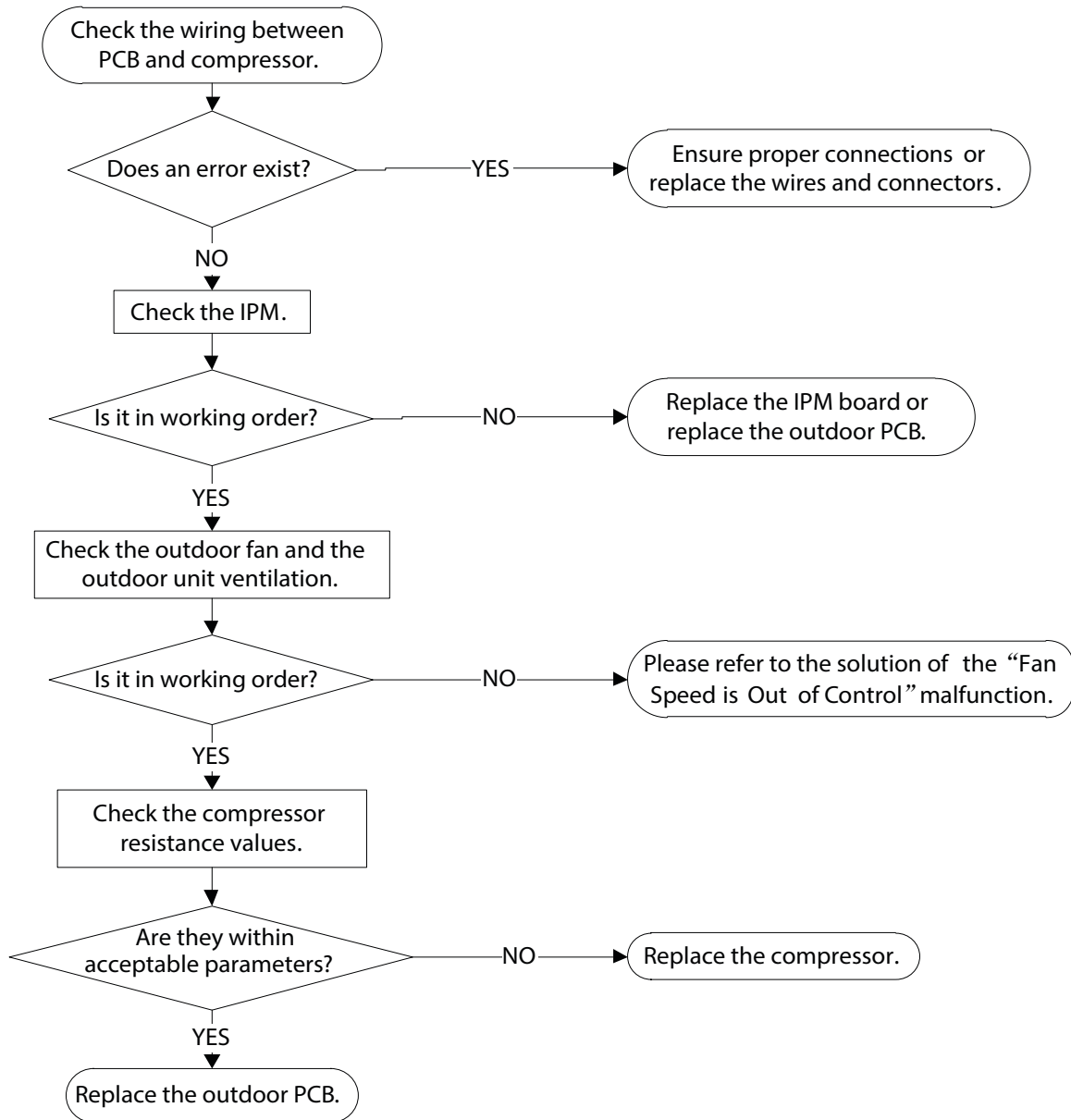


PC 00 (ODU IPM Module Protection Diagnosis and Solution)

Description: When the voltage signal the IPM sends to the compressor drive chip is abnormal, the display LED shows "PC 00" and the AC turn off.

Recommended parts to prepare: Connection wires, IPM module board, Outdoor fan assembly, Compressor, Outdoor PCB

Troubleshooting and repair:



NOTE: For certain models, the outdoor PCB can not be removed separately. In this case, the outdoor electric control box should be replaced as a whole.

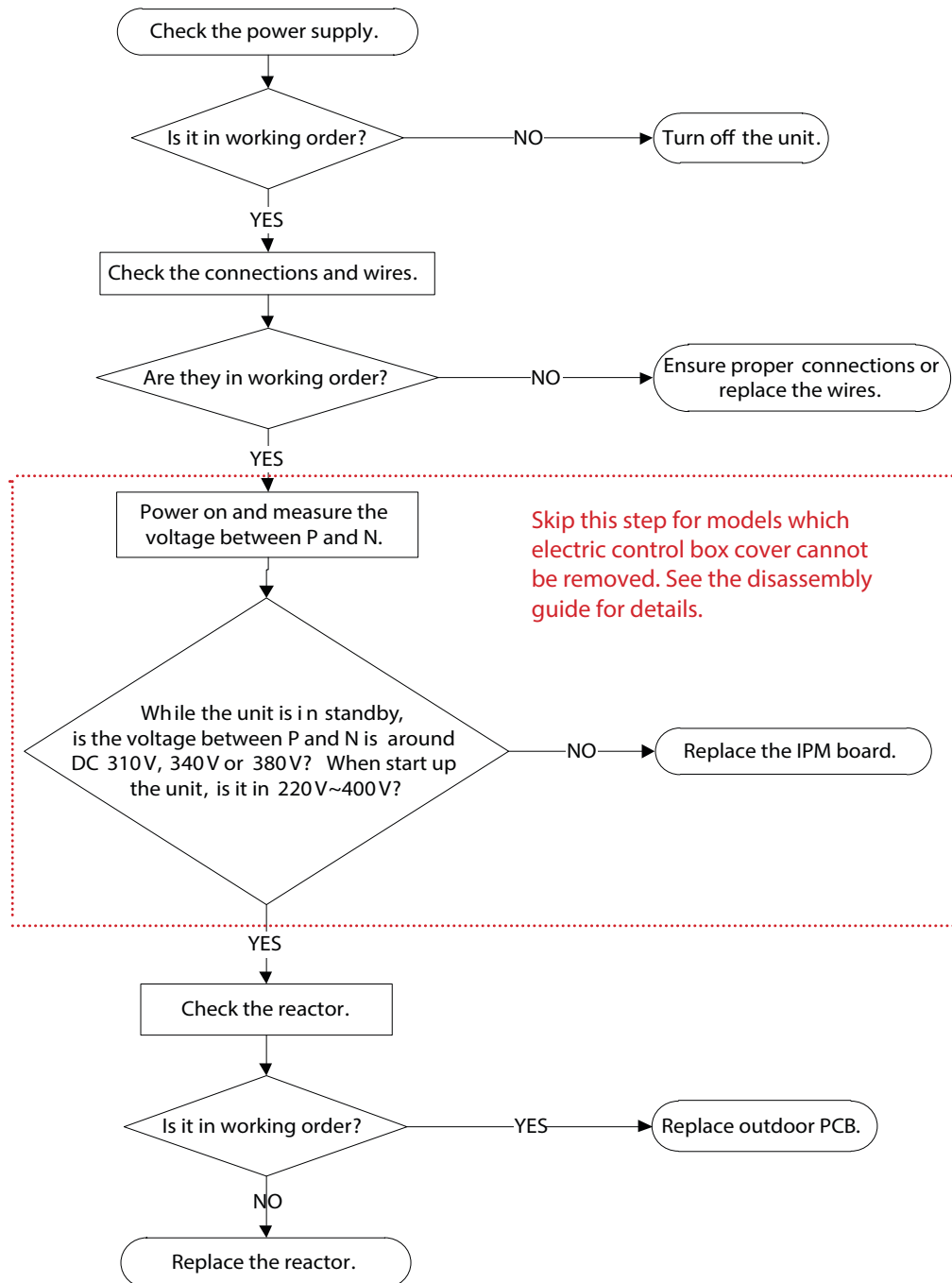
PC 01 (ODU Voltage Protection Diagnosis and Solution)

PC 01 (Over voltage or too low voltage protection) / PC 10 (Outdoor unit low AC voltage protection) / PC 11 (Outdoor unit main control board DC bus high voltage protection) / PC 12 (Outdoor unit main control board DC bus high voltage protection/341 MCE error) Diagnosis and Solution)

Description: Abnormal increases or decreases in voltage are detected by checking the specified voltage detection circuit.

Recommended parts to prepare: Power supply wires, IPM module board, PCB, Reactor

Troubleshooting and repair:



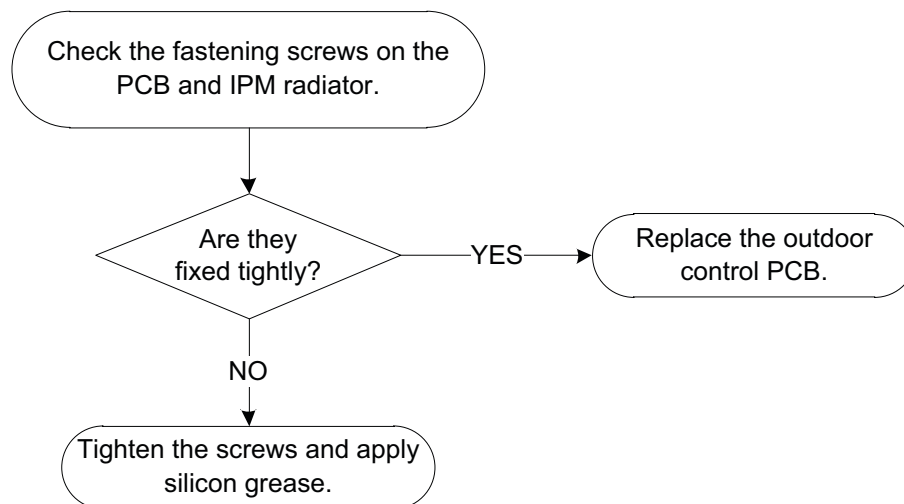
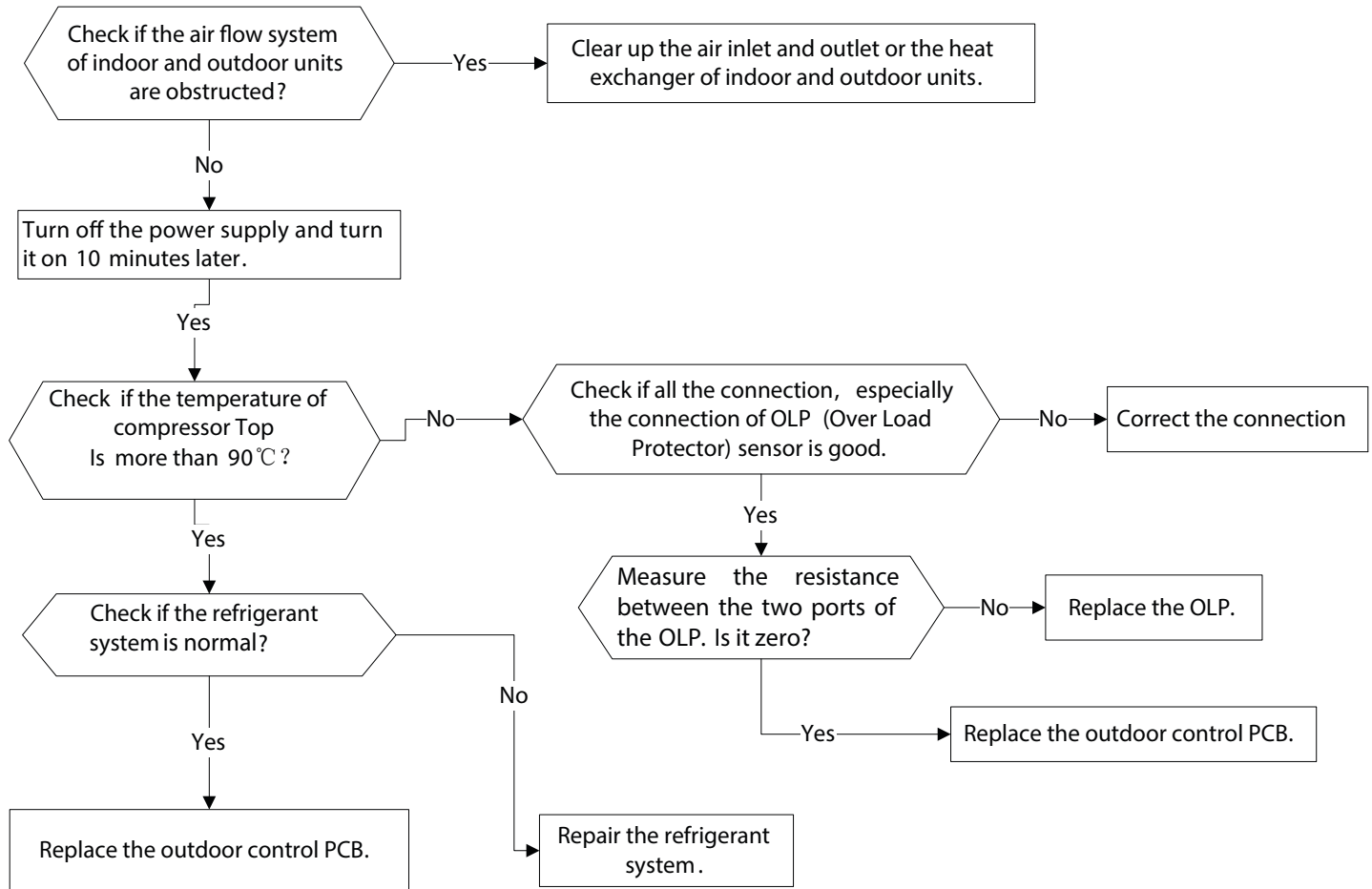
NOTE: For certain models, the outdoor PCB cannot be removed separately. In this case, the outdoor electric control box should be replaced as a whole.

PC 02 / LC 06 (Compressor Top (or IPM) Temp. Protection Diagnosis and Solution)

Description: For some models with overload protection, If the sampling voltage is not 5V, the LED will display the failure. If the temperature of IPM module is higher than a certain value, the LED displays the failure code.

Recommended parts to prepare: Connection wires, Outdoor PCB, IPM module board, High pressure protector, System blockages

Troubleshooting and repair:

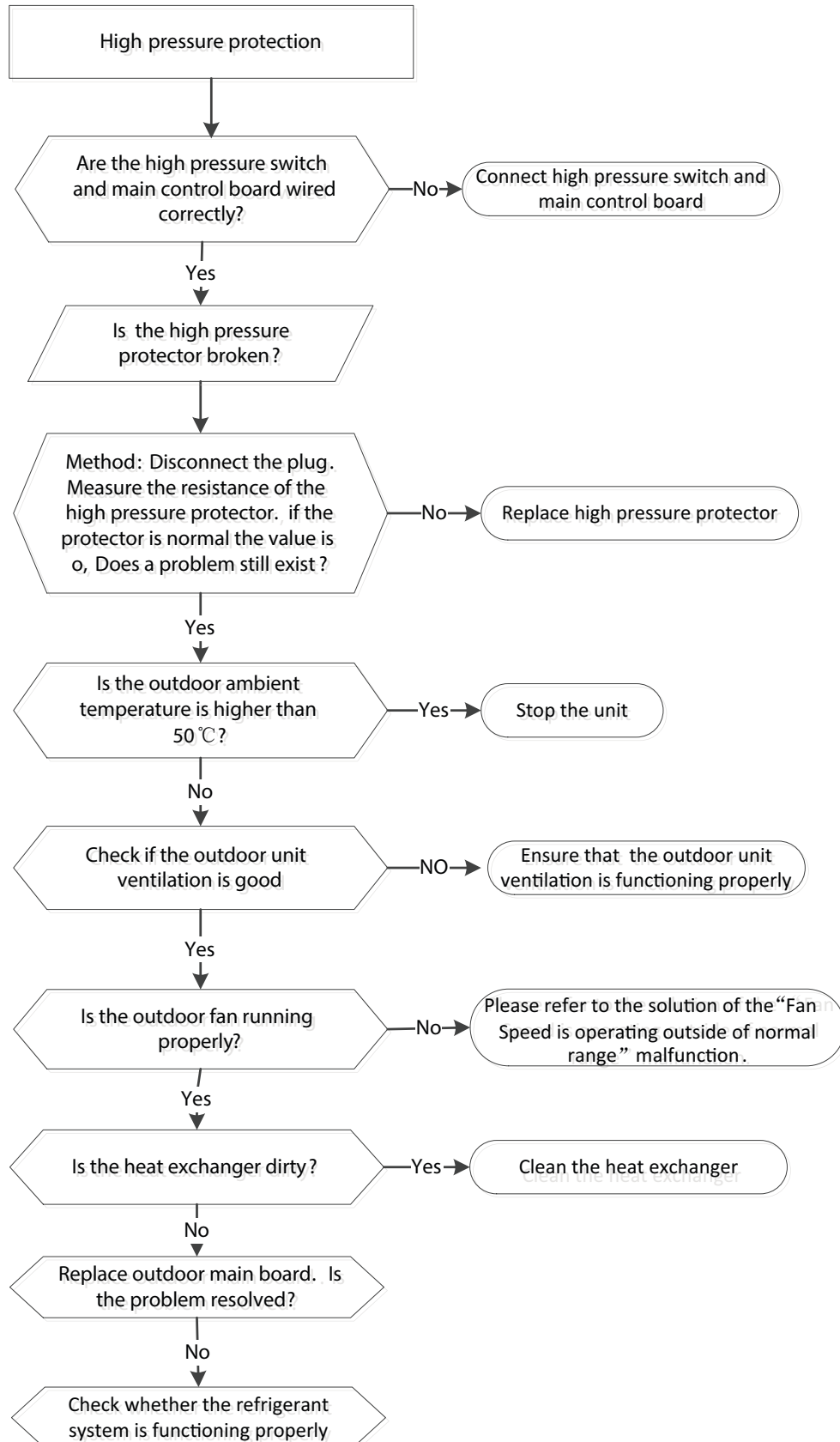


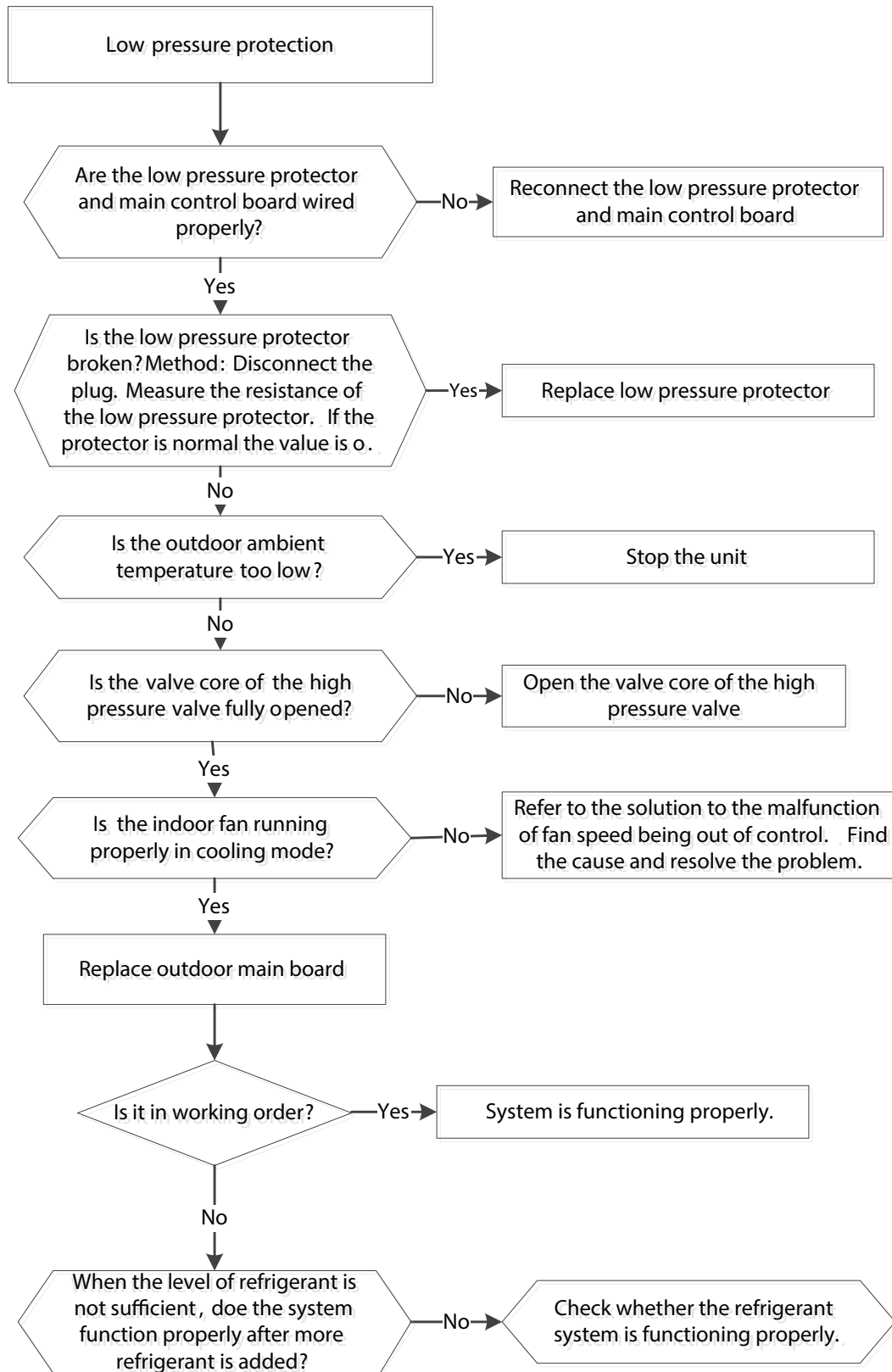
PC 03 Pressure Protection (low or high pressure), PC 30 High Pressure Protection, PC 31 Low Pressure Protection (Diagnosis and Solution)

Description: Outdoor pressure switch cut off the system because high pressure is higher than 4.4 MPa or outdoor pressure switch cut off the system because low pressure is lower than 0.13 MPa, the LED displays the failure code.

Recommended parts to prepare: Connection wires, Pressure switch, Outdoor fan, Outdoor main PCB, Refrigerant

Troubleshooting and repair:



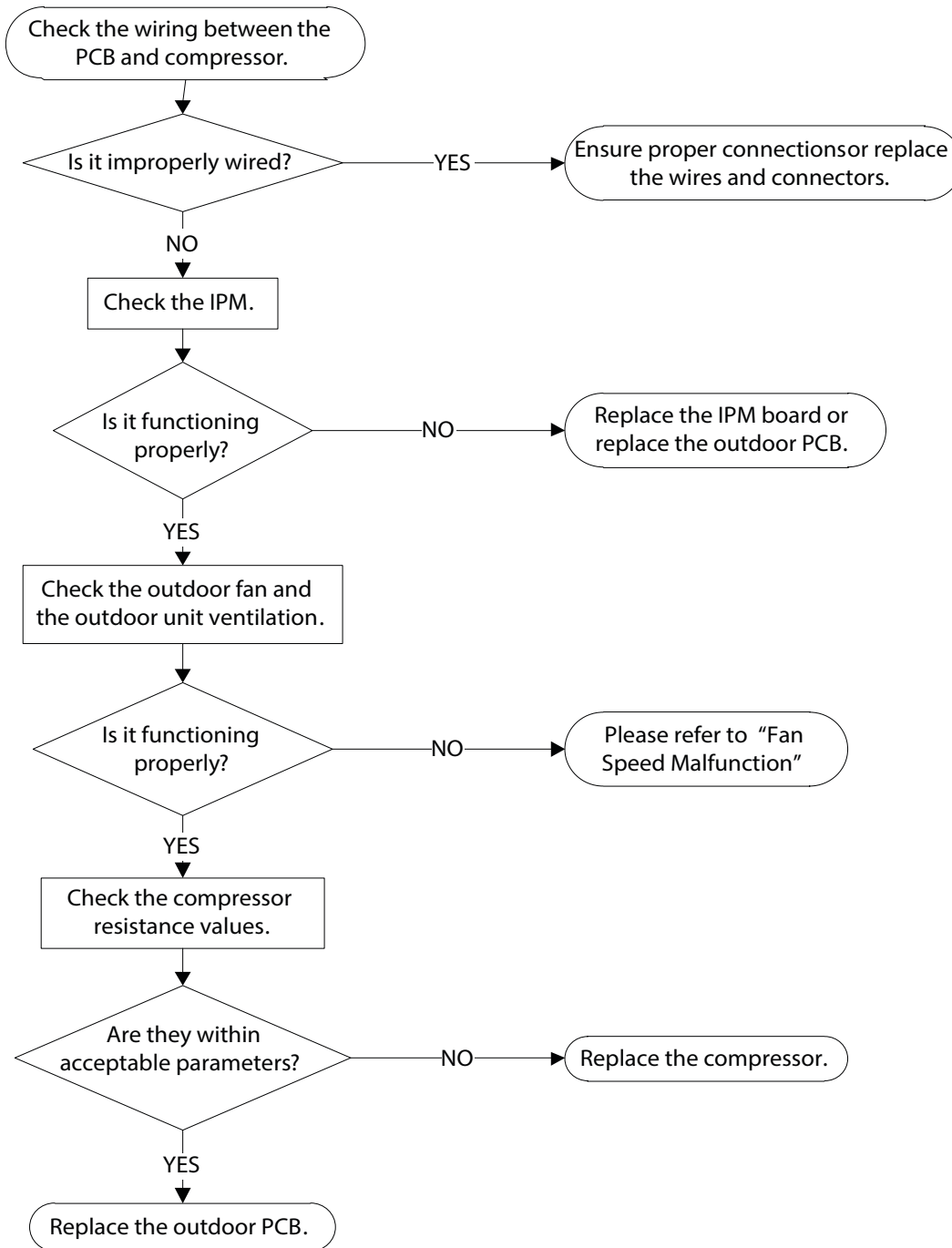


PC 04 (Inverter Compressor Drive Error Diagnosis and Solution)

Description: An abnormal inverter compressor drive is detected by a special detection circuit, including communication signal detection, voltage detection, compressor rotation speed signal detection and so on.

Recommended parts to prepare: Connection wires, IPM module board, Outdoor fan assembly, Compressor, Outdoor PCB

Troubleshooting and repair:

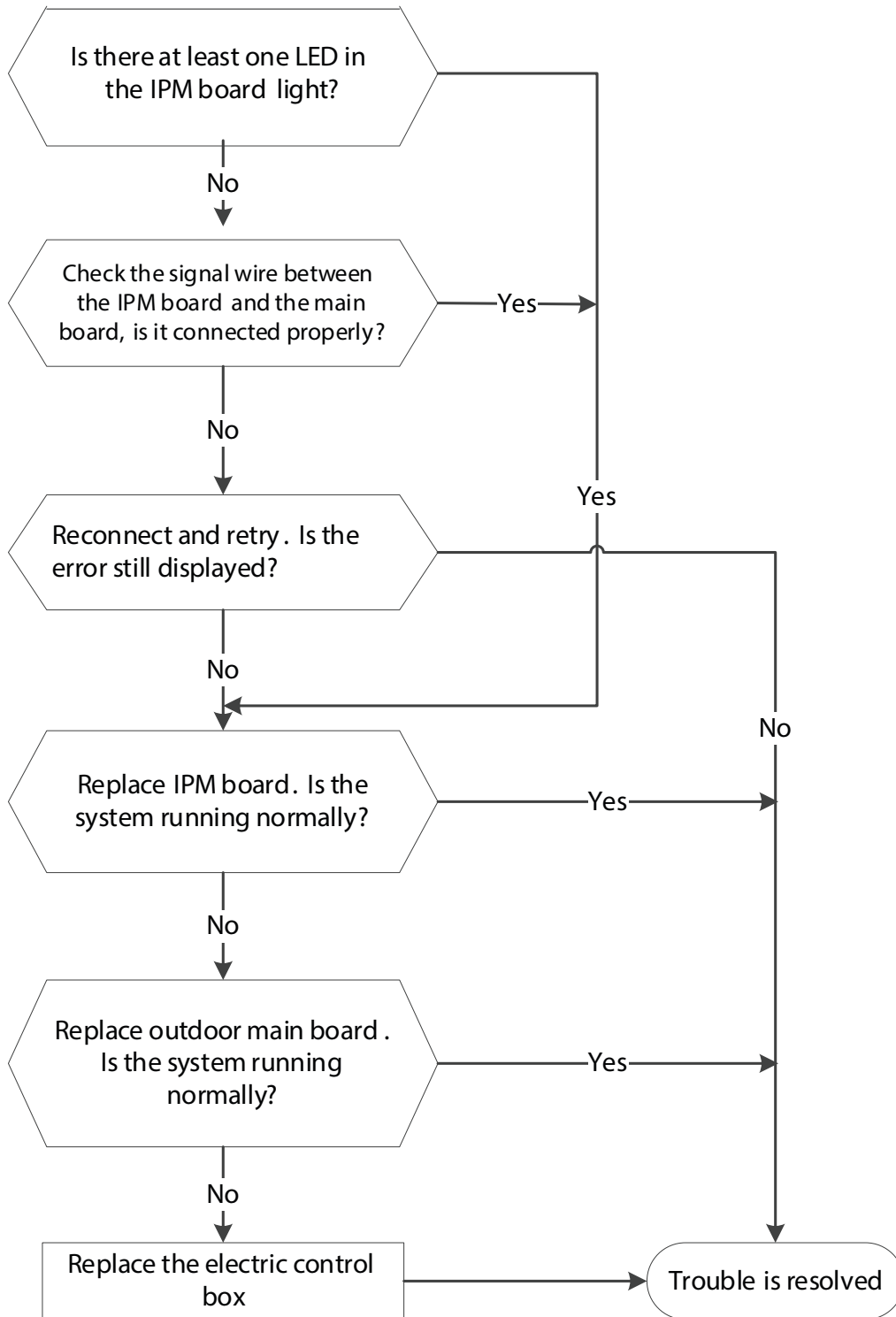


PC 40 (Communication error between ODU main chip and compressor driven chip diagnosis and solution)

Description: The main PCB cannot detect the IPM board.

Recommended parts to prepare: Connection wires, Outdoor PCB, IPM module board, Electric control box

Troubleshooting and repair:

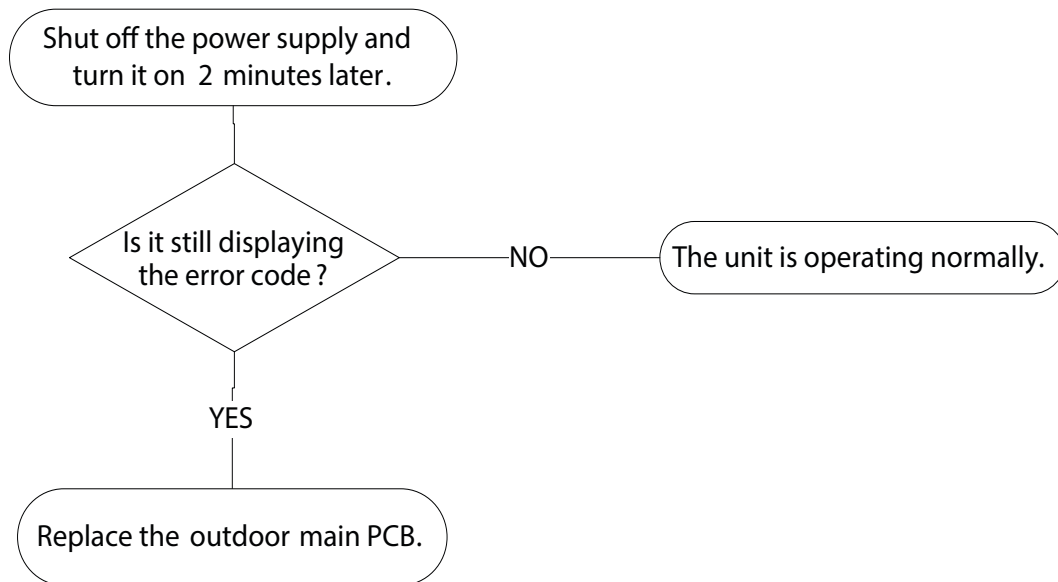


PC 41 (Outdoor compressor current sampling circuit failure diagnosis and solution)

Description: Three-phase sampling offset voltage error, the static bias voltage is normally 2.5V.

Recommended parts to prepare: Outdoor main PCB

Troubleshooting and repair:

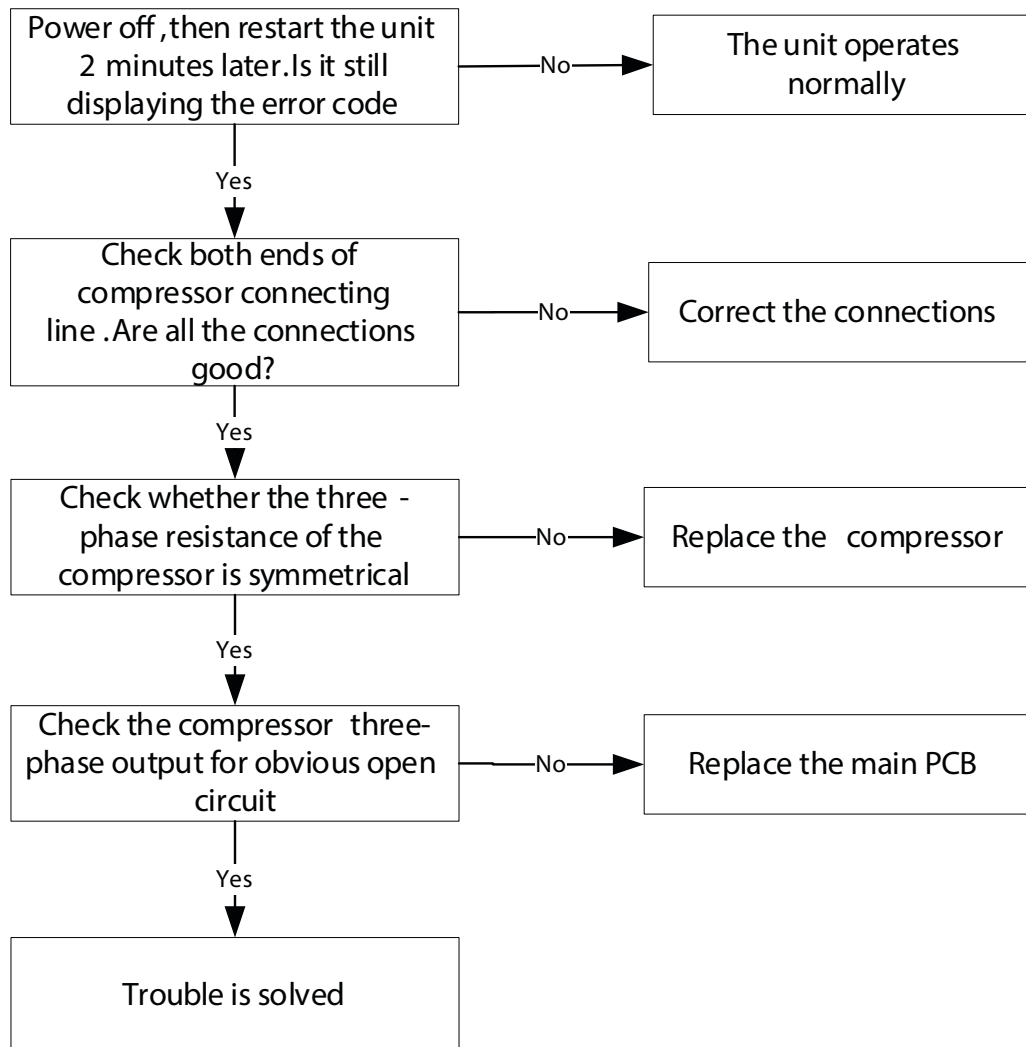


PC 43 (ODU compressor lack phase protection diagnosis and solution)

Description: When the three-phase sampling current of the compressor is abnormal, especially when the current of one or more phases is always small and almost 0, the LED displays the failure code.

Recommended parts to prepare: Connection wire, Compressor, Outdoor PCB

Troubleshooting and repair:

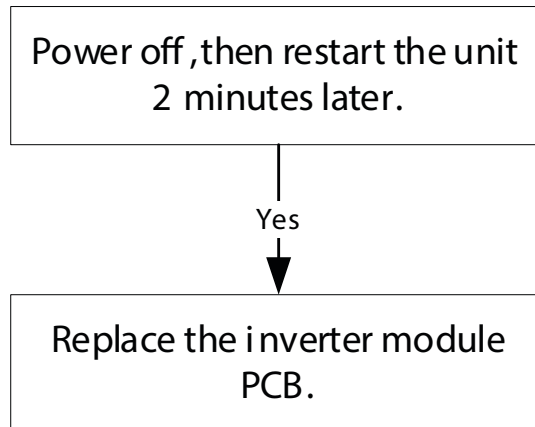


PC 45 (ODU IR chip drive failure diagnosis and solution)

Description: When the IR chip detects its own parameter error, the LED displays the failure code when power on.

Recommended parts to prepare: Inverter module PCB.

Troubleshooting and repair:

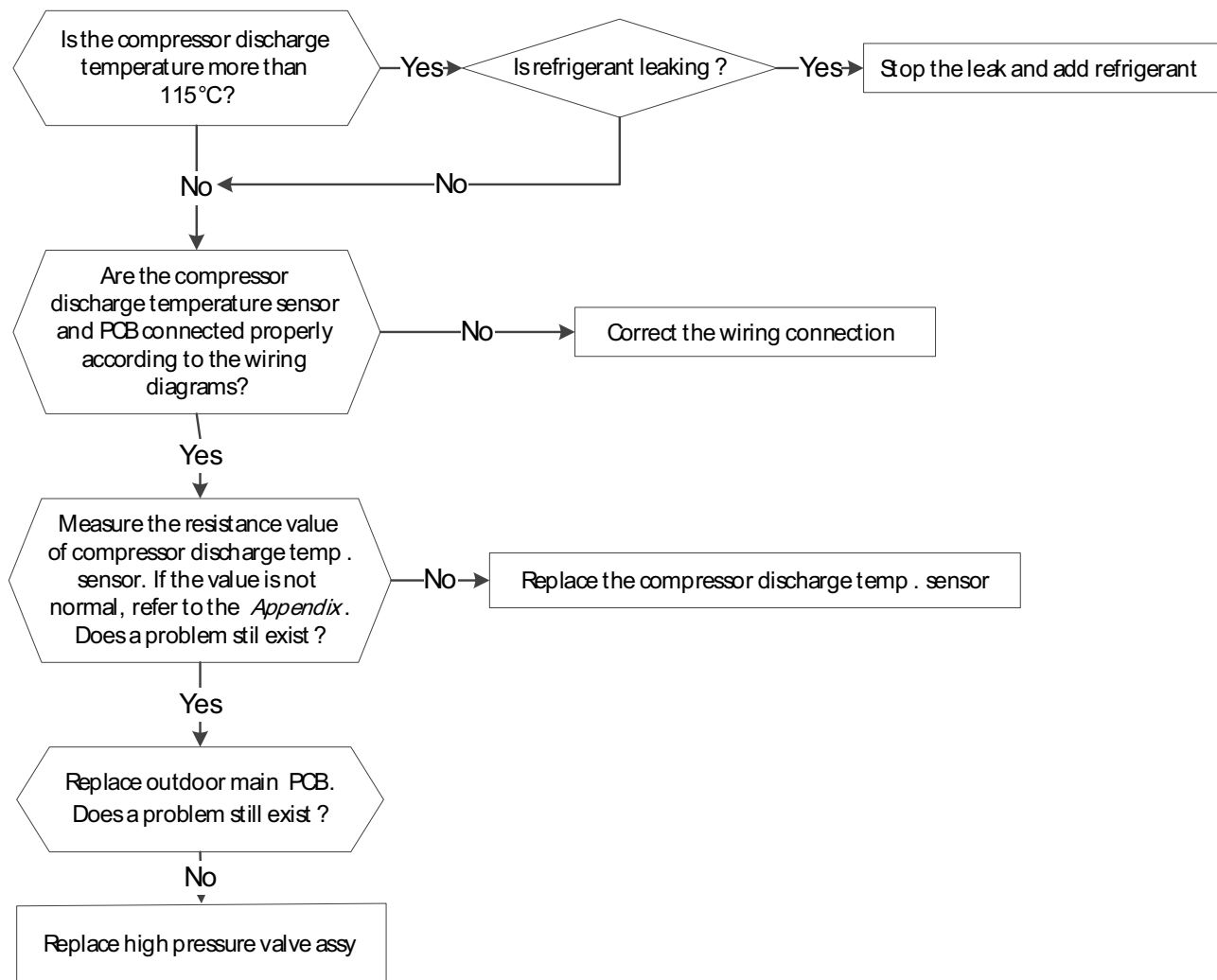


PC 06 (Discharge temperature protection of compressor diagnosis and solution)

Description: When the compressor discharge temperature (TP) is more than 115°C for 10 seconds, the compressor ceases operation and does not restart until TP is less than 90°C

Recommended parts to prepare: Connection wires, Outdoor PCB, Discharge temperature sensor, Refrigerant

Troubleshooting and repair:

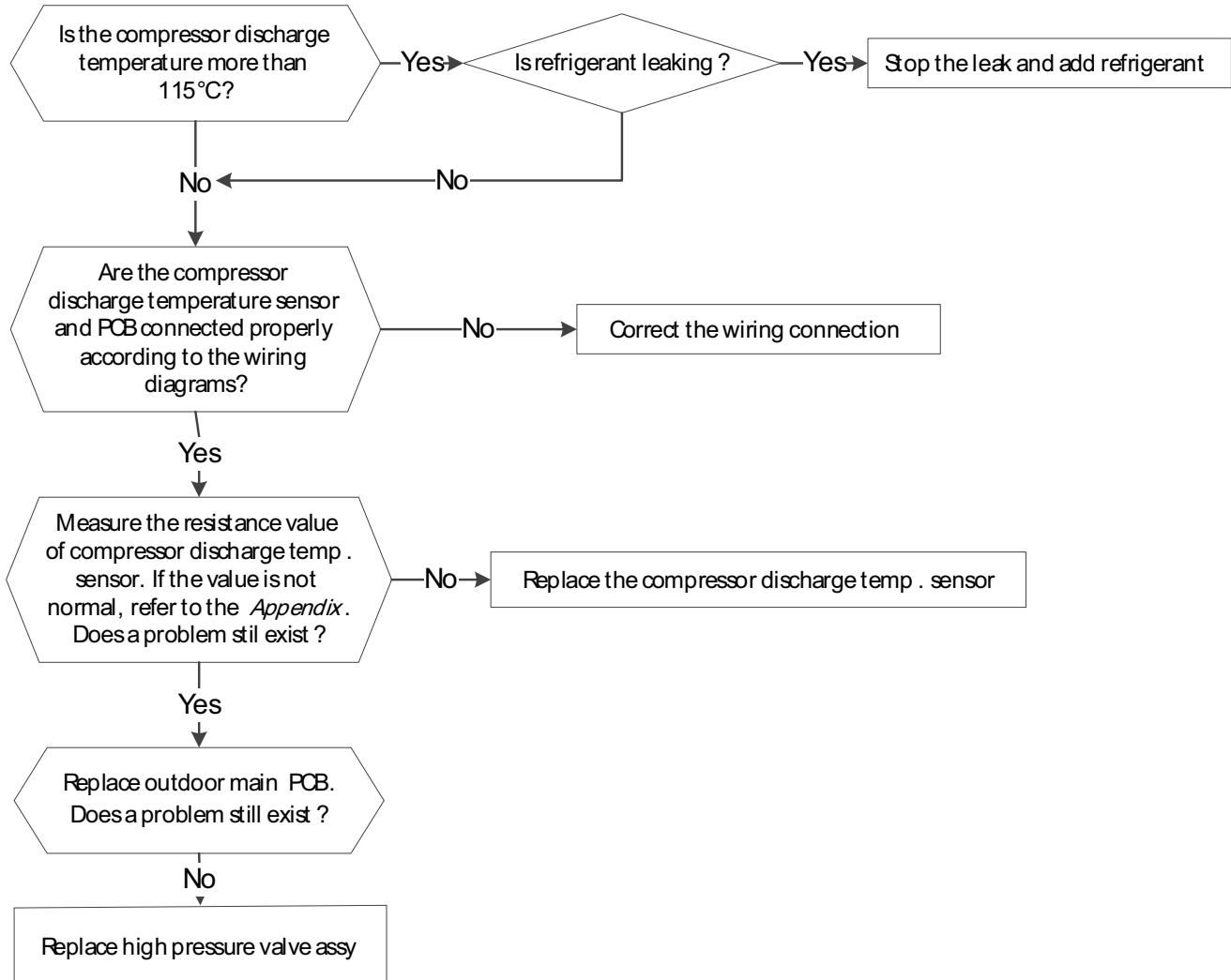


PC 08 (Current overload protection)/ PC 42 (Compressor start failure of outdoor unit)/ PC 44 (ODU zero speed protection) / PC 46 (Compressor speed has been out of control)/ PC 49 (Compressor overcurrent failure)

Description: An abnormal current rise is detected by checking the specified current detection circuit.

Recommended parts to prepare: Outdoor PCB, Connection wires, Bridge rectifier, PFC circuit or reactor, Refrigeration piping system, Pressure switch, Outdoor fan, PM module board

Troubleshooting and repair:

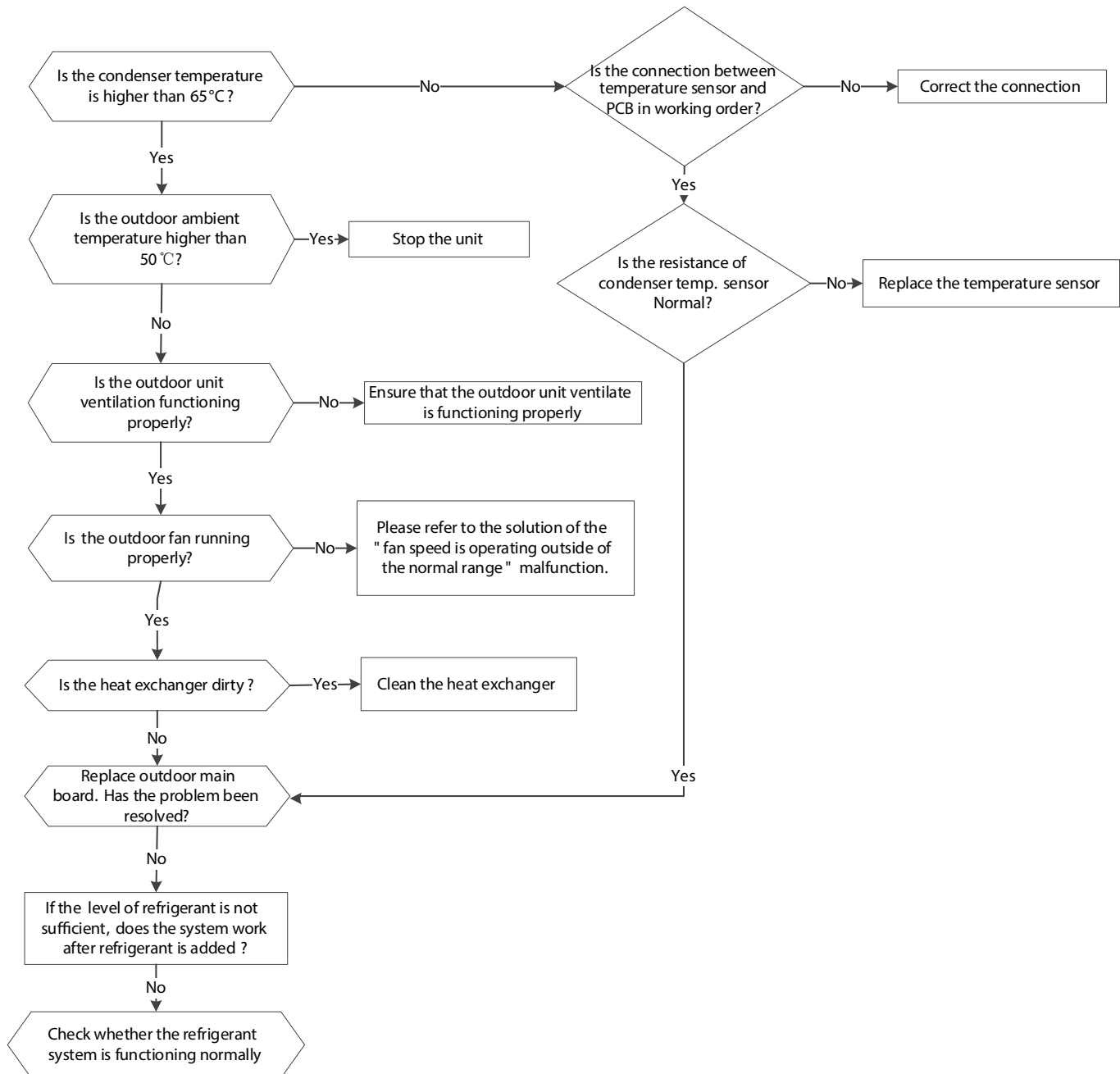


PC 0A (High temperature protection of condenser diagnosis and solution)

Description: The unit will stop when condenser temperature is higher than 65°C, and runs again when it is less than 52°C

Recommended parts to prepare: Connection wires, Condenser temperature sensor, Outdoor fan, Outdoor main PCB, Refrigerant

Troubleshooting and repair:

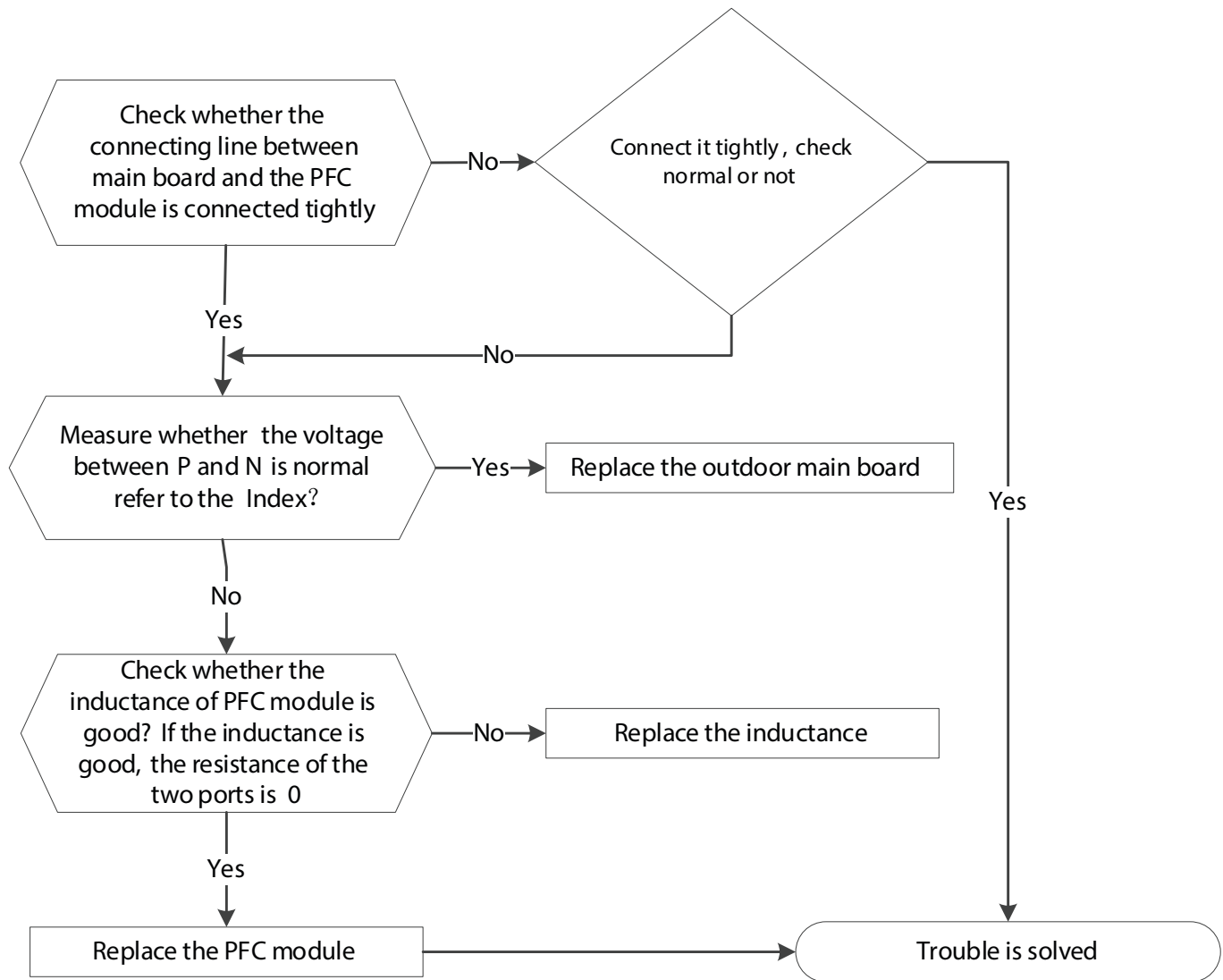


PC 0F (PFC module protection diagnosis and solution)

Description: Outdoor PCB detects PFC signal is low voltage or DC voltage is lower than 340V for 6s when quick check.

Recommended parts to prepare: Connection wires Outdoor PCB, Inductance, PFC circuit or IPM module board

Troubleshooting and repair:

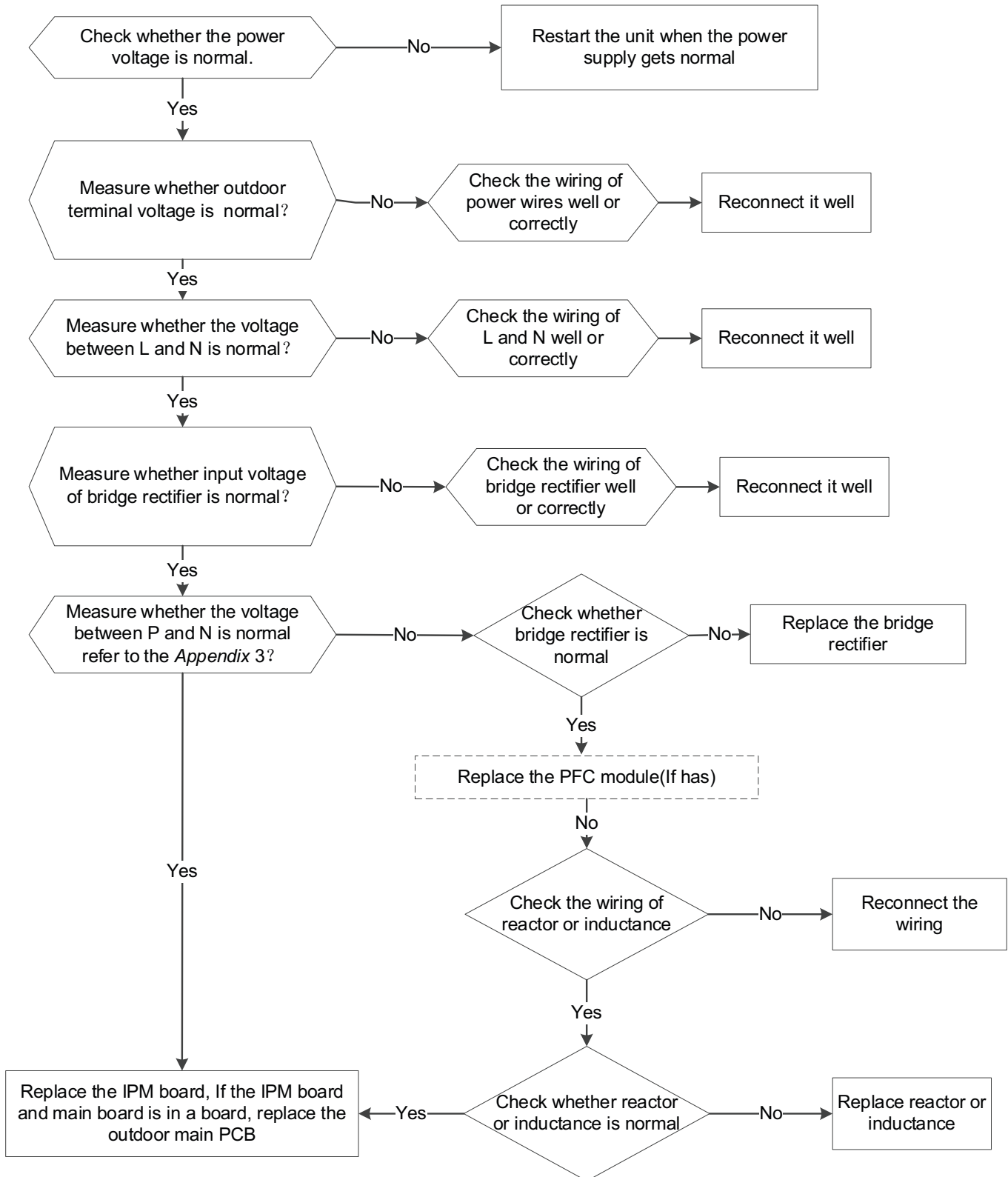


PC 10 (ODU low AC voltage protection)/ PC 11 (ODU main control board DC bus high voltage protection)/ PC 12 (ODU main control board DC bus high voltage protection /341 MCE error) Diagnosis and Solution

Description: Abnormal increases or decreases in voltage are detected by checking the specified voltage detection circuit.

Recommended parts to prepare: Power supply wires, IPM module board, Outdoor PCB, Bridge rectifier, PFC circuit or reactor

Troubleshooting and repair:



PC 0L (Low Ambient Temperature Protection)

Description: It is a protection function. When compressor is off, outdoor ambient temperature(T4) is lower than -35oC. for 10s, the AC will stop and display the failure code.

When compressor is on, outdoor ambient temperature(T4) is lower than -40oC.for 10s, the AC will stop and display the failure code.

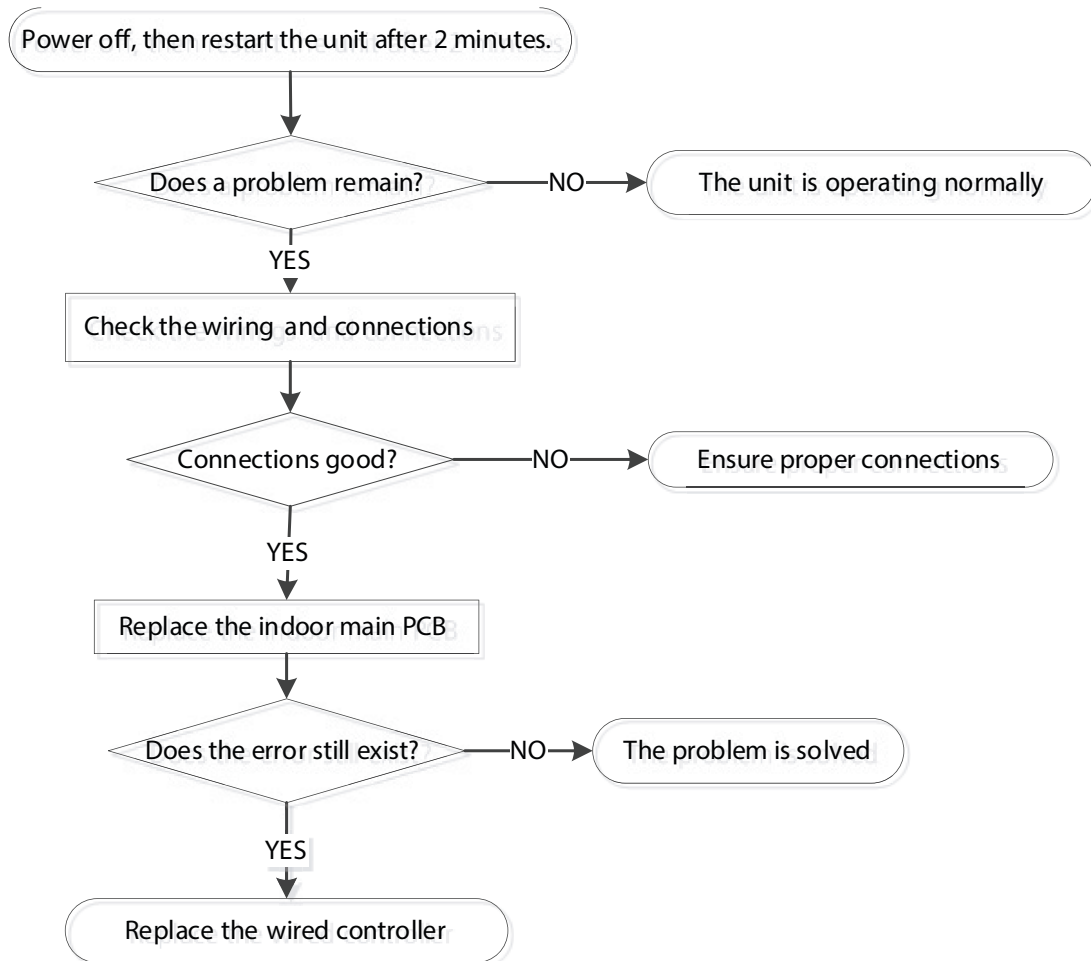
When outdoor ambient temperature(T4) is no lower than -32oC.for 10s, the unit will exit protection

EH b3 (Communication Malfunction Between Wire and Master Control) Diagnosis and Solution

Description: If Indoor PCB does not receive feedback from wired controller, the error displays on the wired controller

Recommended parts to prepare: Connection wires, Indoor PCB, Wired controller

Troubleshooting and repair:

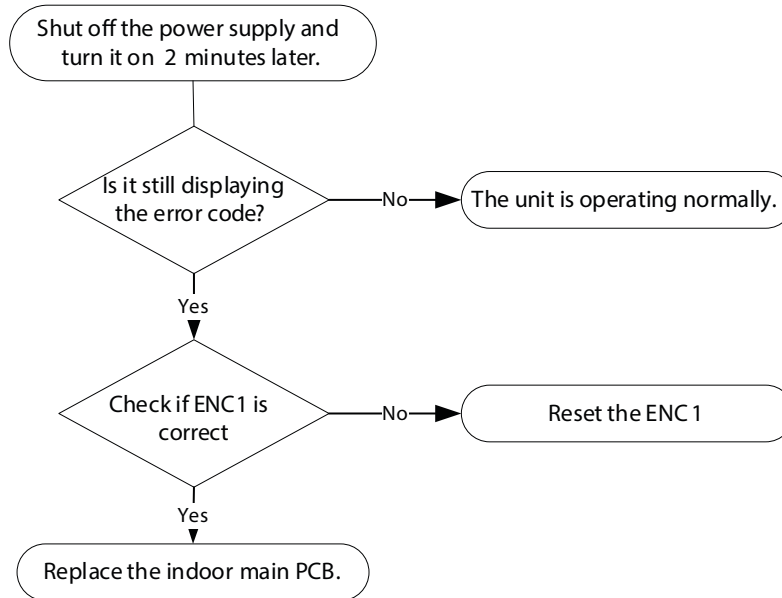


EH bA (Communication Malfunction Between Indoor Unit and External Fan Module)/ EH 3A(External Fan DC Bus Voltage Is Too Low Protection)/ EH 3b (External Fan DC Bus Voltage is Too High) Fault) Diagnosis and Solution

Description: Indoor unit does not receive the feedback from external fan module during 150 seconds. or Indoor unit receives abnormal increases or decreases in voltage from external fan module.

Recommended parts to prepare: Indoor main PCB

Troubleshooting and repair:

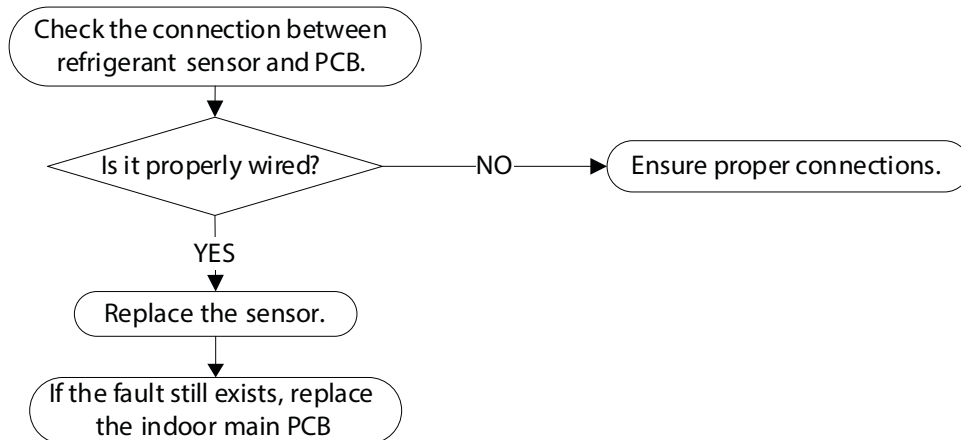


FH CC (Refrigerant Sensor Error) or EH C3(Refrigerant Sensor is Out of Range) Diagnosis and Solution

Description: Indoor unit receives fault signal for 10s or indoor unit does not receive feedback from refrigerant sensor for 150s.

Recommended parts to prepare: Connection wires, Sensors, Indoor main PCB

Troubleshooting and repair:



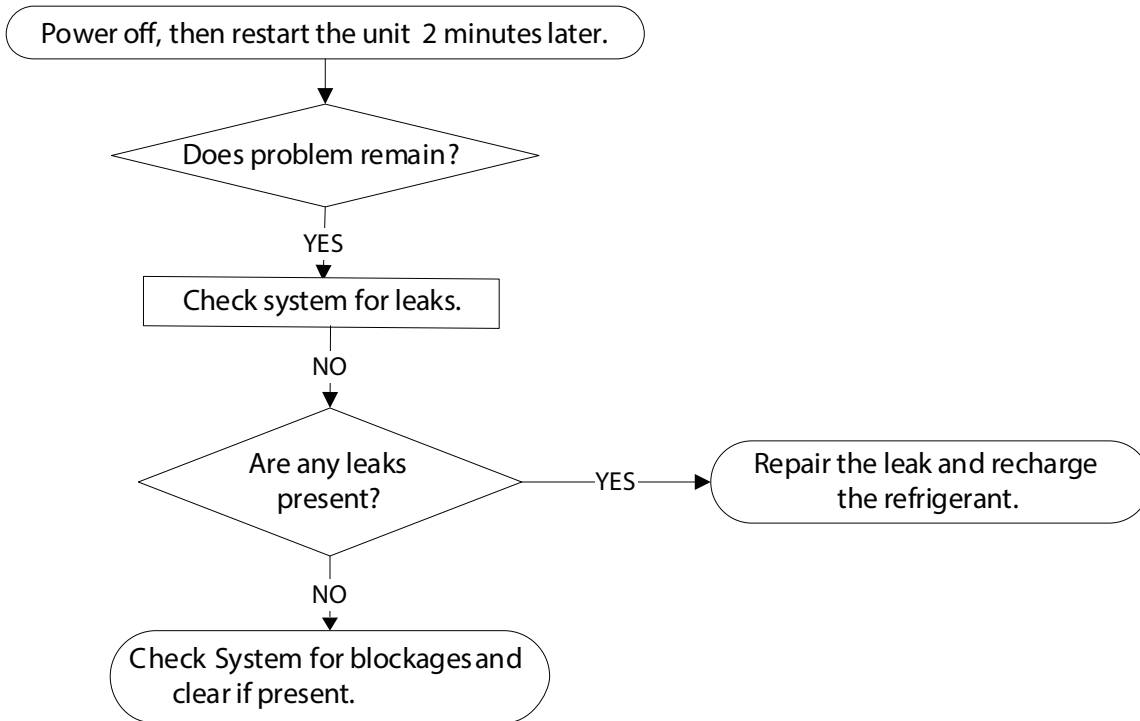
EH C1 (Refrigerant Sensor Detects Leakage) or EH C2 (Refrigerant Sensor is Out of Range and Leakage is Detected) Diagnosis and Solution

Description: The refrigerant sensor detects a concentration higher than or equal to 10%*LFL for 10 seconds or the refrigerant sensor detects a concentration higher than or equal to 20%*LFL or the multi model receives the refrigerant leakage protection fault sent by the outdoor unit.

Multi-zone: Only the buzzer of the indoor unit that detects refrigerant leakage continues to sound the alarm, the shortest sound is 10 seconds, and the longest sound is 5 minutes (you can press any key such as remote control or wire control, APP and so on to eliminate the alarm), and the other non-refrigerant leakage fault indoor unit only displays "ECC1", but the buzzer does not sound.

Recommended parts to prepare: Additional refrigerant

Troubleshooting and repair:

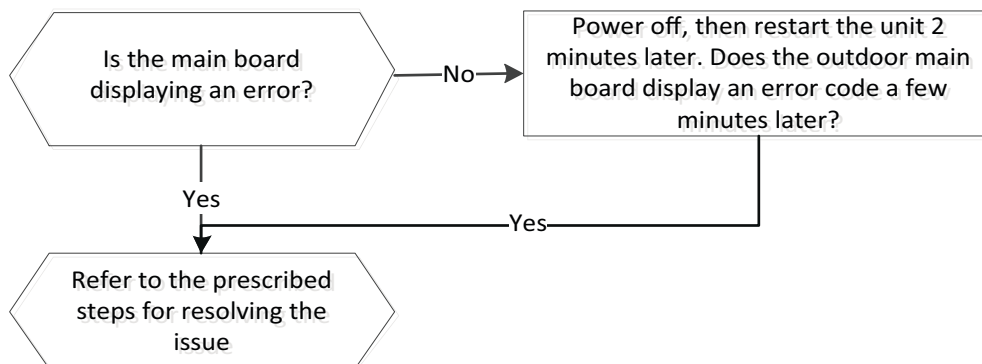


EC 0d (ODU Malfunction Diagnosis and Solution)

Description: The indoor unit detects the outdoor unit in error.

Recommended parts to prepare: Outdoor unit

Troubleshooting and repair:

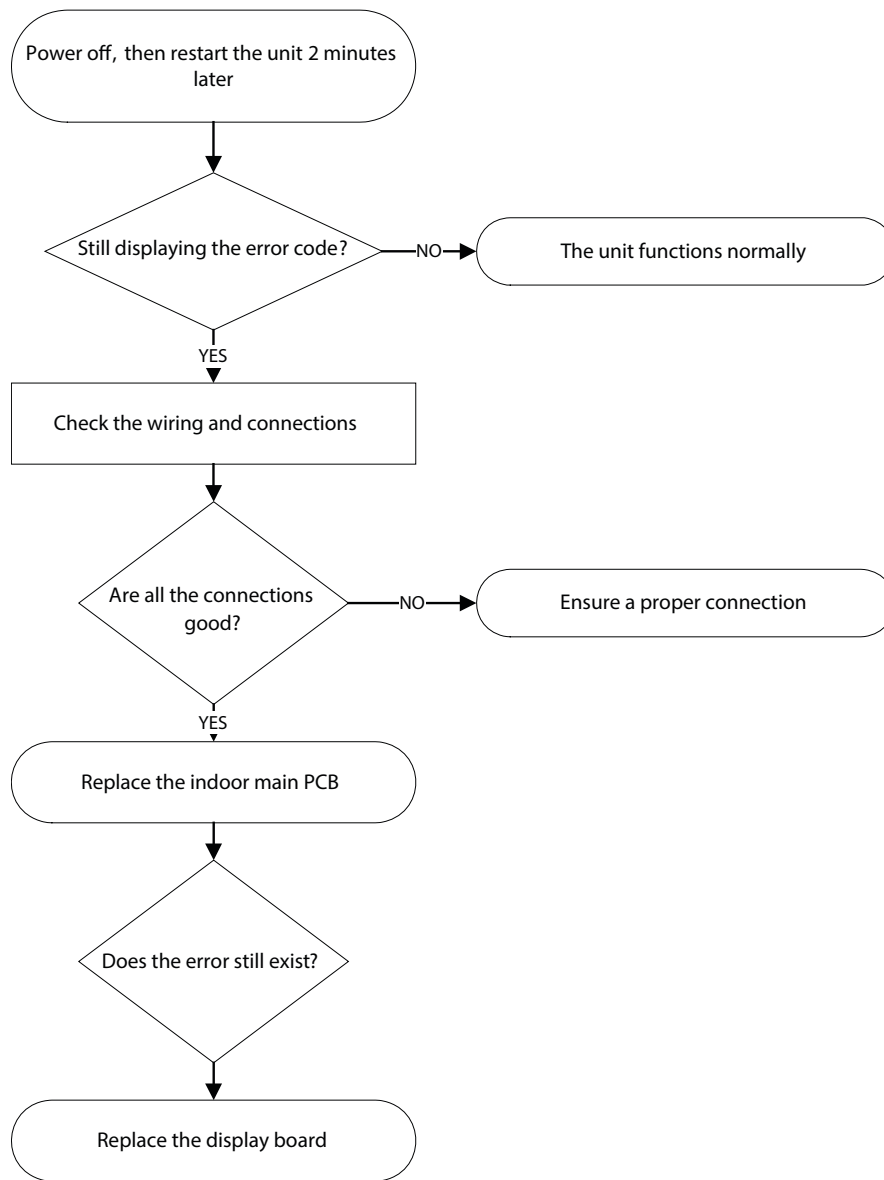


EH 0b(IDU Main Control Board and Display Board Communication Error Diagnosis and Solution)

Description: Indoor PCB does not receive feedback from the display board.

Recommended parts to prepare: Communication wire, Indoor PCB, Display board

Troubleshooting and repair:

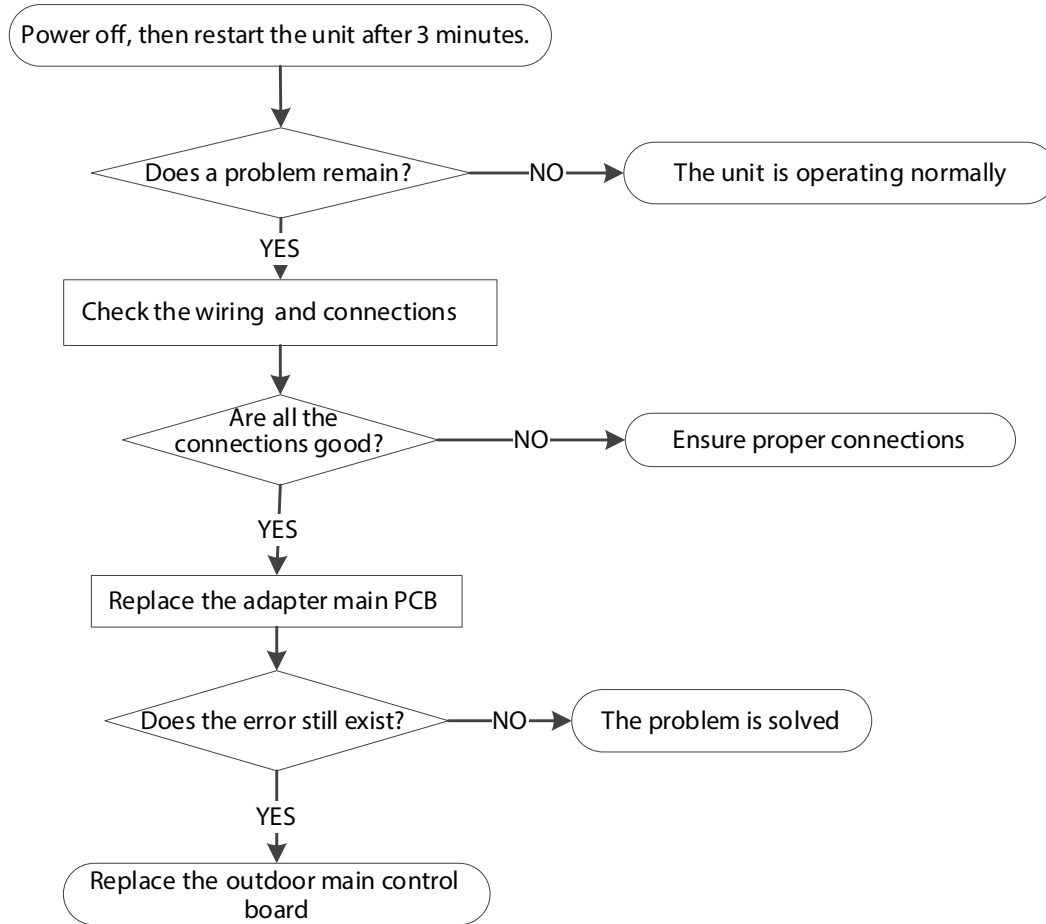


EL 16 (Communication Malfunction Between Adapter Board and Outdoor Main Board Diagnosis and Solution)

Description: The adapter PCB cannot detect the main control board.

Recommended parts to prepare: Connection wires, Adapter board, Outdoor main PCB

Troubleshooting and repair:



FL 09 (Mismatch between the new and old platforms diagnosis and solution)

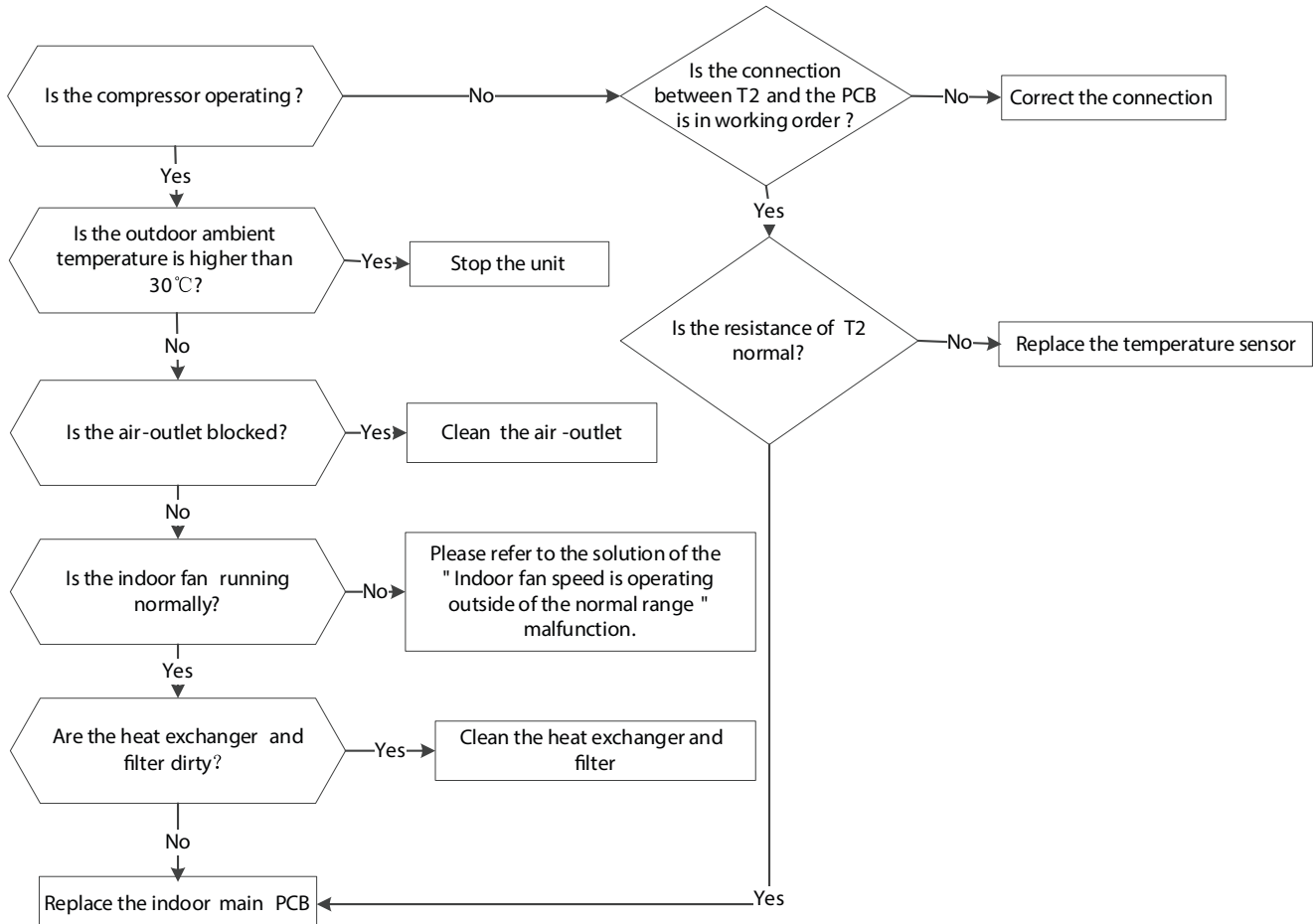
Description: Indoor and outdoor units are mismatched, the LED displays this code. Please replace the matching indoor or outdoor unit.

PH 90 (High temperature protection of evaporator diagnosis and solution)

Description: When evaporator coil temperature is more than 60°C in heating mode, the unit stops. It starts again only when the evaporator coil temperature is less than 52°C.

Recommended parts to prepare: Connection wires, Evaporator coil temperature sensor (T2), Indoor fan, Indoor main PCB

Troubleshooting and repair:

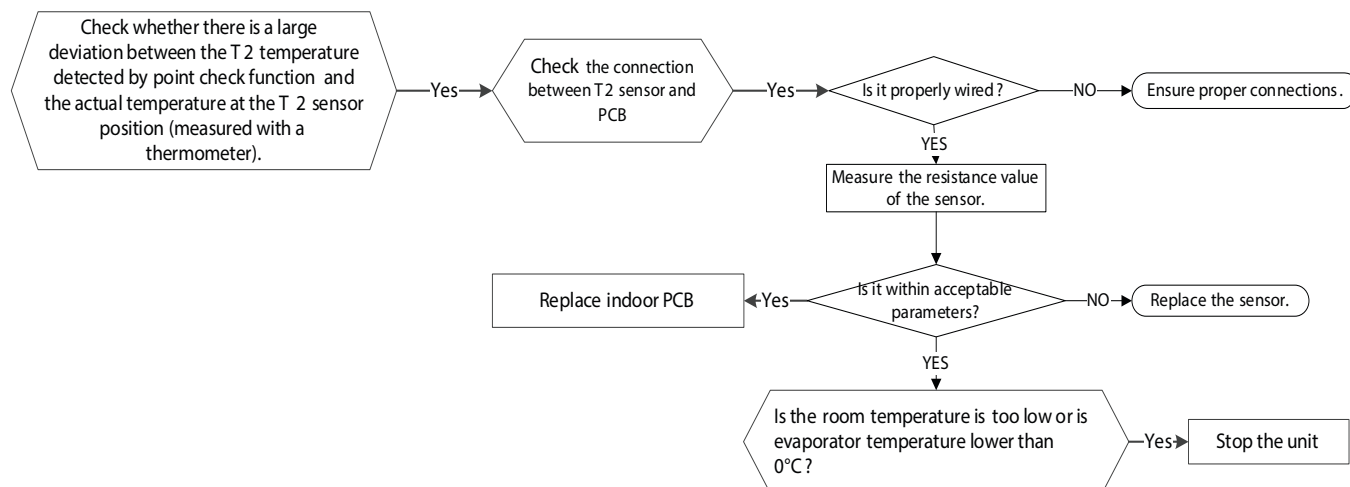


PH 91 (Low temperature protection of evaporator diagnosis and solution)

Description: When evaporator coil temperature is lower than 0°C in cooling mode or drying mode, the unit stops. It starts again only when the evaporator coil temperature is more than 5°C.

Recommended parts to prepare: Connection wires, Evaporator coil temperature sensor (T2), Indoor main PCB

Troubleshooting and repair:



CHECK PROCEDURES

Temperature Sensor Check

WARNING

ELECTRICAL SHOCK HAZARD

Be sure to turn off all power supplies or disconnect all wires to avoid electric shock. Operate after compressor and coil have returned to normal temperature in case of injury.

1. Disconnect the temperature sensor from the PCB.
2. Measure the sensor's resistance value with a multi-meter.
3. Check the corresponding temperature sensor resistance value table (see "Temperature Sensor Resistance Value Table for TP (°C - K)" on page 87 and "Other Temperature Sensors Resistance Value Table (°C - K)" on page 88).

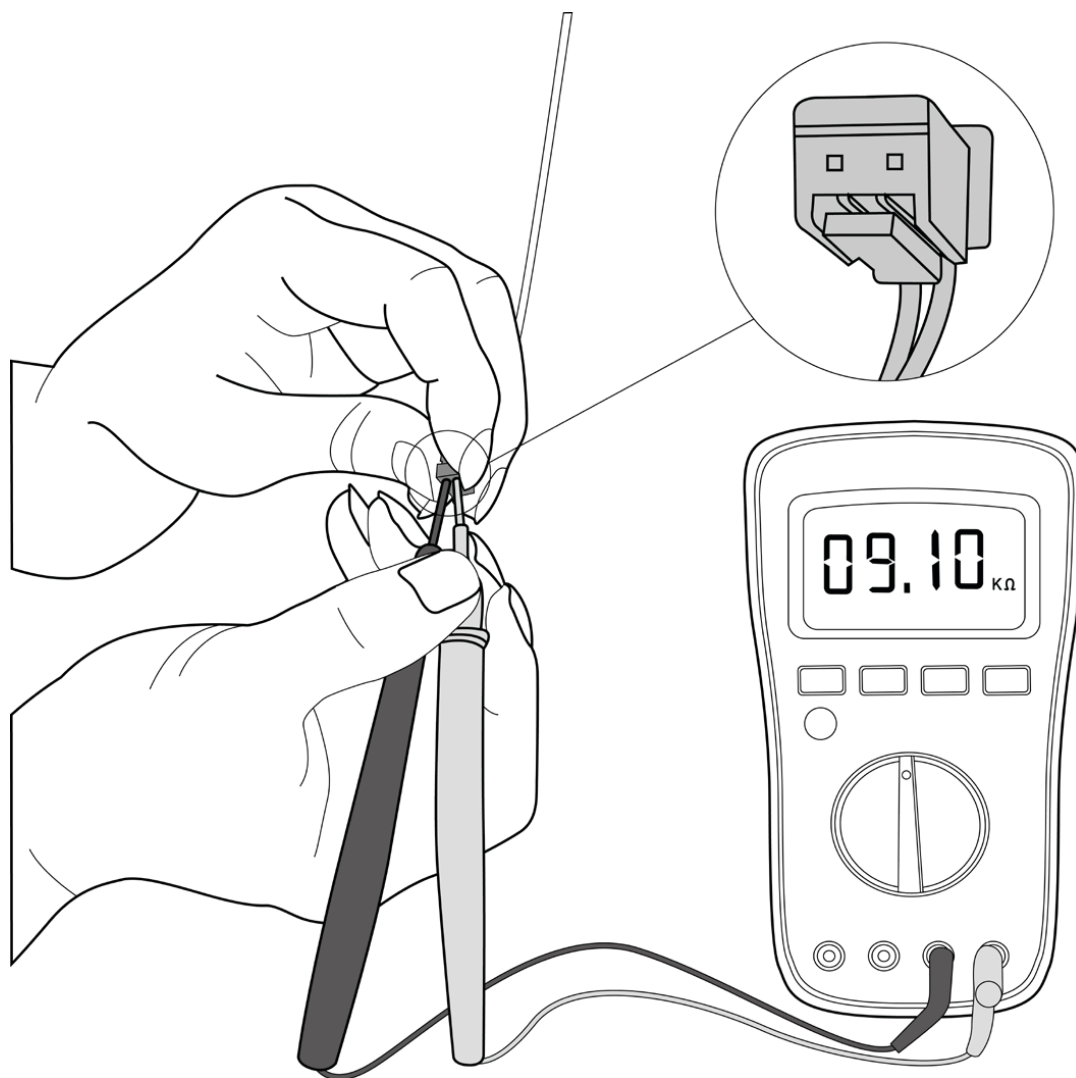


Fig. 31 —Measure the Sensor's Resistance Value

Compressor Check

- 1. Disconnect the compressor power cord from the outdoor PCB.
- 2. Measure the resistance value of each winding using a multi-meter.
- 3. Check the resistance value of each winding in tables 9 through 12:

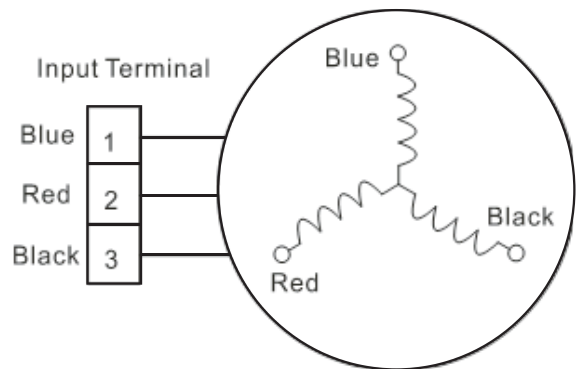


Fig. 32 —Compressor Check

Table 9 – Resistance Values

Resistance Value	KSN140D58UFZ	KTF250D22UMT	KTM240D46UKT2	KTF310D43UMT	MTH550UKPC8FU
Blue-Red	1.86Ω	0.75Ω	1.04Ω	0.65Ω	0.295Ω
Blue-Black					
Red-Black					

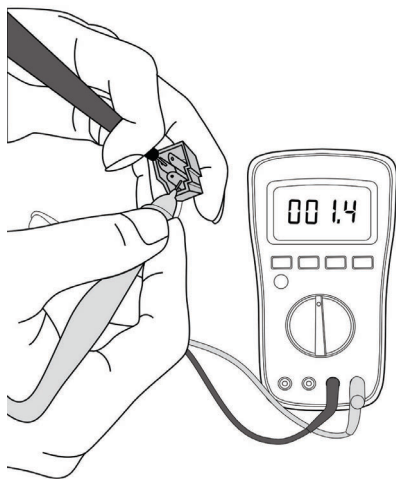


Fig. 33 —Resistance Check

NOTE: The picture and the value are only for reference, actual condition and specific value may vary.

IPM Continuity Check

⚠

WARNING

ELECTRICAL SHOCK HAZARD

Electricity remains in capacitors even when the power supply is off.
Ensure the capacitors are fully discharged before troubleshooting.

1. Turn off outdoor unit and disconnect power supply.
2. Discharge electrolytic capacitors and ensure all energy-storage unit has been discharged.
3. Disassemble outdoor PCB or disassemble IPM board.
4. Measure the resistance value between P and U(V, W, N); U(V, W) and N.

Table 10 – Resistance Value

Digital Tester		Resistance Value	Digital Tester		Resistance Value
(+) Red	(-) Black	∞ (Several MfÇ)	(+) Red	(-) Black	∞ (Several MfÇ)
P	N		U	N	
	U		V		
	V		W		
	W		-		

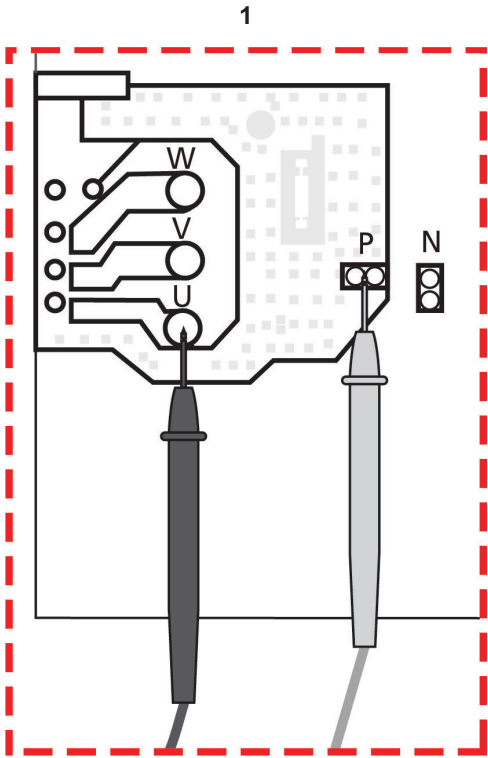


Fig. 34 —Resistance Value

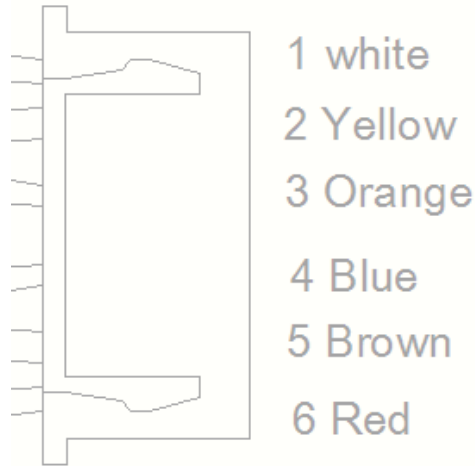
Table 11 – Voltage Range

208-240V (1-phase)		
In Standby		
Around 310VDC		
In Operation		
With passive PFC module	With partial active PFC module	With fully active PFC module
>200VDC	>310VDC	>370VDC

4-Way Valve Check

1. Power on, use a digital tester to measure the voltage, when the unit operates in cooling, it is 0V. When the unit operates in heating, it is about equal to power supply voltage.
If the value of the voltage is not in the range, the PCB must have problems and need to be replaced.
2. Turn off the power, use a digital tester to measure the resistance.
The value should be 1.8~2.5 K Ω .

EXV Check



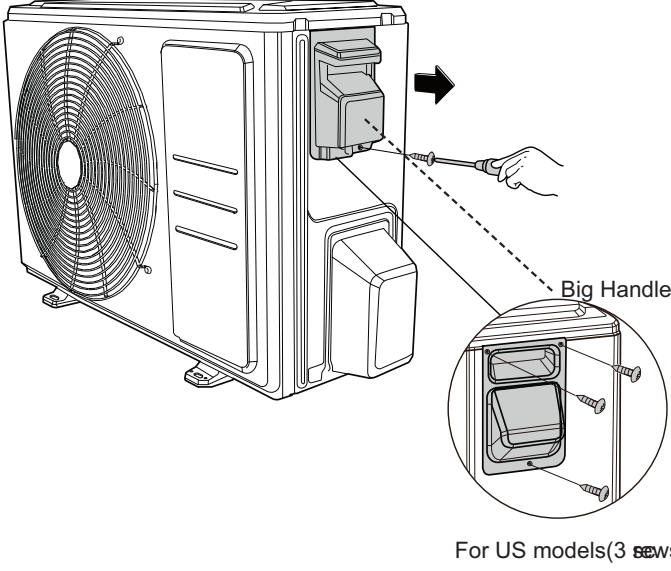
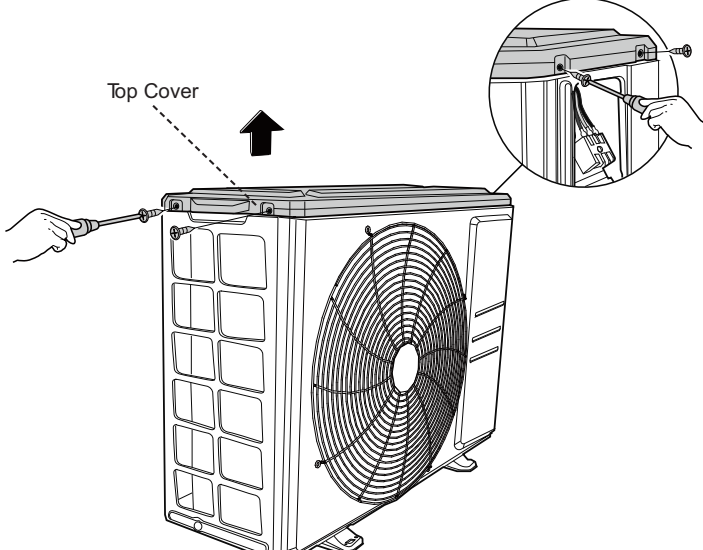
1. Turn off outdoor unit and disconnect power supply.
2. Disconnect the connectors of EXV.
3. Measure the resistance value between Red and Blue (Yellow); Brown and Orange (White).

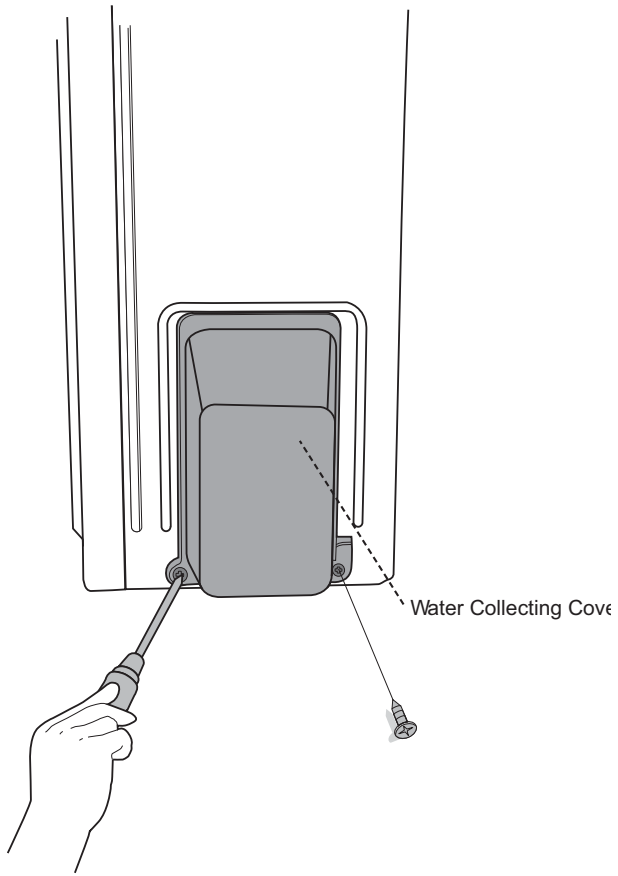
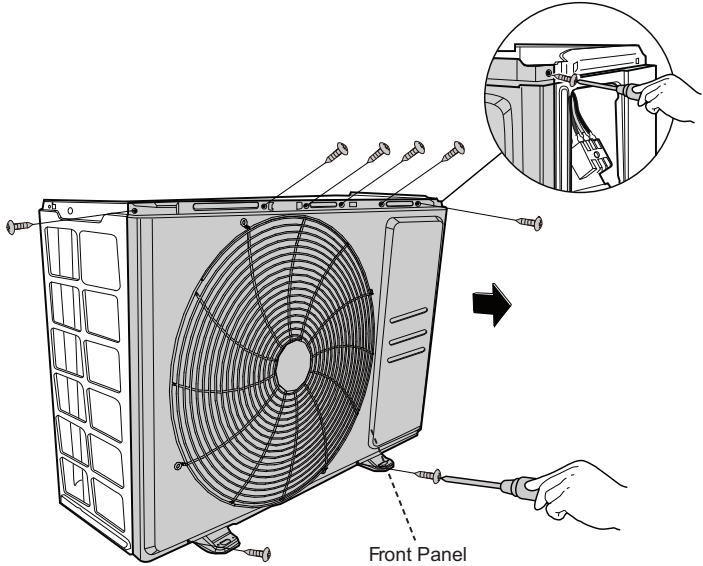
Resistance to EXV coil

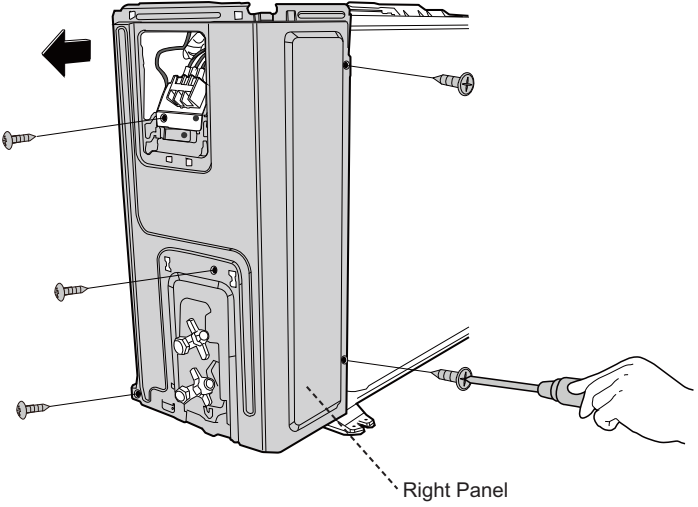
Color of Lead Wire	Normal Value
Redd-Blue	About 500 Ω
Red-Yellow	
Brown-Orange	
Brown-White	

DISASSEMBLY INSTRUCTIONS

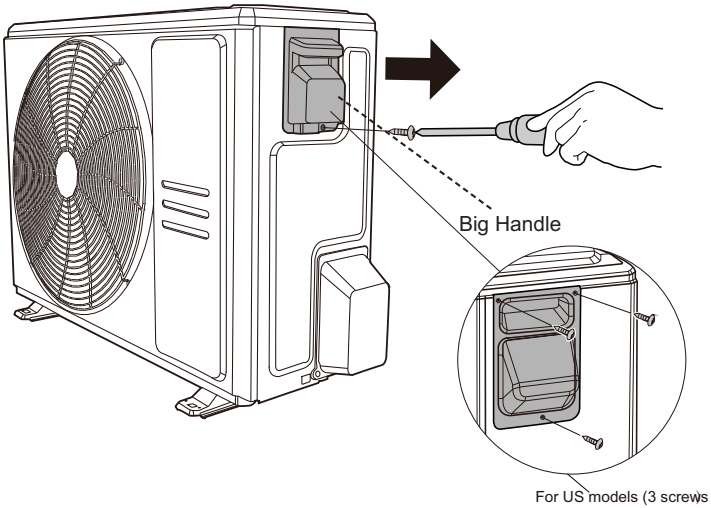
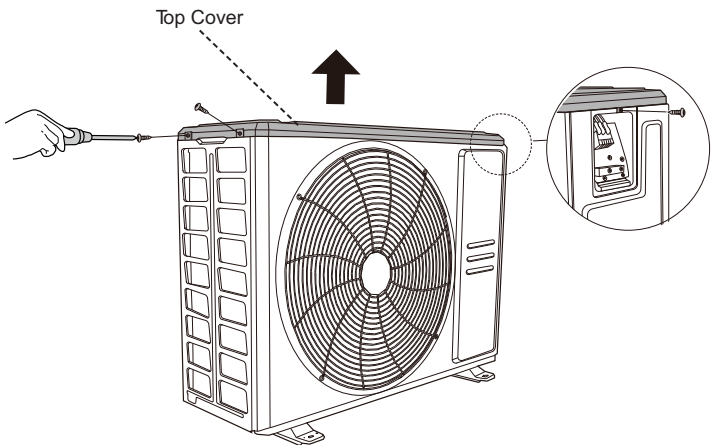
18K Unit Disassembly - Panel Plate

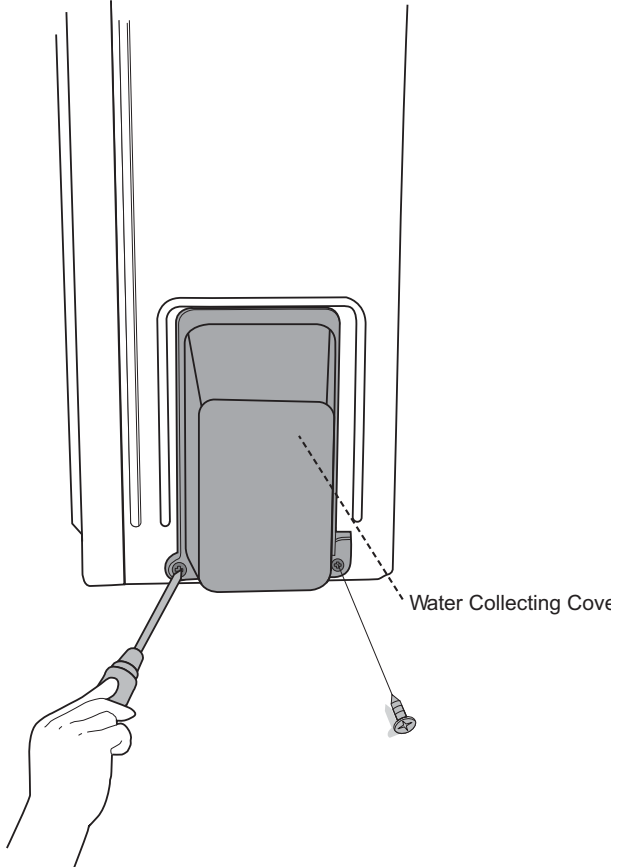
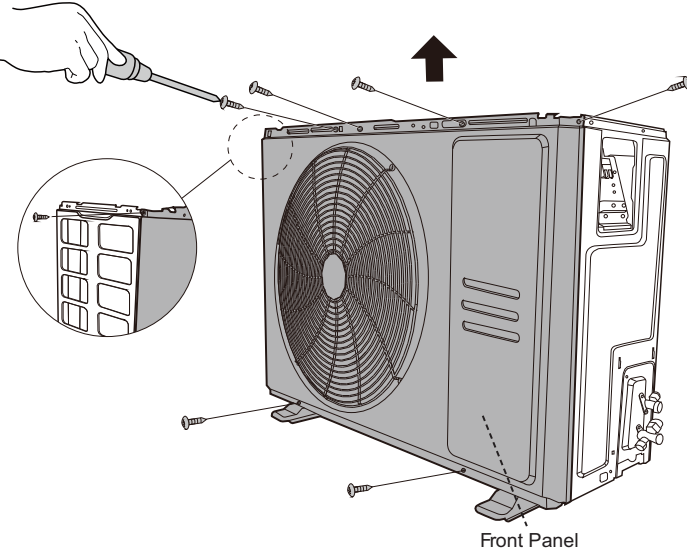
PROCEDURES	ILLUSTRATION
1. Turn off the air conditioner and the power breaker 2. Remove the screw of the big handle and then remove the big handle (3 screws) (see illustration)	 Big Handle For US models(3 screws)
3. Remove the screws of the top cover and then remove the top cover (4 screws). One of the screws is located underneath the big handle. (see illustration)	 Top Cover

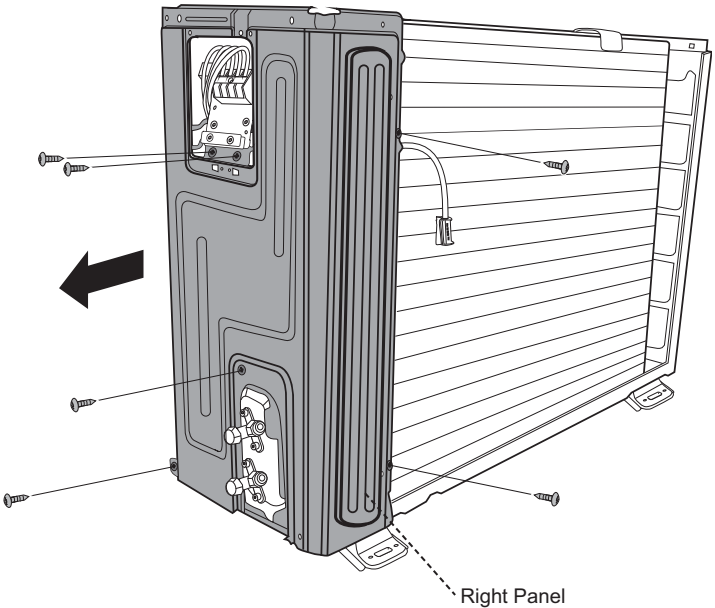
PROCEDURES	ILLUSTRATION
4. Remove the screws of water collecting cover and then remove the water collecting cover (2 screws).(see illustration)	 The illustration shows a hand using a screwdriver to remove a screw from the bottom of the water collecting cover. A dashed line points to the cover, which is labeled "Water Collecting Cover".
5. Remove the screws of the front panel and then remove the front panel (7 screws (on/off models) or 9 screws. (inverter models). (see illustration)	 The illustration shows a hand using a screwdriver to remove a screw from the front panel of the outdoor unit. A dashed line points to the front panel, which is labeled "Front Panel". An inset circular diagram shows a close-up of the top of the unit where the front panel is being detached.

PROCEDURES	ILLUSTRATION
6. Remove the screws of the right panel and then remove the right panel (5 screws) (see illustration)	 The illustration shows a side view of a rectangular electronic device. A hand is using a screwdriver to remove screws from the right panel. A dashed line points to the right panel, labeled 'Right Panel'. An arrow points to the left, indicating the direction of removal. There are five screws shown being removed from the right panel.

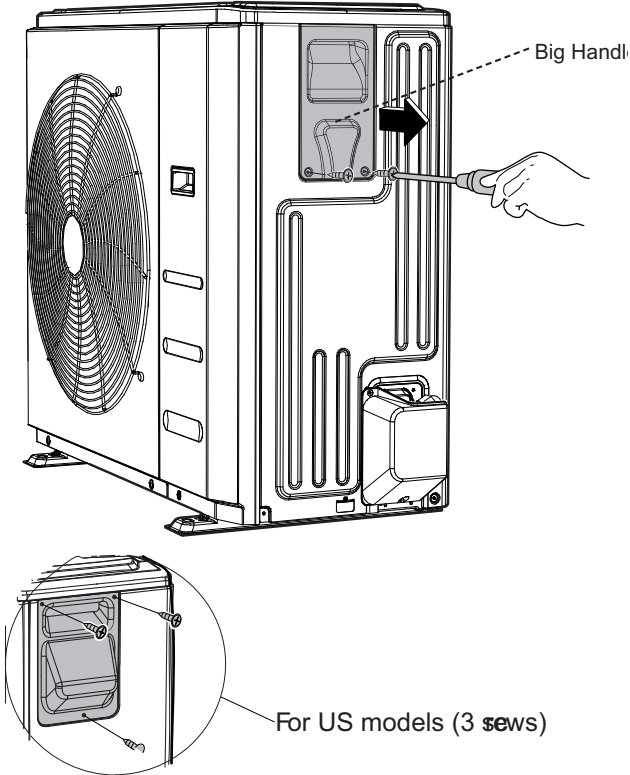
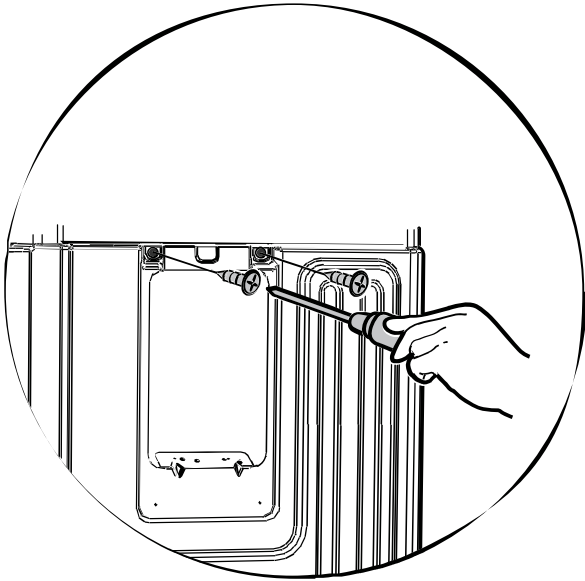
24K Unit Disassembly - Panel Plate

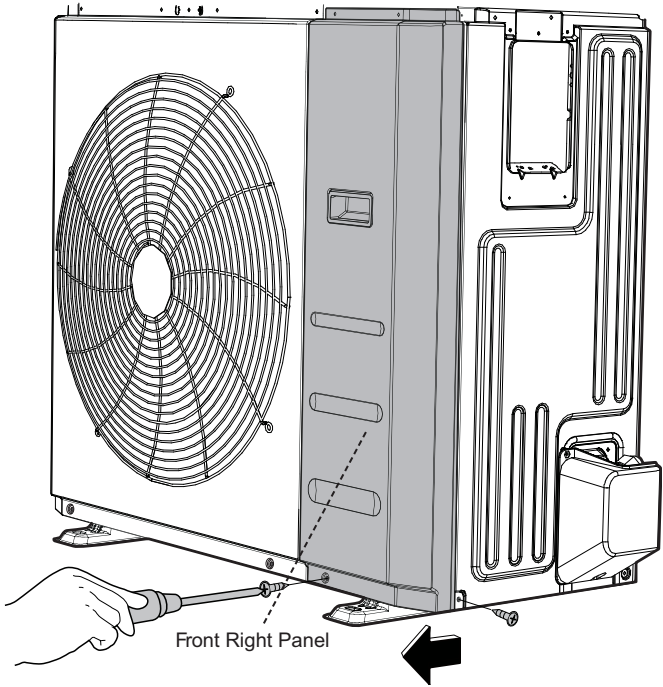
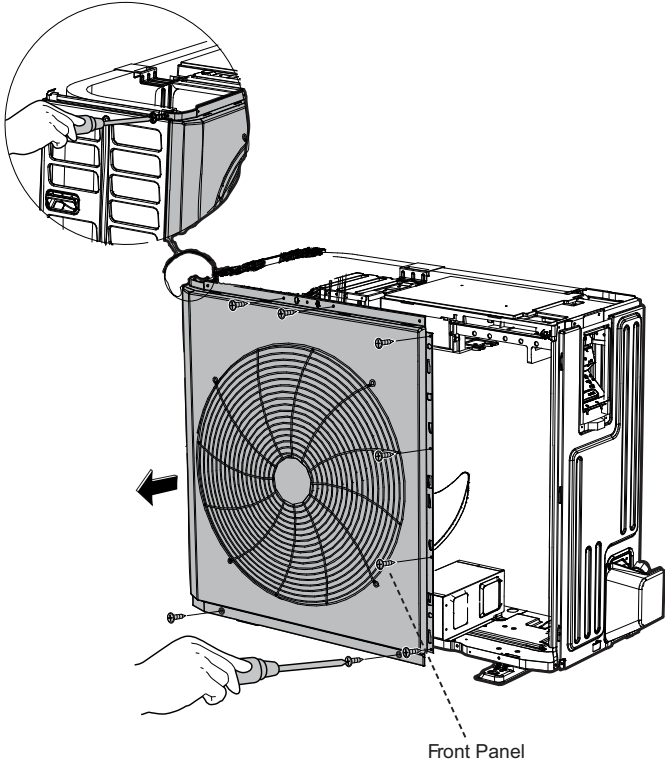
PROCEDURES	ILLUSTRATION
1. Turn off the air conditioner and the power breaker 2. Remove the screw of the big handle and then remove the big handle (3 screws) (see illustration)	 Big Handle For US models (3 screws)
3. Remove the screws of the top cover and then remove the top cover (3 screws). One of the screws is located underneath the big handle. (see illustration)	 Top Cover

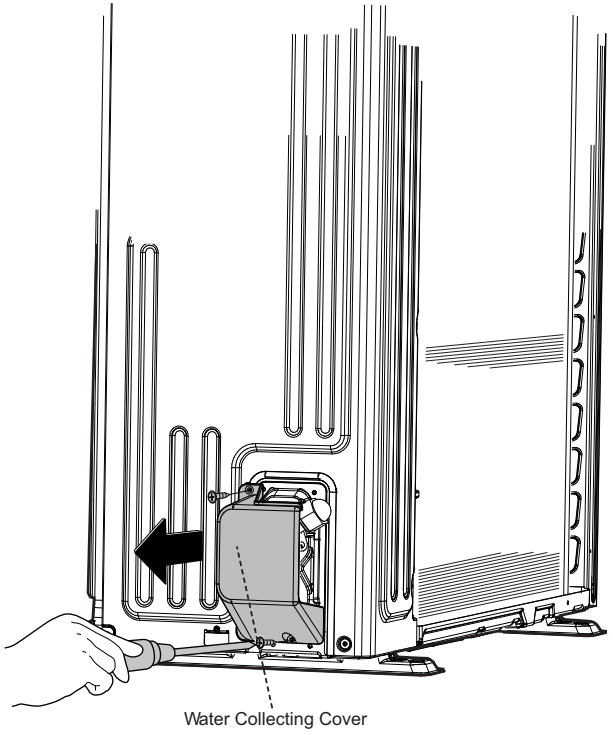
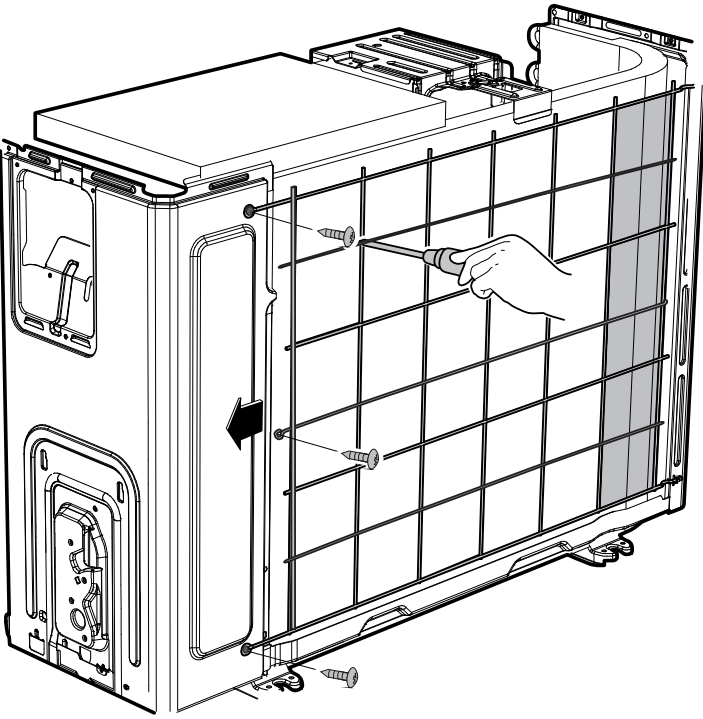
PROCEDURES	ILLUSTRATION
4. Remove the screws of water collecting cover and then remove the water collecting cover (2 screws).(see illustration)	 The illustration shows a hand using a screwdriver to remove a screw from the bottom of a rectangular water collecting cover. The cover is mounted on the side of the outdoor unit. A dashed line points to the cover, and a label 'Water Collecting Cove' is next to it. Another screw is shown being removed from the top of the cover.
5. Remove the screws of the front panel and then remove the front panel (7 screws (on/off models) or 9 screws. (inverter models). (see illustration)	 The illustration shows a hand using a screwdriver to remove a screw from the top of the front panel. The front panel is a large rectangular cover with a circular fan grille on the left. A dashed line points to the panel, and a label 'Front Panel' is next to it. An inset shows a close-up of the front panel with a grid of screws. An upward arrow indicates the direction to lift the panel after the screws are removed.

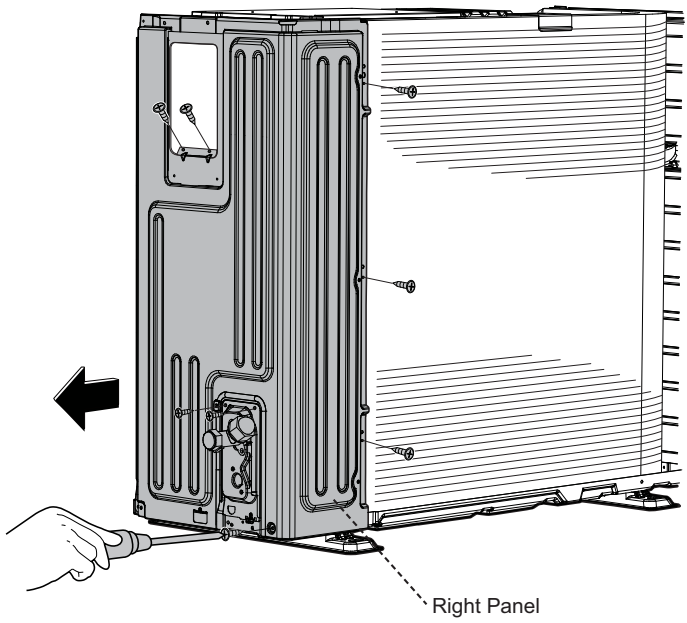
PROCEDURES	ILLUSTRATION
6. Remove the screws of the right panel and then remove the right panel (5 screws) (see illustration)	 Right Panel

30K/36K Unit Disassembly - Panel Plate

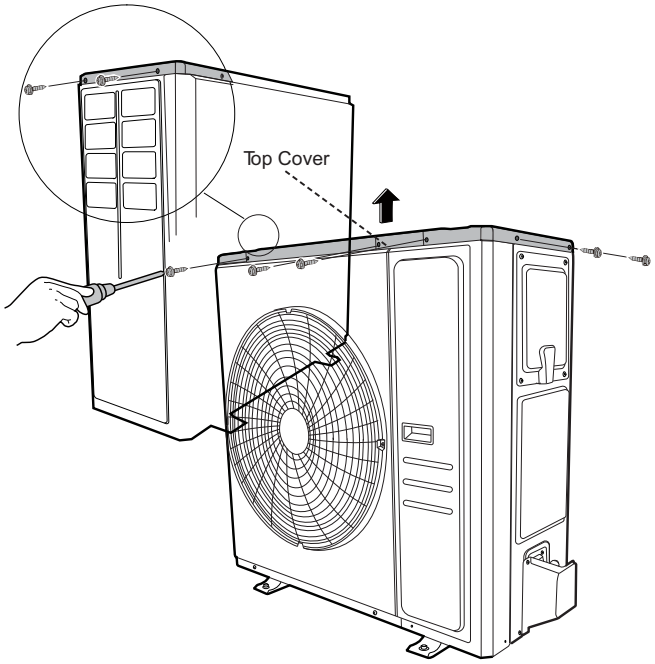
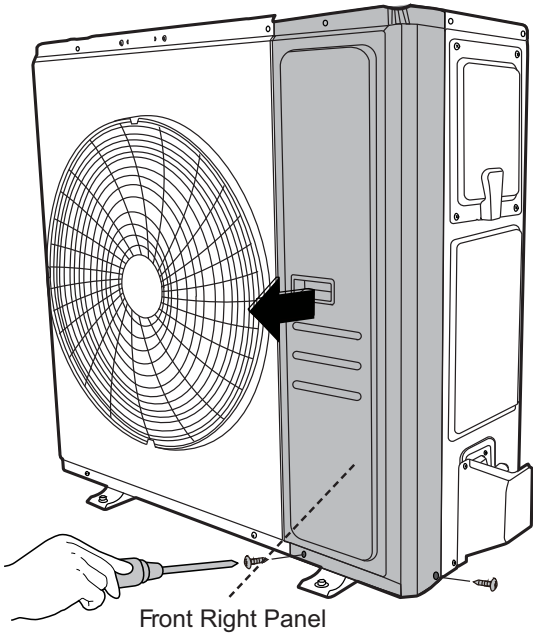
PROCEDURES	ILLUSTRATION
1. Turn off the air conditioner and the power breaker 2. Remove the screw of the big handle and then remove the big handle (2 screws) (see illustration)	 Big Handle For US models (3 screws)
3. Remove the screws of the top cover and then remove the top cover (4 screws). One of the screws is located underneath the big handle. (see illustration)	

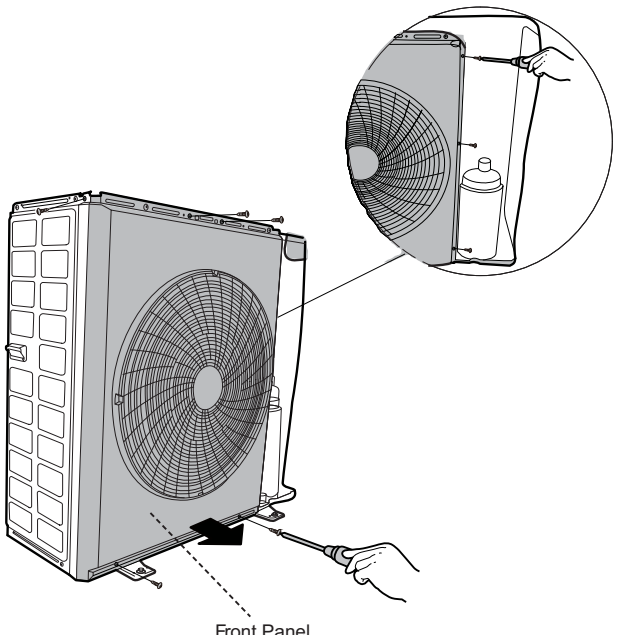
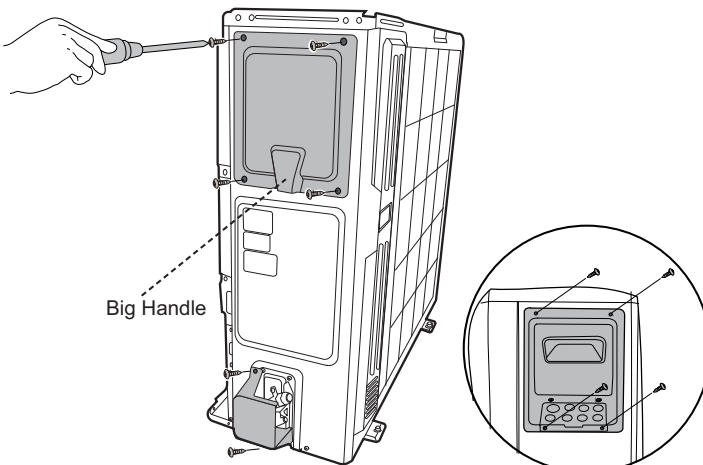
PROCEDURES	ILLUSTRATION
4. Remove the screws of the front right panel and then remove the front right panel (2 screws).(see illustration)	
5. Remove the screws of the front panel and then remove the front panel (9 screws). (see illustration)	

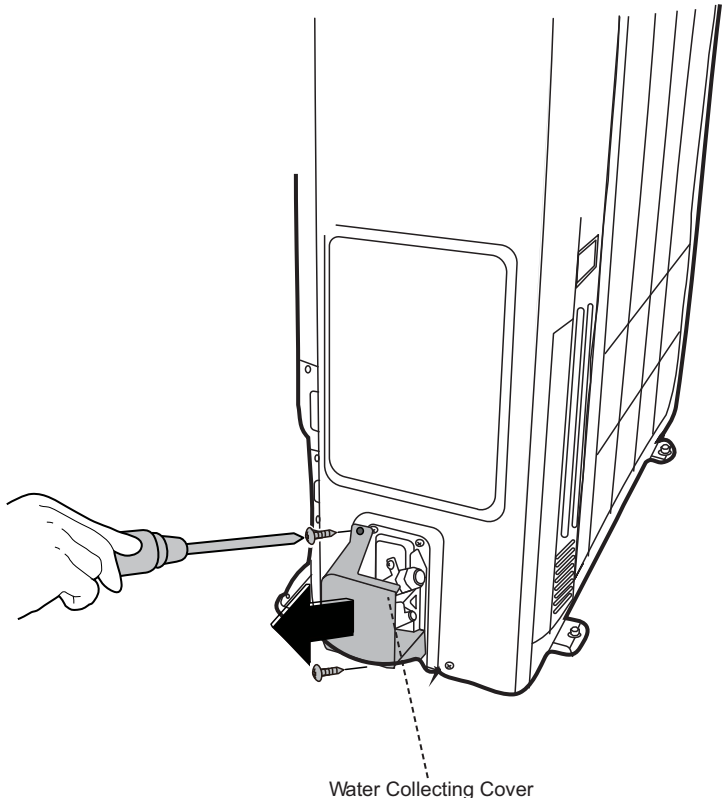
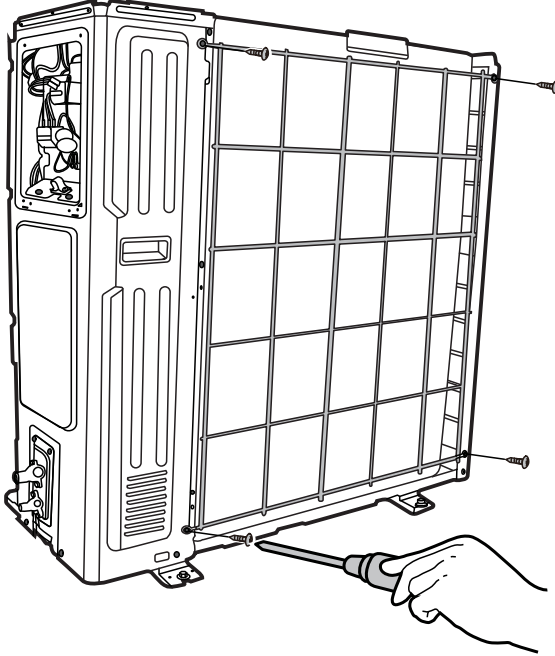
PROCEDURES	ILLUSTRATION
6. Remove the screws of water collecting cover and then remove the water collecting cover (2 screws) (see illustration)	 The illustration shows the rear of a vertical unit with a grid of vertical fins. A hand is using a screwdriver to remove a screw from a rectangular cover located at the bottom center. A dashed line points to this cover with the label 'Water Collecting Cover'. A large black arrow points to the left, indicating the direction of removal.
7. For some models, remove the screws of the rear net and then remove the rear net (3 screws) (see illustration)	 The illustration shows the rear of the unit with a large rectangular grid (the rear net) attached to the back panel. A hand is using a screwdriver to remove a screw from the net. A large black arrow points to the left, indicating the direction of removal.

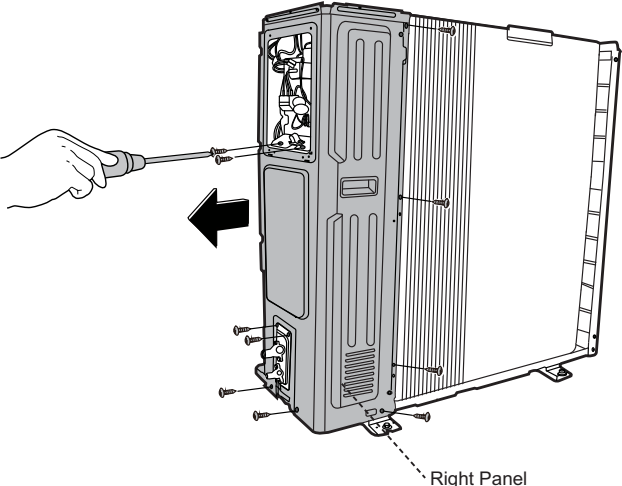
PROCEDURES	ILLUSTRATION
8. Remove the screws of the right panel and then remove the right panel (8 screws)	 The illustration shows a side view of a rectangular electronic device. A hand is using a screwdriver to remove screws from the right panel. A large black arrow points to the left, indicating the direction of removal. A dashed line points to the right panel, which is labeled 'Right Panel'.

48K/60K Unit Disassembly - Panel Plate

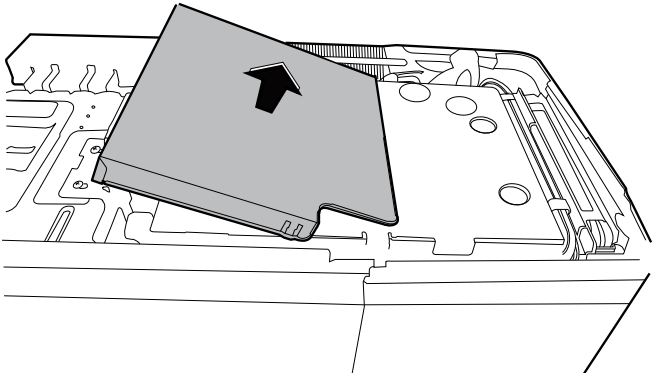
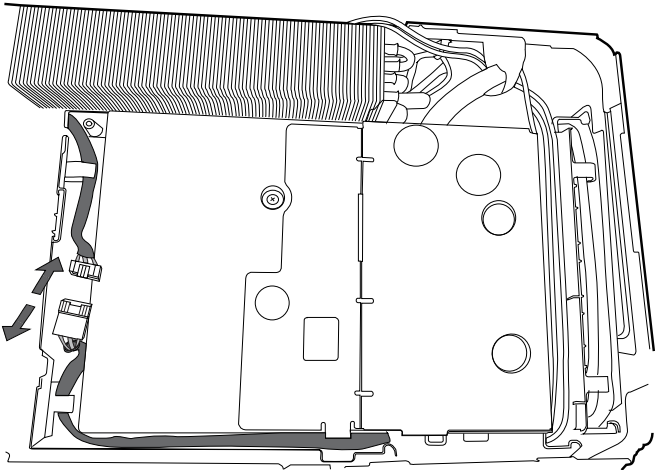
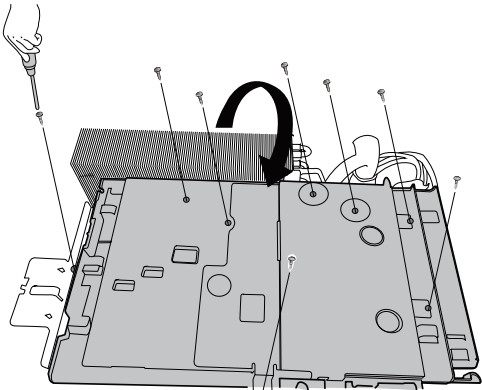
PROCEDURES	ILLUSTRATION
1. Turn off the air conditioner and the power breaker 2. Remove the screws of the top cover and then remove the top cover (7 screws). (see illustration)	 The illustration shows a side view of the air conditioner unit. A hand is using a screwdriver to remove screws from the top cover. The top cover is shown being lifted away from the unit. The label 'Top Cover' points to the cover being removed.
3. Remove the screws of the front right panel and then remove the front right panel (2 screws) (see illustration)	 The illustration shows a side view of the air conditioner unit. A hand is using a screwdriver to remove screws from the front right panel. The front right panel is shown being removed from the unit. The label 'Front Right Panel' points to the panel being removed.

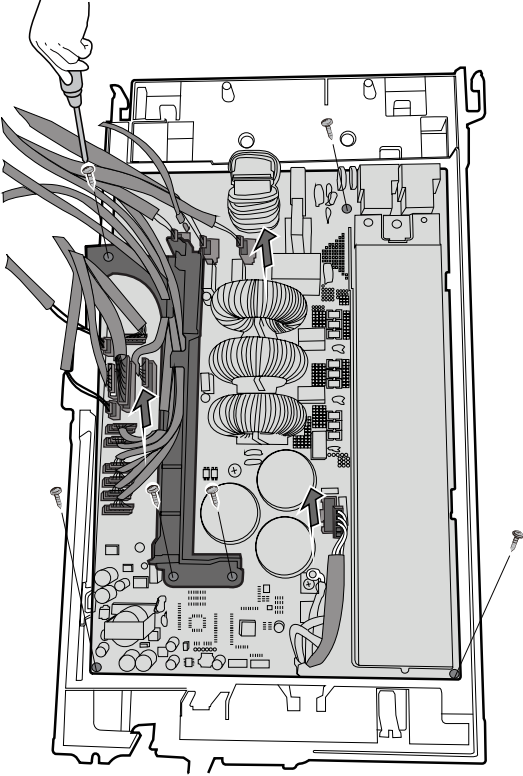
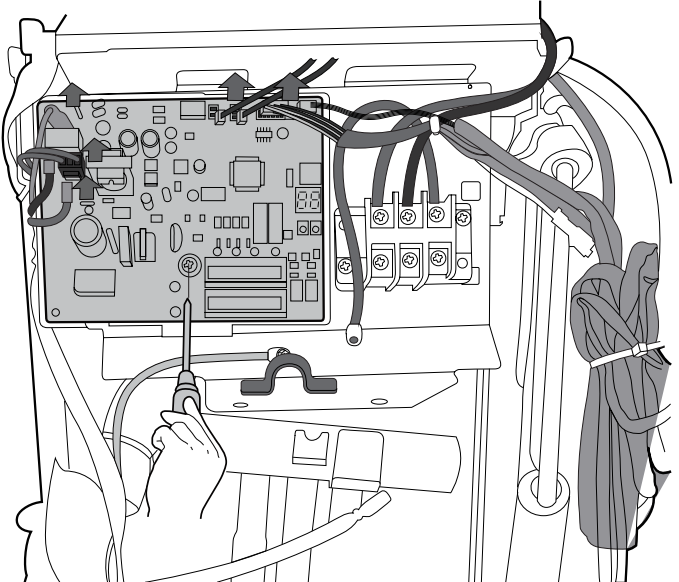
PROCEDURES	ILLUSTRATION
4. Remove the screws of the front panel and then remove the front panel (8 screws).(see illustration)	 Front Panel
5. Remove the screws of the big handle and then remove the big handle (4 screws)	 Big Handle For US models

PROCEDURES	ILLUSTRATION
6. Remove the screws of water collecting cover and then remove the water collecting cover (2 screws) (see illustration)	 Water Collecting Cover
7. Remove the screws of the rear net and then remove the rear net (4 screws) (see illustration)	

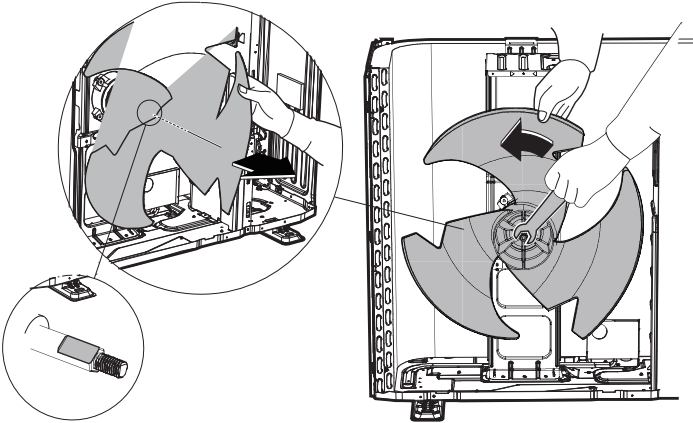
PROCEDURES	ILLUSTRATION
8. Remove the screws of the right panel and then remove the right panel (10 screws) (see illustration)	 The illustration shows a side view of a rectangular electronic device. A hand is using a screwdriver to remove a screw from the top of the right panel. A large black arrow points to the right panel. The right panel is labeled 'Right Panel' with a dashed line. There are 10 screws in total, 5 on the top and 5 on the bottom. The device has a control panel on the left side with a handle and some buttons. The right panel is a vertical panel with a handle and a latch.

All Size Units, Disassembly - Electrical Parts

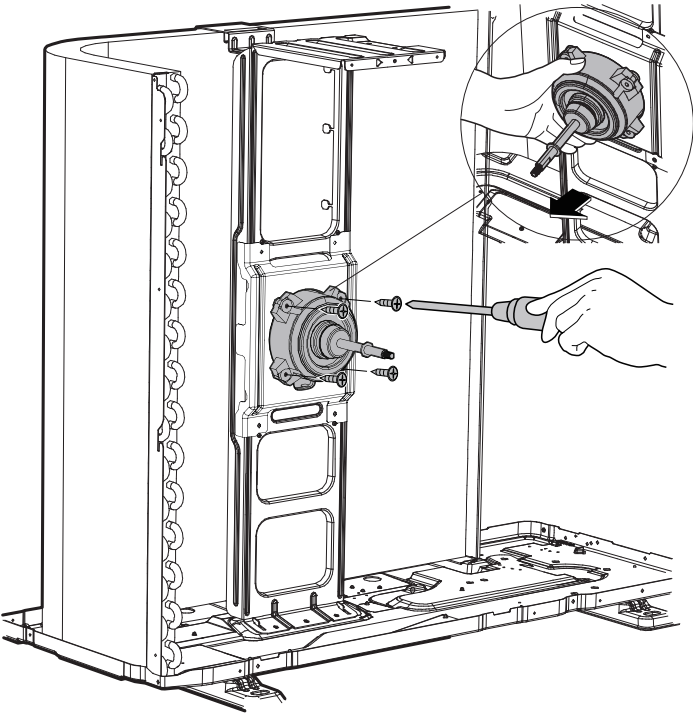
PROCEDURES	ILLUSTRATION
1. Remove the cover of electrical control box.(see illustration)	
2. Disconnect the fan motor connector. (see illustration)	
3. Remove eight fixing screws. (see illustration) 4. Turn over the electronic control box subassembly.	

PROCEDURES	ILLUSTRATION
5. Remove 3 screws and then remove the bracket. (see illustration) 6. Disconnect the connectors from the electronic control board. (see illustration) 7. Remove 3 screws and then remove the electronic control board. (see illustration)	
8. Pull out the connector, remove one screw and then remove the 24V board subassembly on terminal board. (see illustration)	

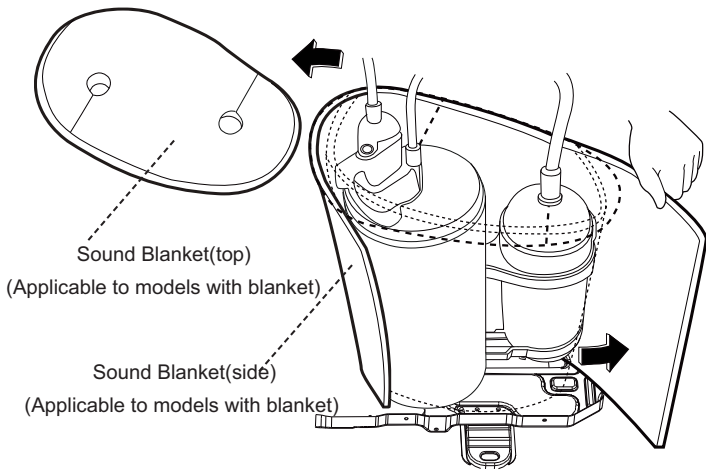
All Size Units, Disassembly - Fan Assembly

PROCEDURES	ILLUSTRATION
1. Remove the nut securing the fan with a spanner. 2. Remove the fan. (see illustration)	

All Size Units, Disassembly - Fan Motor

PROCEDURES	ILLUSTRATION
1. Remove the fixing screws of the fan motor (4 screws) 2. Remove the fan motor. (see illustration)	 A technical line drawing of a refrigerator's internal compartment with the back panel removed. A fan motor is mounted on the back wall. A hand is shown using a screwdriver to remove one of the four screws securing the motor. An inset circular image provides a magnified view of the screwdriver tip turning a screw on the motor's mounting bracket.

All Size Units, Disassembly - Sound Blanket

PROCEDURES	ILLUSTRATION
1. Remove the sound blanket (side and top)) (see illustration)	 The illustration shows a side view of a mechanical unit with two cylindrical components. A hand is shown pulling a rectangular sound blanket away from the side of the unit, indicated by a black arrow. A dashed line outlines the blanket's position. A separate oval diagram shows the top of the unit with two circular openings, with a black arrow pointing to the top of the unit, indicating the removal of the top sound blanket. Labels with dashed lines point to the top and side of the unit. Sound Blanket(top) (Applicable to models with blanket) Sound Blanket(side) (Applicable to models with blanket)

All Size Units, Disassembly - Four-Way Valve (For Heat Pump Models

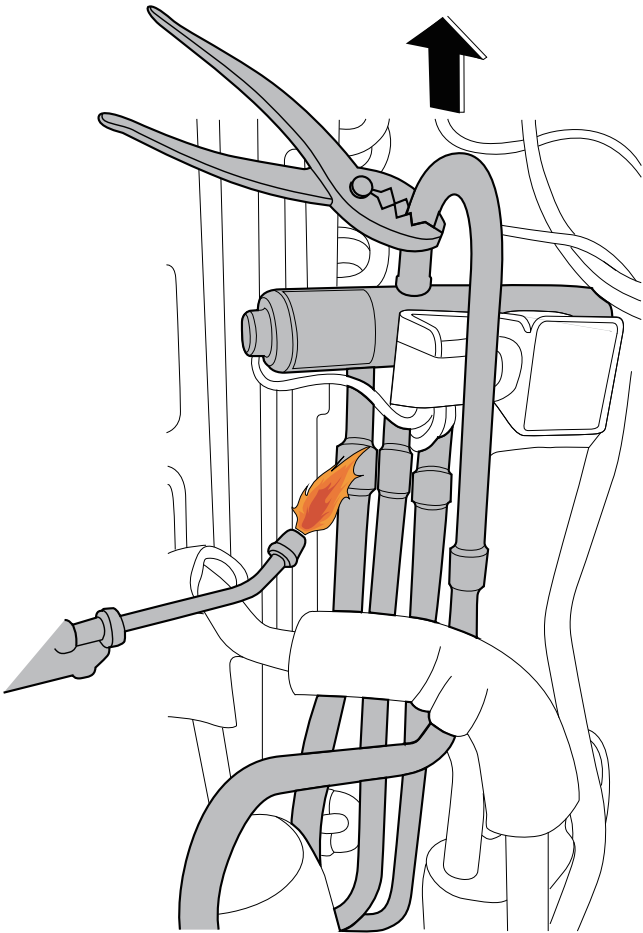


WARNING

FIRE HAZARD

Evacuate the system and confirm that there is no refrigerant left in the system before removing the four-way valve and the compressor. (For R32 & R290, you should evacuate the system with the vacuum pump; flush the system with nitrogen; then repeat the two steps before heating up the brazed parts. The operations above should be implemented by professionals.)

NOTE: Remove the panel plate, connection of four-way valve on PCB (refer to “18K Unit Disassembly - Panel Plate” on page 65, “24K Unit Disassembly - Panel Plate” on page 68, “30K/36K Unit Disassembly - Panel Plate” on page 71, or “48K/60K Unit Disassembly - Panel Plate” on page 75 and “All Size Units, Disassembly - Electrical Parts” on page 79) before disassembling sound blanket.

PROCEDURES	ILLUSTRATION
1. Heat up the brazed parts and then detach the the four-way valve and the piper. (see illustration) 2. Remove the four-way valve assembly with pliers. (see illustration)	

All Size Units, Disassembly - Compressor

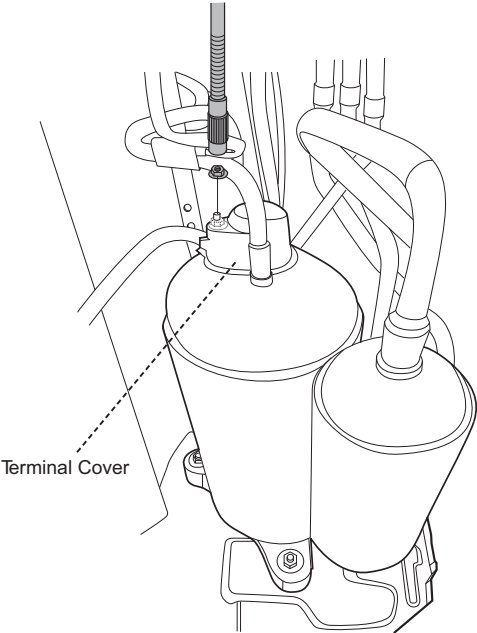
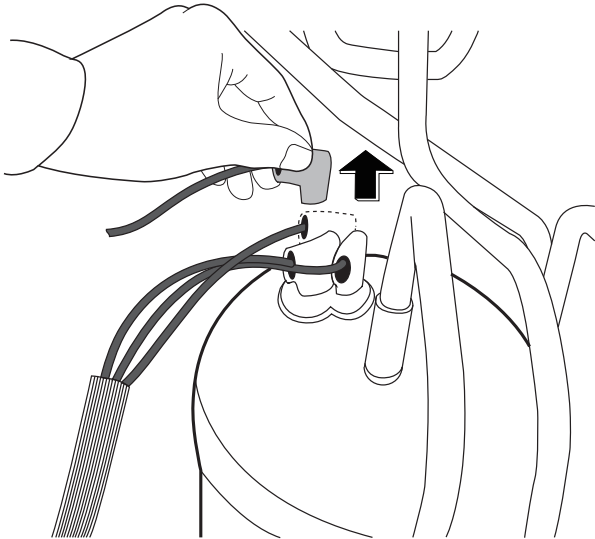


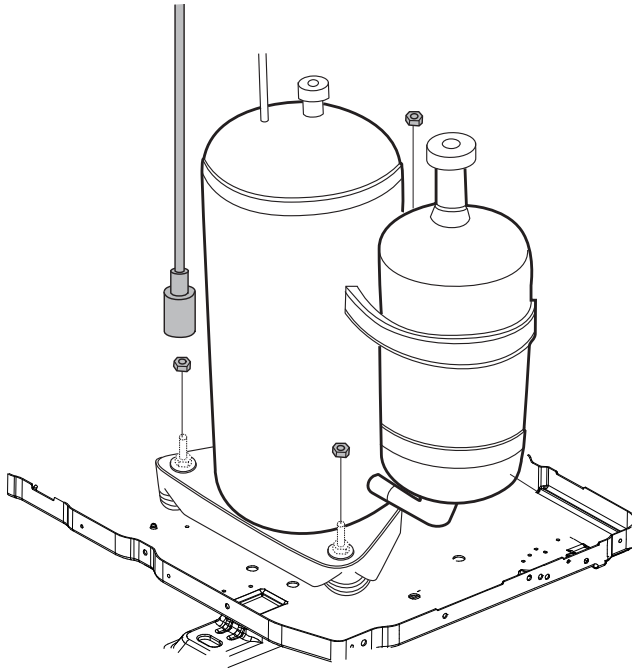
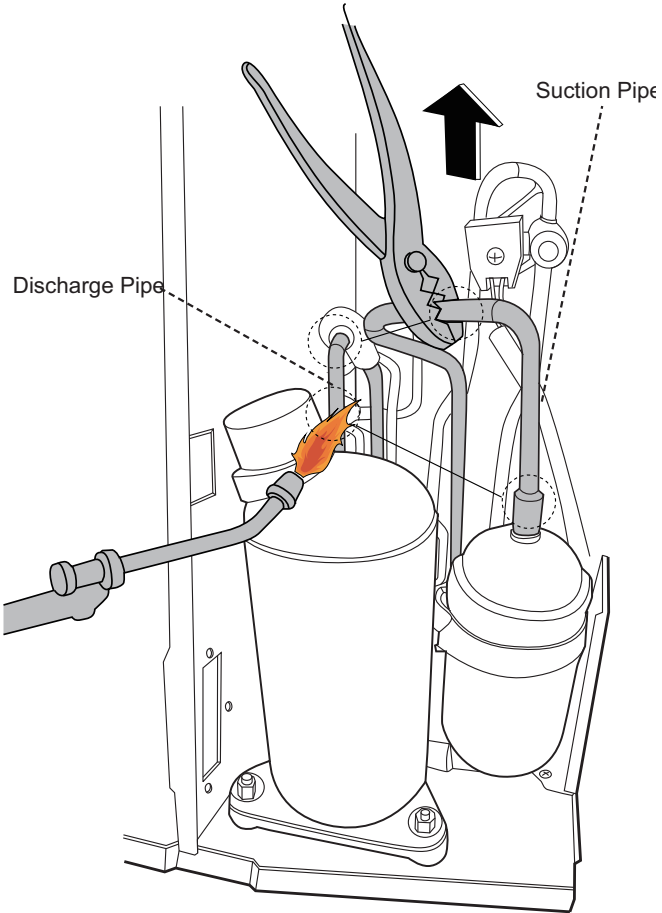
WARNING

EXPLOSION RISK

Evacuate the system and confirm that there is no refrigerant left in the system before removing the four-way valve and the compressor. (For R32 & R290, you should evacuate the system with the vacuum pump; flush the system with nitrogen; then repeat the two steps before heating up the brazed parts. The operations above should be implemented by professionals.)

NOTE: Remove the panel plate, connection of four-way valve on PCB (refer to “18K Unit Disassembly - Panel Plate” on page 65, “24K Unit Disassembly - Panel Plate” on page 68, “30K/36K Unit Disassembly - Panel Plate” on page 71, or “48K/60K Unit Disassembly - Panel Plate” on page 75 and “All Size Units, Disassembly - Electrical Parts” on page 79) before disassembling sound blanket.

PROCEDURES	ILLUSTRATION
1. Remove the flange nut of terminal cover and remove the terminal cover (see illustration)	
2. Disconnect the connectors	

PROCEDURES	ILLUSTRATION
3. Remove the hex nuts and washers securing the compressor, located on the bottom plate. (see illustration)	 This illustration shows the compressor assembly mounted on a base pan. The compressor is a cylindrical unit with two main vertical sections. It is secured to the base pan by four hex nuts and washers, one on each side. A vertical pipe extends from the top of the compressor. The base pan is a flat metal plate with mounting tabs.
4. Heat up the brazed parts and then remove the the discharge pipe and the suction pipe. (see illustration) 5. Lift the compressor from the base pan assembly with pliers. (see illustration)	 This illustration shows the compressor being heated with a torch. The torch flame is applied to the brazed joint where the discharge pipe meets the compressor. A pair of pliers is shown lifting the compressor from the base pan. Labels with dashed lines point to the 'Discharge Pipe' and the 'Suction Pipe'. An upward-pointing arrow indicates the direction of removal for the suction pipe.

APPENDIX**Temperature Sensor Resistance Value Table for TP (°C - K)**

°C	°F	K Ohm	°C	°F	K Ohm	°C	°F	K Ohm	°C	°F	K Ohm
-20	-4	542.7	20	68	68.66	60	140	13.59	100	212	3.702
-19	-2	511.9	21	70	65.62	61	142	13.11	101	214	3.595
-18	0	483	22	72	62.73	62	144	12.65	102	216	3.492
-17	1	455.9	23	73	59.98	63	145	12.21	103	217	3.392
-16	3	430.5	24	75	57.37	64	147	11.79	104	219	3.296
-15	5	406.7	25	77	54.89	65	149	11.38	105	221	3.203
-14	7	384.3	26	79	52.53	66	151	10.99	106	223	3.113
-13	9	363.3	27	81	50.28	67	153	10.61	107	225	3.025
-12	10	343.6	28	82	48.14	68	154	10.25	108	226	2.941
-11	12	325.1	29	84	46.11	69	156	9.902	109	228	2.86
-10	14	307.7	30	86	44.17	70	158	9.569	110	230	2.781
-9	16	291.3	31	88	42.33	71	160	9.248	111	232	2.704
-8	18	275.9	32	90	40.57	72	162	8.94	112	234	2.63
-7	19	261.4	33	91	38.89	73	163	8.643	113	235	2.559
-6	21	247.8	34	93	37.3	74	165	8.358	114	237	2.489
-5	23	234.9	35	95	35.78	75	167	8.084	115	239	2.422
-4	25	222.8	36	97	34.32	76	169	7.82	116	241	2.357
-3	27	211.4	37	99	32.94	77	171	7.566	117	243	2.294
-2	28	200.7	38	100	31.62	78	172	7.321	118	244	2.233
-1	30	190.5	39	102	30.36	79	174	7.086	119	246	2.174
0	32	180.9	40	104	29.15	80	176	6.859	120	248	2.117
1	34	171.9	41	106	28	81	178	6.641	121	250	2.061
2	36	163.3	42	108	26.9	82	180	6.43	122	252	2.007
3	37	155.2	43	109	25.86	83	181	6.228	123	253	1.955
4	39	147.6	44	111	24.85	84	183	6.033	124	255	1.905
5	41	140.4	45	113	23.89	85	185	5.844	125	257	1.856
6	43	133.5	46	115	22.89	86	187	5.663	126	259	1.808
7	45	127.1	47	117	22.1	87	189	5.488	127	261	1.762
8	46	121	48	118	21.26	88	190	5.32	128	262	1.717
9	48	115.2	49	120	20.46	89	192	5.157	129	264	1.674
10	50	109.8	50	122	19.69	90	194	5	130	266	1.632
11	52	104.6	51	124	18.96	91	196	4.849	?	?	?
12	54	99.69	52	126	18.26	92	198	4.703	?	?	?
13	55	95.05	53	127	17.58	93	199	4.562	?	?	?
14	57	90.66	54	129	16.94	94	201	4.426	?	?	?
15	59	86.49	55	131	16.32	95	203	4.294	?	?	?
16	61	82.54	56	133	15.73	96	205	4.167	?	?	?
17	63	78.79	57	135	15.16	97	207	4.045	?	?	?
18	64	75.24	58	136	14.62	98	208	3.927	?	?	?
19	66	71.86	59	138	14.09	99	210	3.812	?	?	?

Other Temperature Sensors Resistance Value Table (°C - K)

°C	°F	K Ohm	°C	°F	K Ohm	°C	°F	K Ohm	°C	°F	K Ohm
-20	-4	115.266	20	68	12.6431	60	140	2.35774	100	212	0.62973
-19	-2	108.146	21	70	12.0561	61	142	2.27249	101	214	0.61148
-18	0	101.517	22	72	11.5	62	144	2.19073	102	216	0.59386
-17	1	96.3423	23	73	10.9731	63	145	2.11241	103	217	0.57683
-16	3	89.5865	24	75	10.4736	64	147	2.03732	104	219	0.56038
-15	5	84.219	25	77	10	65	149	1.96532	105	221	0.54448
-14	7	79.311	26	79	9.55074	66	151	1.89627	106	223	0.52912
-13	9	74.536	27	81	9.12445	67	153	1.83003	107	225	0.51426
-12	10	70.1698	28	82	8.71983	68	154	1.76647	108	226	0.49989
-11	12	66.0898	29	84	8.33566	69	156	1.70547	109	228	0.486
-10	14	62.2756	30	86	7.97078	70	158	1.64691	110	230	0.47256
-9	16	58.7079	31	88	7.62411	71	160	1.59068	111	232	0.45957
-8	18	56.3694	32	90	7.29464	72	162	1.53668	112	234	0.44699
-7	19	52.2438	33	91	6.98142	73	163	1.48481	113	235	0.43482
-6	21	49.3161	34	93	6.68355	74	165	1.43498	114	237	0.42304
-5	23	46.5725	35	95	6.40021	75	167	1.38703	115	239	0.41164
-4	25	44	36	97	6.13059	76	169	1.34105	116	241	0.4006
-3	27	41.5878	37	99	5.87359	77	171	1.29078	117	243	0.38991
-2	28	39.8239	38	100	5.62961	78	172	1.25423	118	244	0.37956
-1	30	37.1988	39	102	5.39689	79	174	1.2133	119	246	0.36954
0	32	35.2024	40	104	5.17519	80	176	1.17393	120	248	0.35982
1	34	33.3269	41	106	4.96392	81	178	1.13604	121	250	0.35042
2	36	31.5635	42	108	4.76253	82	180	1.09958	122	252	0.3413
3	37	29.9058	43	109	4.5705	83	181	1.06448	123	253	0.33246
4	39	28.3459	44	111	4.38736	84	183	1.03069	124	255	0.3239
5	41	26.8778	45	113	4.21263	85	185	0.99815	125	257	0.31559
6	43	25.4954	46	115	4.04589	86	187	0.96681	126	259	0.30754
7	45	24.1932	47	117	3.88673	87	189	0.93662	127	261	0.29974
8	46	22.5662	48	118	3.73476	88	190	0.90753	128	262	0.29216
9	48	21.8094	49	120	3.58962	89	192	0.8795	129	264	0.28482
10	50	20.7184	50	122	3.45097	90	194	0.85248	130	266	0.2777
11	52	19.6891	51	124	3.31847	91	196	0.82643	131	268	0.27078
12	54	18.7177	52	126	3.19183	92	198	0.80132	132	270	0.26408
13	55	17.8005	53	127	3.07075	93	199	0.77709	133	271	0.25757
14	57	16.9341	54	129	2.95896	94	201	0.75373	134	273	0.25125
15	59	16.1156	55	131	2.84421	95	203	0.73119	135	275	0.24512
16	61	15.3418	56	133	2.73823	96	205	0.70944	136	277	0.23916
17	63	14.6181	57	135	2.63682	97	207	0.68844	137	279	0.23338
18	64	13.918	58	136	2.53973	98	208	0.66818	138	280	0.22776
19	66	13.2631	59	138	2.44677	99	210	0.64862	139	282	0.22231

System Pressure Table-R454B

Pressure			Temperature		Pressure			Temperature	
Kpa	bar	PSI	°C	°F	Kpa	bar	PSI	°C	°F
58.196	0.58	8.44	-60	-76	935.23	9.35	135.64	8	46.4
61.517	0.62	8.92	-59	-74.2	963.75	9.64	139.78	9	48.2
64.988	0.65	9.43	-58	-72.4	992.93	9.93	144.01	10	50
68.615	0.69	9.95	-57	-70.6	1022.8	10.23	148.34	11	51.8
72.402	0.72	10.50	-56	-68.8	1053.3	10.53	152.76	12	53.6
76.354	0.76	11.07	-55	-67	1084.5	10.85	157.29	13	55.4
80.478	0.80	11.67	-54	-65.2	1116.4	11.16	161.91	14	57.2
84.776	0.85	12.30	-53	-63.4	1149	11.49	166.64	15	59
89.256	0.89	12.95	-52	-61.6	1182.3	11.82	171.47	16	60.8
93.923	0.94	13.62	-51	-59.8	1216.3	12.16	176.40	17	62.6
98.781	0.99	14.33	-50	-58	1251.1	12.51	181.45	18	64.4
103.84	1.04	15.06	-49	-56.2	1286.6	12.87	186.60	19	66.2
109.1	1.09	15.82	-48	-54.4	1322.8	13.23	191.85	20	68
114.56	1.15	16.61	-47	-52.6	1359.9	13.60	197.23	21	69.8
120.25	1.20	17.44	-46	-50.8	1397.7	13.98	202.71	22	71.6
126.15	1.26	18.30	-45	-49	1436.3	14.36	208.31	23	73.4
132.28	1.32	19.18	-44	-47.2	1475.7	14.76	214.02	24	75.2
138.64	1.39	20.11	-43	-45.4	1515.9	15.16	219.85	25	77
145.24	1.45	21.06	-42	-43.6	1557	15.57	225.82	26	78.8
152.09	1.52	22.06	-41	-41.8	1598.9	15.99	231.89	27	80.6
159.18	1.59	23.09	-40	-40	1641.6	16.42	238.09	28	82.4
166.54	1.67	24.15	-39	-38.2	1685.2	16.85	244.41	29	84.2
174.15	1.74	25.26	-38	-36.4	1729.7	17.30	250.86	30	86
182.04	1.82	26.40	-37	-34.6	1775	17.75	257.43	31	87.8
190.2	1.90	27.59	-36	-32.8	1821.3	18.21	264.15	32	89.6
198.65	1.99	28.81	-35	-31	1868.4	18.68	270.98	33	91.4
207.39	2.07	30.08	-34	-29.2	1916.5	19.17	277.95	34	93.2
216.42	2.16	31.39	-33	-27.4	1965.6	19.66	285.08	35	95
225.76	2.26	32.74	-32	-25.6	2015.5	20.16	292.31	36	96.8
235.41	2.35	34.14	-31	-23.8	2066.5	20.67	299.71	37	98.6
245.37	2.45	35.59	-30	-22	2118.4	21.18	307.24	38	100.4
255.67	2.56	37.08	-29	-20.2	2171.3	21.71	314.91	39	102.2
266.29	2.66	38.62	-28	-18.4	2225.2	22.25	322.73	40	104
277.25	2.77	40.21	-27	-16.6	2280.2	22.80	330.70	41	105.8
288.56	2.89	41.85	-26	-14.8	2336.1	23.36	338.81	42	107.6
300.22	3.00	43.54	-25	-13	2393.2	23.93	347.09	43	109.4
312.24	3.12	45.28	-24	-11.2	2451.3	24.51	355.52	44	111.2
324.63	3.25	47.08	-23	-9.4	2510.4	25.10	364.09	45	113
337.39	3.37	48.93	-22	-7.6	2570.7	25.71	372.84	46	114.8
350.54	3.51	50.84	-21	-5.8	2632.1	26.32	381.74	47	116.6
364.08	3.64	52.80	-20	-4	2694.7	26.95	390.82	48	118.4
378.02	3.78	54.83	-19	-2.2	2758.3	27.58	400.04	49	120.2
392.37	3.92	56.91	-18	-0.4	2823.2	28.23	409.46	50	122
407.13	4.07	59.05	-17	1.4	2889.3	28.89	419.04	51	123.8

System Pressure Table-R454B (Continued)

Pressure			Temperature		Pressure			Temperature	
Kpa	bar	PSI	°C	°F	Kpa	bar	PSI	°C	°F
422.31	4.22	61.25	-16	3.2	2956.5	29.57	428.79	52	125.6
437.92	4.38	63.51	-15	5	3025	30.25	438.72	53	127.4
453.98	4.54	65.84	-14	6.8	3094.7	30.95	448.83	54	129.2
470.47	4.70	68.23	-13	8.6	3165.7	31.66	459.13	55	131
487.43	4.87	70.69	-12	10.4	3238.1	32.38	469.63	56	132.8
504.84	5.05	73.22	-11	12.2	3311.7	33.12	480.30	57	134.6
522.73	5.23	75.81	-10	14	3386.7	33.87	491.18	58	136.4
541.1	5.41	78.48	-9	15.8	3463	34.63	502.25	59	138.2
559.95	5.60	81.21	-8	17.6	3540.7	35.41	513.52	60	140
579.31	5.79	84.02	-7	19.4	3619.9	36.20	525.00	61	141.8
599.16	5.99	86.90	-6	21.2	3700.5	37.01	536.69	62	143.6
619.54	6.20	89.85	-5	23	3782.7	37.83	548.61	63	145.4
640.43	6.40	92.88	-4	24.8	3866.3	38.66	560.74	64	147.2
661.86	6.62	95.99	-3	26.6	3951.5	39.52	573.10	65	149
683.82	6.84	99.18	-2	28.4	4038.3	40.38	585.69	66	150.8
706.34	7.06	102.44	-1	30.2	4126.8	41.27	598.52	67	152.6
729.41	7.29	105.79	0	32	4217	42.17	611.60	68	154.4
753.06	7.53	109.22	1	33.8	4309	43.09	624.95	69	156.2
777.28	7.77	112.73	2	35.6	4402.9	44.03	638.56	70	158
802.08	8.02	116.33	3	37.4	4498.7	44.99	652.46	71	159.8
827.48	8.27	120.01	4	39.2	4596.5	45.97	666.64	72	161.6
853.49	8.53	123.78	5	41	4696.5	46.97	681.15	73	163.4
880.11	8.80	127.64	6	42.8	4798.9	47.99	696.00	74	165.2
907.35	9.07	131.60	7	44.6	4904.1	49.04	711.25	75	167